Class 10
Physics

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1. Light Reflection

Solutions

LEVEL-I

1. Correct option: (D) Light cannot travel through a vacuum is the wrong statement.

That "Light cannot travel through a vacuum" is untrue. It is true that light may pass through a vacuum. In reality, it is known that light can travel through empty spaces, including the vacuum of space. This is so because electromagnetic waves, which make up light, may pass across empty space.

Therefore, D) Light cannot travel through vacuum is the wrong statement.

2. Correct option: (C) Because only a portion of the incident light is reflected and quite a large portion goes into water.

Because just a small fraction of the incident light is reflected and a sizable percentage travels into the water, the images of clouds and trees in water are always darker than they are in reality.

A portion of the light is reflected at the surface when it travels from one medium (like air) into another (like water), while the remaining component is transmitted into the water. Since just a portion of the incident light is reflected, the image seems less bright than it would be in reality. The reflected light is what allows us to see the image.

3. Correct option: (C) Convex.

In a mirror or window pane, if your face seems larger than usual, it means that the reflecting surface is convex. The surface of a convex mirror or window pane bulges outward, making the reflected image look larger. This results from the way light rays interact with a convex surface, where they reflect and diverge.

4. Correct option: (B) Convex mirror.

From a point source of light, a convex mirror can create a parallel beam of light. When light rays from a point source diverge and hit a convex mirror, the reflection causes the light to spread out and give the impression that it is originating from a single virtual point, or focal point, behind the mirror. The reflected light appears parallel as a result, producing a parallel beam.

A flat mirror would reflect light without changing its direction, whereas a concave mirror would converge light rays onto a focal point.

Therefore, the correct answer is B) Convex mirror.

5. Correct option: (A) Under no circumstances is right answer.

The reflected picture in a convex mirror is always virtual, upright, and scaled down. No matter how far the real object is from the mirror, it never creates an inverted image of it.

Option B) When the object is very far from the mirror is wrong because, even in this case, a convex mirror will still provide a virtual, erect, and diminished picture of the thing.

Option C) When the object is at a distance equal to the radius of the mirror is wrong because it doesn't alter how a convex mirror creates an image.

Option D) Because a convex mirror does not actually have a focal length, Option D) When the object is at a distance equal to the focal length of the mirror is similarly inaccurate. A virtual point behind the mirror serves as the convex mirror's focal point.

So, A) Under no circumstances is the right response.

6. Correct option: (D) Same for all lights.

A spherical mirror's focal length is constant for all light colours, regardless of whether it is concave or convex. The curvature of the mirror surface and the mirror's geometry, not the wavelength or colour of the light, define the focal length.

Therefore, regardless of whether the light is red, blue, white, or another colour, a spherical mirror will always have the same focal length.

7. Correct option: (C) For both convex and concave mirrors.

The relationship between a spherical mirror's focal length (f) and radius of curvature (r) is represented by the equation $f = \frac{r}{2}$. This equation applies to mirrors with both convex and concave surfaces.

The focal point and the centre of curvature are situated on the same side as the incoming light rays in a concave mirror, when the reflecting surface is concave inward. The formula $f = \frac{r}{2}$ is correct since the focal length in this instance is positive.

The focal point and the centre of curvature are on the side away from the incoming light rays in a convex mirror, when the reflecting surface is curved outward. For a convex mirror, the focal length is negative, but its absolute value is still equal to half of the radius of curvature. Therefore, convex mirrors can also be described by the equation $f = \frac{r}{2}$.

As a result, both convex and concave mirrors may be described by the equation $f = \frac{r}{2}$, making option C) the right response.

8. Correct option: (B) Concave mirror.

Concave mirrors provide actual as well as virtual, upright and inverted, lowered same-sized and amplified pictures depending on the position of the object relative to that main axis. Such a concave mirror produces a virtual and magnified image when an object lies between the focus and the pole. Only a concave mirror like that, along with a convex lens, could create an enlarged image of an object. Despite the fact that both a convex mirror and a concave lens consistently produce distorted images of objects.

9. Correct option: (B) Convex

Convex mirrors must be utilised in the rearview mirror because they produce smaller but still upright pictures. The cars you see are not inverted because concave mirrors would not provide an inverted image. Convex mirrors make everything appear smaller, allowing us to see more behind us when seated in the car.

Compared to the convex mirror, the plane mirror takes up less space while producing images of things that are the same size.

10. Correct option: (D) Zero

When an object is positioned near the concave mirror's center of curvature, the mirror's image is also created there. As a result, the distance between the object and the image created by the concave mirror is zero, which is the minimal distance.

11. Correct option: (C) +5D.

Given: Focal length of convex mirror, $f = 20\text{cm} = 0.2\text{m}$

In this question, we have to calculate the power of the mirror.

Power of mirror is given by as,

$$P = \frac{1}{f}$$

Substitute the value of f , we get,

$$P = \frac{1}{0.2}$$

$$P = 5D$$

So, the power of the mirror is $+5D$. The correct option is (C) $+5D$.

12. Correct option: (D).

The focal length of a concave mirror does not depend on colour of light. Instead of being affected by the colour or wavelength of the light, a spherical mirror's focal length is influenced by its geometry and the curvature of the mirror surface. The focal length of a concave mirror is given by $f = \frac{R}{2}$ where r is the radius of curvature of the mirror. Therefore, the radius of the mirror's curvature affects the focal length of a concave mirror. Therefore, claim (i) is untrue.

If a spherical mirror is immersed in a liquid, its focal length will remain unchanged. The focal length of the mirror is unaffected by a change in medium, such as submerging the mirror in liquid. However, the change in medium may affect the speed of light. The focal length will remain constant as long as the mirror's form and curvature stay the same. Therefore, statement (ii) is true

13. Correct option: (C) $\frac{(n+1)}{n} f$ **.**

Given: The focal length of the mirror is f .

The image must be inverted and the magnification ratio must be negative because it is a real image.

Magnification= $-\frac{v}{u}$

Here, $u = d$ is the distance of the object and image distance is v .

Now, $-n = \frac{-v}{d}$

$v = nd$

According to mirror formula,

$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

$\frac{1}{f} = \frac{1}{nd} + \frac{1}{d}$

$d = \frac{(n+1)f}{n}$

So, the correct option is (C) $\frac{(n+1)}{n} f$.

14. Correct option: (B) $\sqrt{x_1 x_2}$ **.**

The focal length of the mirror is f .

Now, $u = -(x_1 + f), v = -(x_2 + f)$

Using the mirror formula,

$$\frac{1}{f}=\frac{1}{u}+\frac{1}{v}$$

$$\frac{-1}{x_2+f}=\frac{-1}{x_1+f}=-\frac{1}{f}$$

$$\frac{x_1+x_2+2f}{(x_1+f)(x_2+f)}=\frac{1}{f}$$

$$(x_1+x_2+2f)f=(x_1+f)(x_2+f)$$

$$(x_1+x_2)f+2f^2=x_1x_2+(x_1+x_2)f+f^2$$

$$f^2=x_1x_2$$

$$f=\sqrt{x_1x_2}$$

So, The correct option is (B) $\sqrt{x_1x_2}$.

15. I need more information.(A)

16. Correct option: (C) 1.60cm **(Concave).**

Given: Distance of the object, $u=-60\text{m}$.

Magnification, $m=+4$

In this question we have to find radius of curvature.

The magnification is given by as,

$$m=\frac{-v}{m}$$

$$4=\frac{\text{v}}{0.60}$$

$$\text{v}=4\times0.60$$

$$\text{v}=2.4\text{m}$$

Now using mirror formula,

$$\frac{1}{f}=\frac{1}{v}+\frac{1}{u}$$

$$\frac{2}{R}=\frac{1}{2.4}-\frac{1}{0.60}=-1.25$$

$$\frac{2}{R}=-1.25$$

$$R=-\frac{2}{1.25}$$

$$R=-1.6\text{cm}$$

We determine that the given mirror has a radius of curvature of -1.6cm. By convention, we know that the given mirror is concave because the radius was discovered to be negative.

The correct option is (C) 1.60cm (Concave).

17. Correct option: (D) It is a concave mirror of focal length 24 cm

Given: The distance between the object and the image is 90 cm.

Spherical mirror forms image four times then object.

$$\text{Image size}(v)=4\times\text{object size}(u)$$

$$v=4u$$

Now, the distance between the object and the image is 90 cm.

$$v+u=90\text{cm}$$

Put the value of v ,we get,

$4u+u=80$

$5u=90$

$u=18\text{cm}$

$v+18=90$

$v=72\text{cm}$

The image is erect therefore,

$v=72\text{cm}$

u=-18cm

Now using mirror formula,

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

$$\frac{1}{72}-\frac{1}{-18}=\frac{1}{f}$$

$$f=-24\text{cm}$$

So, the mirror is concave with focal length 24cm.

The correct option is (D) It is a concave mirror of focal length 24 cm

18. Correct option: (A) Virtual, erect and enlarged three times.

Given: Convex mirror focal length, $f=-15\text{cm}$.

Object distance, $u=-10\text{cm}$

The object is a virtual object since the incident rays are convergent behind the mirror, and the image created by this virtual object is virtual and inverted as $u>f$.

According to mirror formula,

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

$$\frac{1}{v}=\frac{1}{f}-\frac{1}{u}$$

Substitute the given data we get,

$$\frac{1}{v}=\frac{1}{-15}-\frac{1}{10}$$

$v=30\text{cm}$

Now, magnification of convex mirror,

$$m=\frac{-v}{u}=\frac{-30}{-10}=+3$$

$m=3$

As a result, the image created will be Virtual, erect and enlarged three times and located 30 metres away.

The correct option is (A) Virtual, erect and enlarged three times.

19. Correct option: (D)

Given: focal length of the mirror, $f=1\text{m}$.

Object distance, $u=-1.5\text{m}$

Using the mirror formula,

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$$

$$\frac{1}{v}=\frac{1}{f}+\frac{1}{u}$$

$$\frac{1}{v}=\frac{1}{1}+\frac{1}{-1.5}$$

$v=3\text{m}$

I need more information.

20. Correct option: (D) Between P and F.

The focal length (F) and centre of curvature (C) of a concave mirror are used to relate the positions of the object and the image.

The image that forms when an object is positioned between a concave mirror's focal point (F) and centre of curvature (C) is virtual, upright, and magnified. On the same side of the mirror as the item, it is also situated.

In the example given, the object is said to be between the focal point (F) and the centre of curvature (C) when it is positioned so that the image generated is twice the size of the object. This is so that a magnified and virtual image may be produced.

It is implied that the object has moved farther away from the mirror but is still between the focal point (F) and the centre of curvature (C) when it is moved away from the mirror to a new position and it is discovered that the size of the image is once more double that of the object.

Therefore, the object lies between the focal point (F) and the center of curvature (C), which corresponds to option D) Between P (pole) and F (focal point).

The correct option is (D) Between P and F.

21. Correct option: (D) Virtual and inverted.

Given:

The object distance, $u=30\text{cm}$

The focal length, $f=20\text{cm}$

Using the mirror formula,

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

$$\frac{1}{v}+\frac{1}{30}=\frac{1}{20}$$

$v=60\text{cm}$

So, the image is virtual $(v>0)$.

The magnification is given by as,

$$m=\frac{-v}{m}=\frac{-60}{30}=-3(<0)$$

So, the image in inverted.

The correct option is (D) Virtual and inverted

22. Correct option: (B) 15cm.

Given: The object distance, $u=-20cm$

Magnification, $m=3$

The magnification is given by as,

$u = -20cm$

$m = 3$

$m = \frac{-v}{u}$

$3 = \frac{-v}{-20}$

$v = -60\text{cm}$

Now using mirror formula ,

$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

$\frac{1}{f} = \frac{1}{-60} - \frac{1}{v}$

$f = -15cm$

So, the focal length is 15cm.

The correct option is (B) 15cm.

23. Correct option: (C) Virtual and at $\frac{20}{3}$ cm from the mirror.

Given: The object distance, $u = -20\text{cm}$

The focal length, f =+10cm

Using the mirror formula,

$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

$\frac{1}{+10} = \frac{1}{v} + \frac{1}{-20}$

$v = \frac{-20}{3}$

So, the image formed by as mirror is Virtual and at $\frac{20}{3}$ cm from the mirror.

The correct option is (C) Virtual and at $\frac{20}{3}$ cm from the mirror.

24. Correct option: (A) $v = 30\text{cm}$ **; m=3 (real, inverted)**

Given: The object distance, $u = -10\text{cm}$

Focal length, $f = \frac{-15}{2}\text{cm}$ (Half of the radius of curvature)

Now according to mirror formulas,

$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$

$\frac{1}{v} = -\frac{2}{15} + \frac{1}{10}$

$\frac{1}{v} = \frac{1}{-30}$

$v = -30\text{cm}$

The image is real and it appears in front of the mirror at a distance of 30 cm.

Magnification is given by as,

$m = \frac{-v}{u}$

$m = \frac{30}{-10}$

$m = -3$

This image is three times magnified and inverted.

The correct option is (A) $v = 30\text{cm}$; m=3 (real, inverted)

25. Correct option: (D) $\frac{180}{11}\text{cm}$.

Given: Magnification, $m = -4.5$

Object distance is , $u = -20\text{cm}$

Magnification is given by as,

$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$

$\frac{1}{v} = \frac{1}{-20} - \frac{1}{-20}$

$\frac{1}{v} = 0$

$v = \infty$

Now using mirror formula,

$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$\frac{1}{-90} + \frac{1}{-20} = \frac{1}{f}$

$\frac{1}{f} = -\left(\frac{1}{90} + \frac{1}{20}\right)$

$\frac{1}{f} = \frac{-110}{1800}$

$\frac{1}{f} = \frac{-11}{180}$

$f = \frac{180}{11}\text{cm}$

So, the focal length is $\frac{180}{11}\text{cm}$.

The correct option is (D) $\frac{180}{11}\text{cm}$.

26. Correct option: (C) A convex mirror .

Because convex mirrors are diverging in nature and all rays passing through them appear to meet at a point but do not actually converge, a diminished virtual image can only be created in a convex mirror.

So, the correct option is (C) A convex mirror .

27. Correct option: (C) 0.55cm.

Given: The object distance, $u = -1m$=100cm

The focal length, f=-10cm (half of radius of curvature)

Object hight, $h_1 = 5\text{cm}$

Now using mirror formula,

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{-10} = \frac{1}{v} - \frac{1}{100}$$

$$\frac{1}{v} = \frac{1}{100} - \frac{1}{10}$$

$$\frac{1}{v} = \frac{1-10}{100}$$

$$\frac{1}{v} = \frac{-9}{100}$$

$$v = -\frac{100}{9}\text{cm}$$

The magnification is given by as,

$$m = \frac{-v}{m} = \frac{h_2}{h_1}$$

$$-\frac{\left(-\frac{100}{9}\right)}{100} = \frac{h_2}{5}$$

$$h_1 = \frac{5}{9}$$

$$h_1 = 0.55\text{cm}$$

So, the size of image is 0.55cm.

The correct option is (C) 0.55cm.

28. Correct option: (A) 50cm

Given: The focal length, $f = 50\text{cm}$

Magnification, m=2

The magnification is given by as,

$$m = \frac{v}{u} = 2$$

$$v = 2u$$

Now using mirror formula,

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{2u} + \frac{1}{u}$$

$$u = \frac{3f}{2}$$

$$u = \frac{3\times 50}{3}$$

$$u = 75\text{cm}$$

So, the object is placed at 75cn from the mirror.

The correct solution is (A) 75cm

29. Correct option: (C) 2cm.

Given: The object hight, $O = -6\text{cm}$

Focal length $= -f$

The object distance, $u = -4f$

The magnification (m) by a spherical mirror is given by,

$$m = \frac{I}{O} \quad \text{.....(1)}$$

Where, I =height of the image, O =height of the object

As follows give the magnifications in terms of focal length (f)and image distance (u):

$$m = \frac{f}{f-u} \text{.....(2}$$

From the equation (1) and (2) we get,

$$\frac{I}{O} = \frac{f}{f-u}$$

$f = 50\text{cm}$ $\quad \frac{I}{6} = \frac{-f}{-f-(-4f)}$

m=2

$$I = -2\text{cm}$$

So, the length of the image is 2cm.

The correct option is (C) 2cm.

30. Correct option: (D) None of these

As is well known, mirrors are used to reflect light that strikes them. The following are the laws of reflection:

(1) The angle of the incident and reflected light beams is equal. As a result, the angle of incidence and the angle of light reflection are equal.

(2) The plane in which the incident ray, the reflected ray, and the normal at the point of incident ray all reside is the same.

Now that we are aware of these rules of reflection, it is evident that the mirror is not impacted by dipping when reflecting light. A concave mirror's convergence is unaffected by any medium because it is also not possible for the light to be refracted in this situation. Convergence (or power) for a mirror is independent of the mirror medium.

So, The correct option is (D) None of these

31. Correct option: (D) 15 cm behind the mirror

Given: focal length, $f = 30\text{cm}$

Object distance, $u = -30\text{cm}$

Now, using the mirror formula,

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} + \frac{1}{-30} = \frac{1}{30}$$

$$\frac{1}{v} = \frac{1}{30} + \frac{1}{30}$$

$$v = 15\text{cm}$$

The image is generated 15cm behind the mirror, in accordance with the sign convention in mirrors.

The correct option is (D) 15 cm behind the mirror

32. Correct option: (D) Always.

A convex mirror constantly creates an is erect, diminished, and virtual. Consequently, which of the following scenarios will result in an image that is erect, diminished, and virtual when using a convex mirror with a focal length of f .

LEVEL-II

1. Correct option: (A) Real, inverted and same in size.

Given:

$u = -40\text{cm}$

$f = -20\text{cm}$

Using the mirror formula,

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{-20} - \frac{1}{-40}$$

$v = -40\text{cm}$

The image is real.

Magnification is given by as,

$$m = \frac{-v}{u} = \frac{-(-40)}{-40} = -1$$

$m = -1$

So, the image is) Real, inverted and same in size.

The correct option is (A) Real, inverted and same in size.

2. Correct option: (D) is wrong statement.

Since light rays going through convex mirrors always diverge and never actually come together at a spot, they can only reflect virtual images.

In these situations, we must stretch the rays and make it appear as though they are meeting behind the mirror. Thus, the photographs are never real.

The option (D) is wrong statement. Real images can never be created by convex mirrors.

3. Correct option: (D) 15 cm.

Given : The object is twice as far as the image, from the lens.

So, $u = 2v$

Using mirror formula,

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$\frac{1}{10} = \frac{1}{v} + \frac{1}{2v}$$

$$\frac{1}{10} = \frac{2v}{3}$$

$$\frac{2v}{3} = 10$$

$v = 15\text{cm}$

$u = 2 \times v$

$u = 2 \times 15$

$u = 30\text{cm}$

The distance of the object from the mirror is 15cm.

The correct option is (D) 15 cm.

4. Correct option: (B) Equal to 30 cm

Because the focal length of a mirror depends on its geometry and radius of curvature, both of which are constant, it does not change when submerged in water.

The correct option is (B) Equal to 30 cm

5. Correct option: (A) 8cm.

Let us consider distance is u .

Given: $v+u=120$ .....(1)

Magnification is , $\frac{v}{u}=16$ .....(2)

Now, from equation (1) and (2),

$\frac{120+9u}{u}=16$

$15u=120$

$u=8\text{cm}$

The distance of the flower from the mirror is 8cm.

The correct option is (A) 8cm.

6. Correct option: (A) 20 cm.

Given: The focal length of the mirror, $f=20\text{cm}$.

In the case of real image,

Let's consider object distance is u_1 and then the image distance is u_2 .

Using mirror formula,

$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$

$\frac{1}{-2u_1}-\frac{1}{u_1}=\frac{-1}{20}$

$u_1=30\text{cm}$

In the case of Virtual image,

The object distance is $-u_1$ and then the image distance is $-u_2$.

Using the mirror formula,

$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$

$\frac{1}{2u_1}+\frac{1}{-u_1}=\frac{-1}{20}$

$u_2=10\text{cm}$

The distance between two positions of object is given as,

$u=30-10$

$u=10\text{cm}$

Thus the distance between the two positions of object is 20cm.

The correct option is (A) 20 cm.

7. Correct option: (B) Convex

The man requires the image created by the mirror to be the same height as his actual height, which is 6 metres, in order to see his full height in the mirror. The observed image in the question, however, is 2 metres hig

It is clear that a convex mirror was utilised because the viewed image is smaller than the man's actual height. The virtual images created by a convex mirror are always smaller than the actual thing. The image created by a convex mirror is smaller in size but upright.
The correct option is (B).

8. Correct option: (C) It minimises spherical aberration and provides a parallel beam of light

Automobile headlights employ parabolic reflectors because they have the ability to reduce spherical aberration. When light rays from a point source do not converge to a single focal point, spherical aberration, an optical distortion, results. It may result in loss of image quality and blurring.
The reflector's parabolic form aids in concentrating the light's rays from the source to a single focal point. By ensuring that the light rays converge into a parallel beam and reduce spherical aberration, a more concentrated and effective beam of light is produced.
Option A) Using parabolic reflectors is not primarily done to produce a broad beam of light. The focus is largely on getting a sharp beam.
Option B) Increasing light intensity has more to do with the type and strength of the light source being used than it does with the reflector's form.
Option D) The reflector's design is not directly related to removing colour effects from the light beam. By choosing light sources with the required colour qualities or by using the proper filters, colour impacts can be reduced.
Therefore, The correct answer is (C) It minimises spherical aberration and provides a parallel beam of light.

9. Correct options: (B,C,D)

Explanation:
A) A convex mirror always produces a virtual image, which is created by the apparent junction of light beams that appear to diverge from a fictitious image point behind the mirror. As a result, statement A is untrue.
B) A convex mirror's image can actually create a virtual image. The image appears to be behind the mirror when the reflected light rays diverge. In contrast to the item, it is tiny and upright. As a result, statement B is true.
C) A concave mirror's image can be amplified. A concave mirror can produce magnified or diminished pictures, depending on the object's position in relation to the focal point and the centre of curvature. Therefore, claim C is true.
D) A concave mirror's image can be reversed. The concave mirror creates a real and inverted image when the object is positioned beyond the focal point. As a result, statement D is true.
Therefore, the correct statements are B), C), and D).

10. Correct option: (A) plane mirror.

The only option presented that creates a virtual and erect image of a real object in all directions is a plane mirror. The reflected rays from a virtual image point behind a flat mirror appear to diverge when an object is placed in front of it. This virtual representation always forms on the same side of the mirror as the actual object and is the same size as the real thing.
Both option B convex mirror and option C concave mirror create virtual images, but depending on how an object is positioned in relation to the mirror, the images created by these mirrors can either be erect or inverted.
Therefore, The correct option is (A) plane mirror.

11. Correct options: (A) and (C).

Explanation:

A) The focal length is dependent on the light's wavelength: This is untrue. Whether a curved mirror is concave or convex, its shape, curvature, and geometry define its focal length; the wavelength of light has no bearing on these factors. Regardless of the colour or wavelength of the light, the focus length stays constant.

B) Its focal length depends on the nature of the medium in which the mirror is placed: This statement also not correct. A curved mirror's focal length is defined by its shape and curvature, not by the properties of the medium it is put in. Regardless of the media in which the mirror is housed, the focal length stays constant as long as the mirror's shape and curvature do.

C) Image formed by it is free from spherical aberration: This statement is correct. Spherical aberration is minimised by using curved reflectors, particularly parabolic mirrors. When parallel light rays are reflected or bent by a curved surface, spherical aberration, an optical distortion, may result. A curved mirror can reduce or completely get rid of spherical aberration by employing an appropriate form, like a parabolic curve.

D) Image formed by it is free from chromatic aberration: This statement is not correct. Spherical aberration is intended to be reduced by curved mirrors, especially parabolic mirrors. Spherical aberration is an optical distortion that can happen when parallel light rays are reflected or bent by an irregular surface. A curved mirror can reduce or even completely remove spherical aberration by employing the proper shape, such as a parabolic curve.

Therefore, the correct answers are A) Its focal length depends on the wavelength of light and B) Its focal length depends on the nature of the medium in which the mirror is placed.

12. Correct options: (A) It can form real image and (B) It can form virtual image

A) It can create a real image: Under some circumstances, a concave mirror can create a real image. A true inverted picture is created in front of the mirror when the object is positioned outside of the mirror's focus point.

B) It can create a virtual image: Concave mirrors can create virtual images, yes. A virtual upright picture is created behind the mirror when the object is positioned between it and the focus point.

The resulting image is always inverted, hence this claim is untrue. Depending on the object's position in relation to the concave mirror, the image produced by the mirror can be either inverted or upright. A true inverted image is created if the object is positioned outside of the focal point. A virtual erect picture is created if the object is positioned between the mirror and the focus point.

D) Image created is always erect: This claim is also untrue. According to the object's placement, a concave mirror can produce an image that is either erect or inverted. A virtual erect picture is created if the object is positioned between the mirror and the focus point.

Therefore, the correct option are (A) It can form real image and (B) It can form virtual image.

13. Correct options: (A) $\frac{f}{2}$ and (B) $\frac{3f}{2}$.

Case:1 Image is virtual.

Let's consider, $u = -x, v = +2x$

Using mirror formula,

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

$\frac{1}{2x}-\frac{1}{x}=\frac{1}{-f}$

$x=\frac{f}{2}$

Case:2 Image is real. $u=-y, v=-2y$

Again using the mirror formula,

$\frac{1}{u}+\frac{1}{v}=\frac{1}{f}$

$\frac{1}{-2y}-\frac{1}{y}=\frac{1}{-f}$

$\frac{3}{2y}=\frac{1}{f}$

$y=\frac{3}{2}f$

So, the correct option are (A) $\frac{f}{2}$ and (B) $\frac{3f}{2}$.

14. Correct options: (A) pole ,(C) Radius of curvature and (D) Principal axis.

The geometry of the mirror determines the pole, radius of curvature, and principal axis.

After reflection, a parallel light beam to the principal axis will meet at the focus. The position of the image would shift after reflection if the beam were not paraxial.

Rays parallel to the principal axis can only be focussed at one place after reflection, according to the Paraxial Approximation, if they are close to the principal axis. What happens if distant parallel rays that strike a spherical mirror are impacted upon it. The principal axis will not in this instance be intersected by the reflected rays at a single location, but rather at multiple points. There is no such thing as focus as a result. We may now state that the beams' paraxiality is necessary for the definition of focus.

Additionally, Pole is the mirror's midpoint; regardless of whether the rays are paraxial or not, this point remains constant. Even this definition is irrespective of whether the rays are near the major axis or not: the radius of curvature is the radius of the sphere, of which the mirror is a part. The principal axis is the line connecting the pole and the centre of curvature; if light rays are not near the principal axis, the pole and centre of curvature do not move, and hence, the principal axis does not change either. Thus, we draw the conclusion that only the definition of focus alters.

So, The correct option are (A) pole ,(C) Radius of curvature and (D) Principal axis.

15. Correct option: (A) At a distance 60 cm in front of mirror,

Given: The object distance is, $u=-30\text{cm}$

The focal length, $f=-20\text{cm}$

Using mirror formula,

$\frac{1}{f}=\frac{1}{v}+\frac{1}{v}$

$\frac{1}{-20}=\frac{1}{v}+\frac{1}{-30}$

$\frac{1}{v}=\frac{1}{-20}+\frac{1}{30}$

$v=-60\text{cm}$

As a result, the image generated is in front of the mirror because we can see that the distance that results is negative.

So, the distance of the mirror is 60 cm in front of mirror.
The correct option is (A) At a distance 60 cm in front of mirror.

16. Correct option: (B) 2.

The object distance is, $u = -30\text{cm}$
The image distance, $v = -60\text{cm}$
The magnification is given by,

$$m = \frac{-v}{u}$$

$$m = -\frac{-60}{-30}$$

$$m = -2$$

So, the magnification 2.
The correct option is (B) 2.

17. Correct option: (A) The image will be moved to infinity .

Object Distance in this case, $u = -20\text{cm}$
Focal length, $f = -20\text{cm}$
Using mirror formula,

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{-20} - \frac{1}{-20}$$

$$\frac{1}{v} = 0$$

$$v = \infty$$

Therefore, the image will be moved to infinity .
The correct option is (A) The image will be moved to infinity .

18. Correct option: (B) 20/3 cm behind the mirror .

Given: The object distance, $u = -20\text{cm}$
The focal length, $f = 10\text{cm}$
Using the mirror formula,

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{v}$$

$$\frac{1}{10} = \frac{1}{v} + \frac{1}{-20}$$

$$\frac{1}{v} = \frac{1}{10} + \frac{1}{20}$$

$$v = \frac{20}{3}\text{cm}$$

The image is created on the side of the mirror that is opposite from the object, according to the positive value for the image distance (v).
So, The image formed at $\frac{20}{3}\text{cm}$ in behind the mirror.
The correct option is (B) 20/3 cm behind the mirror .

19. Correct option: (A) 1/3.

The object distance, $u = -20\text{cm}$

Image distance is, $v = \frac{20}{3}\text{cm}$

Therefore magnification is given by as,

$$m = \frac{-v}{u}$$

$$m = \frac{-\frac{20}{3}}{-20}$$

$$m = \frac{1}{3}$$

The correct option is (A) 1/3.

20. Correct option: (C) 0.167cm.

The magnification is given by as,

$$m = \frac{-v}{u} \qquad \text{.....(1)}$$

Also, $m = \frac{h_i}{h_0}$ .....(2)

From equation(1) and (2) we get,

$$\frac{-v}{u} = \frac{h_i}{h_o}$$

$$\frac{h_i}{0.5} = \frac{1}{3}$$

$$h_i = 0.167\text{cm}$$

The height of the image formed is 0.167cm.

The correct option is (C) 0.167cm.

21. Correct Matching is: a-Q; b-S; c-P; d-T

Explanation:

Magnification by lens is given by as, $m = \frac{f}{f-u}$.

Where, f is focal length and u is object distance.

For convex lens $f = +f$ and for concave lens $f = -f$.

(a) When an object is placed at focal length Before a convex mirror:

$$u = f$$

$$m = \frac{f}{f-(-f)}$$

$$m = \frac{f}{2f}$$

$$m = \frac{1}{2} = 0.5$$

(b) When an object is placed at centre of Curvature before a concave mirror.

$u = -2f$

$m = \dfrac{f}{-f-(-2f)}$

$m = -\dfrac{f}{f}$

$m = -1$

(c) When an object is placed at focus before a concave mirror .

$u = -f$

$m = \dfrac{f}{-f-(-f)}$

$m = -\dfrac{-1}{0}$

$m = -\infty$

(d) When an object is placed at radius of Curvature before a convex mirror.

$u = -2f$

$m = \dfrac{f}{f-(-f)}$

$m = -\dfrac{f}{3f}$

$m = \dfrac{1}{3}$

Therefore, the correct matching is are: a-Q, b-S, c-P, d-T

22. Correct Matching is: a-Q, b-P,S, c-T, d-R

The magnification in a plane mirror is always +1, meaning that the image created is the same size as the object and is neither enlarged nor shrunk.

The height of a virtual image should be considered positive whereas the height of a real image should be considered negative. The presence of virtual and upright images is indicated by a positive magnification magnitude. The sign of a real and inverted image is a magnification that is negative.

23. Correct Answer is: 2

Given: The object distance, $u = -5\text{cm}$

The focal length, $f = -10\text{cm}$

Using the mirror formula,

$\dfrac{1}{f} = \dfrac{1}{v} + \dfrac{1}{u}$

$\dfrac{1}{v} = \dfrac{1}{f} - \dfrac{1}{u}$

$\dfrac{1}{v} = \dfrac{1}{-10} - \dfrac{1}{-5}$

$v = 10\text{cm}$

The image appears behind of the mirror at a distance of 10 cm. The magnification is given by,

$m = \dfrac{-v}{u}$

$m = \dfrac{-10}{-5}$

$m = +2$

+2 magnification reveals a virtual erect image.Then the lateral magnification of object is 2.

2. Light Refraction

LEVEL-I

1. Correct option: (A). $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$.

The formula for refraction at curved surface is given by,

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

Where, μ_2 =Refractive index of the medium In which the refaction takes place

μ_1 =Refractive index of the medium, from where the light is incident on the surface

R = The of the curvature of the surface

2. Correct option: (D) It is the ratio between height of image to height of the object

The image's height to the object's height is the ratio used to determine a lens' magnification. The ratio of picture distance to object distance is another definition of it.

So, the correct option (D)

3. Correct option: (D) centre of curvature.

The lens's imaginary glass sphere's centre of curvature is referred to as the centre of curvature. It is located at a location that lies on the normal vector and is placed at a distance from the curve equal to the radius of curvature. It is the location at infinity where there is no curvature.

4. Correct option: (C) Principal focus.

The principal focus of a lens is a point on its principal axis where, after passing through the lens, light rays parallel to its principal axes converge.

primary focus:

Concave lens: After refraction through the lens, rays parallel to the principal axis converge at the principal focus of a concave lens.

Convex lens: The principal focus of a concave lens is a point where, following refraction through the lens, rays parallel to the principal axis appear to diverge.

The focal length (f) of a lens is the distance it has between its optical centre and its main focus.

5. Correct option: (B) undeviated.

A light ray travelling through the optical centre experiences no refraction, therefore it travels straight through. This is due to the fact that it is perpendicular to the curved surface of the lens as it passes through the optical centre. As a result, both the angle of refraction and the angle of incidence are zero.

6. Correct option: (B) When the object is in between pole and focus

An image that results from placing an object between a convex lens' pole and focus is virtual, upright, and enlarged. Since the light beams do not truly converge to a point, even though the image is produced on the same side as the item, it cannot be projected onto a screen. Instead, they give the illusion of diverging from a point behind the lens.

In the case of a convex lens, Option A) "When the object is at the centre of curvature" does not result in a virtual image. The image created when the object is positioned in the centre of the curve is accurate and upside-down.

The image produced by Option C) "When the object is at the focus" is not clear. In this instance, the light rays are either parallel or divergent, which prevents the development of a distinct image.

The result of Option D) "When the object is beyond the centre of curvature" is a genuine, inverted image rather than a virtual image.

7. Correct option: (B) $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$.

The lens formula is given by as,

$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$

Where,

$O=\sqrt{I_1 I_2}$

$O=\sqrt{8\times 2}$

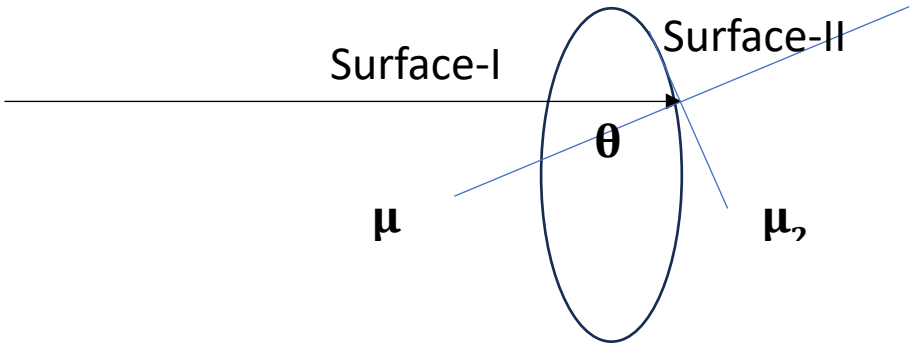
$O=4\text{cm}$

$v=$ distance of image

$u=$ Distance of object

$f=$ Focal length

8. Correct option: (C) $\mu_1=\mu<\mu_2$ **.**



At surface-I, Ray is undeviating.

So, there is no refraction at surface-I

$r_1=r$

At surface-II

$\left(\sin\theta_{\text{I}}\right)r_1=r\sin\theta_{\text{II}}$

$(sin\,\theta_I)r_1=r\,sin\,\theta_{\text{II}}$

$\frac{r_1}{r_2}=\frac{sin\,\theta_{\text{II}}}{sin\,\theta_I}$

$r_2>r_1$

9. Correct option: (C).

There is NO lateral inversion in a convex lens; only transverse inversion.

Convex lenses can provide an upside-down, inverted image of an object, but not one that is laterally inverted.

So, Correct option: (C).

10. Correct option: (A) 2.5 cm.

Given:

$\mu_1=1.5$

$u=-3\text{cm}$

$\mu_2=1$

$R=-5\text{cm}$

Using the lens for refraction through curved surface,

$$\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R}$$
$$\frac{1}{v}-\frac{1.5}{-3}=\frac{1-1.5}{-5}$$
$$\frac{1}{v}+0.5=0.1$$
$$\frac{1}{v}=-0.4$$
$$v=-\frac{10}{4}$$
$$v=-2.5\text{cm}$$

At 2.5cm the surface will the bubble appear.

11. The correct option (B) f_1+f_2.

Let's consider the distance between the lens is d.

Now, power is given by as,

$$p=\frac{1}{f}=\frac{1}{f_1}+\frac{1}{f_2}-\frac{d}{f_1f_2}=0$$
$$\frac{d}{f_1f_2}=\left|\frac{1}{f_1}+\frac{1}{f_2}\right|=\left|\frac{f_1+f_2}{f_1f_2}\right|$$
$$d=|f_1+f_2|$$

The distance between lens is f_1+f_2.

12. The correct option (C) $(n+1)f$.

Image formed by convex lens is real and inverted, so

$$m=\frac{v}{u}=-\frac{1}{n}$$
$$v=\frac{-u}{n}$$

The lens formula for refraction through the curved surface is given by as,

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$$
$$\frac{-n}{u}-\frac{-1}{u}=\frac{1}{f}$$
$$\frac{-1}{u}(n+1)=\frac{1}{f}$$
$$u=-(n+1)f$$

So, the distance of the object from the lens is $(n+1)f$.

13. The correct option (A) $\left(\frac{m+1}{m}\right)f$.

Real image is always inverted , so $m=-m$

We know that,

Magnification is given by as,

$$-m=\frac{v}{u} \qquad \text{.....(1)}$$

The lens formula for refraction through the curved surface is given by as,

$$\frac{1}{f}=\frac{1}{v}-\frac{1}{u} \qquad \text{.....(2)}$$

Where, u =Distance of the object

v =Distance of the image

From equation(1) and (2) we get

$$\frac{1}{f}=\frac{1}{-mu}-\frac{1}{u}$$

$$\frac{1}{f}=\left(\frac{-1}{m}-1\right)\frac{1}{u}$$

$$\frac{1}{f}=\left(\frac{-1-m}{m}\right)f$$

$$u=\left(\frac{-1-m}{m}\right)f$$

$$u=-\left(\frac{m+1}{m}\right)f$$

The object's position is indicated by this expression; however, the distance has a positive value.

So, the object distance is $\left(\frac{m+1}{m}\right)f$.

14. Correct option: (D) 3.

The lens is made of three distinct materials, and the focus length of the lens varies depending on the substance. As a result, three distinct images will be created as the beam converges at these three spots.

15. Correct option: (A) 30 cm left.

Given:

$\mu_1(\text{air})=1$

$\mu_2(\text{glass})=1.5$

$u=-1.5\text{cm}$

R=30cm

Using the lens formula,

$$\frac{\mu_2-\mu_1}{R}=\frac{\mu_2}{v}-\frac{\mu_1}{u}$$

$$\frac{1.5-1}{30}=\frac{1.5}{v}-\frac{1}{1.5}$$

$v=-30\text{cm}$

So, the image would be forms at 30cm left.

16. Correct option: (B) The full image will be formed but it will be less bright.

As a result of several rays from the object, an image was created. As a result, an image will develop if any portion of the lens is covered. Compared to an image created with a lens wrapped, the brightness will be lower. Consequently, the whole image will form even if the centre of the lens is wrapped with black paper, but it will be less brilliant.

So, Correct option: (B) The full image will be formed but it will be less bright

17. The correct option (C) 4cm.

As we know, the height of object can be obtained as:

$O = \sqrt{I_1 I_2}$

$O = \sqrt{8 \times 2}$

$O = 4\text{cm}$

So, the height of object is 4cm,

18. Correct option: (D) 10cm.

Given:

$R = -10\text{cm}$

$\mu - 1.5$

The focal length of the combination of lenses is given by as,

$$f = \frac{R}{2(\mu - 1)}$$

$$f = \frac{-10}{2(1.5 - 1)}$$

$$f = -10\text{cm}$$

The convex lens focal length is 10cm.

19. Correct option: (C) 1.3D.

Given:

$R_1 = 1.6$

$P = 4D$

Using the formula of lens maker,

$$P = (\mu - 1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$$

Substitute the values we get,

$$P = (\mu - 1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right] \text{.....(1)}$$

$$4 = (1.6 - 1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$$

$$\left[\frac{1}{R_1} - \frac{1}{R_2}\right] = \frac{4}{0.6}$$

$$\left[\frac{1}{R_1} - \frac{1}{R_2}\right] = \frac{20}{3} \text{.....(2)}$$

We know that refractive index of water is 1.33.

Therefore, refractive index of glass with respect to water= $\frac{1.6}{1.33}$.

So, the power of lens in water using equation (1) and (2) we get,

$$P = \left(\frac{1.6}{1.33} - 1\right)\left(\frac{20}{3}\right)$$

$P = 1.3533\text{D}$

So, the power in water is 1.33D.

20. Correct option: (A) Only one image will be formed by the lens.

Only the intensity of the image is reduced when a lens is painted as instructed since the darkened area does not refract the incident light to create an image.

While the transparent object still refracts like a biconvex lens to produce a single image which given by as,

$f_2 \quad \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

This lens system will only produce one image because the optic axes of both components coincide. Their distinct visual axes would have resulted in the formation of two images.
So, Correct option: (A) Only one image will be formed by the lens.

21. Correct option: (B) Move to the right.

Separation between the lenses reduces the system's power. Therefore, the focus length increases.

22: Correct option: (B) 32cm, 16cm

Given:

$f = -24\text{cm}$

Magnification is given by as,

$$\text{m} = \frac{f}{f-u}$$

If, $m = +3$,then

$$3 = \frac{-24}{-24-u}$$

$$-24-u = -8$$

$$u + 24 = -8$$

$$u = 8 - 24$$

$$u = -16\text{cm}$$

Now, if m=-3 ,then

$$-3 = \frac{-24}{-24-u}$$

$$u + 24 = -8$$

$$u = -32\text{cm}$$

The object distance is 32cm, 16cm.

23. Correct option: (B) Convex, 50cm.

Let's consider focal length of the convex lens be, f_1 and focus lens of the concave lens is f_2

Given:

Power of convex lens, $P_1 = +6\text{dioptre}$

Power of concave lens, $P_2 = -4\text{dioptre}$

$\mu_1 = 1.0$
$\mu_2 = 1.5$
$R = 20\text{cm}$
$u = \infty$

Using the power of lens= $\frac{1}{f}$

Where, f =focal length

Let focal length of the combination be F ,
Therefore,

$$\frac{1}{F}=\frac{1}{f_1}+\frac{1}{f_2}$$

$$\frac{1}{F}=6+(-4)$$

$$F=50\text{cm}$$

F turns out to be positive, hence the two lenses combined form a convex lens.

24. Correct option: (A) Parallel to the principal axis

Given:

$f_1 = 1m$

$f_2 = -0.25m$

d=0.75m

Using formula of power of combination of lens,

$$P=\frac{1}{f_1}+\frac{1}{f_2}=\frac{d}{f_1 f_2}$$

Substitute the value we get,

$$P=\frac{1}{1}+\frac{1}{-0.25}=\frac{0.75}{(1)(-0.25)}$$

$$P=1-4+3$$

$$p=-3+3$$

$$P=0$$

The incident parallel beam of light will remain parallel after leaving the system since there is no power in the system.

25. Correct option: (C) 109.1cm.

$u=-100\text{cm}$

$v=-120\text{cm}$

The focal length is given by as,

$$\frac{1}{F}=\frac{1}{u}+\frac{1}{v}$$

$$\frac{1}{F}=\frac{1}{-100}+\frac{1}{-120}$$

$$\frac{1}{F}=\frac{-11}{600}$$

Now,

$$\frac{1}{F}=\frac{1}{f_L}+\frac{2}{f_m}$$

Or $f_1=2F$

Or,

$$f_1=2\times\frac{600}{11}$$

$$f_1=\frac{1200}{11}$$

$$f_1=109.1\text{cm}$$

So, the focal length of the lens is 109.1cm.

26. Correct option: (A) 6cm from S_1.

Given: The focal length, $f = 9\text{cm}$

Here, one of the images will be real and the other will be virtual. Assume for the moment that S_1 real image is S_2 is a virtual .

Using lens formula for refraction through curved surface.

27. Correct option: (D) $2f$.

Given: The correct option (D) $2f$.

For part A,

As can be seen from the following calculation, the focal length remains unchanged when cut horizontally.

$R_1 = R_2 = R$

Using the lens maker formula we have,

$$\frac{1}{f} = (\mu - 1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$$

$$\frac{1}{f} = (\mu - 1)\left[\frac{1}{R} - \frac{1}{(-R)}\right]$$

$$\frac{1}{f} = \frac{2(\mu - 1)}{R} \text{.....(1)}$$

The lens is divided in half as it is sliced vertically; as a result, $R_2 = 0$

$$\frac{1}{f_2} = (\mu - 1)\left(\frac{1}{R_1}\right)$$

$$\frac{1}{f_2} = (\mu - 1)\left(\frac{1}{R}\right)$$

$$f_2 = \frac{R}{(\mu - 1)} \text{.....(2)}$$

From equation (1) and (2),

$$\frac{f_2}{f} = \frac{\frac{R}{(\mu - 1)}}{\frac{R}{2(\mu - 1)}}$$

$$\frac{f_2}{f} = 2$$

$$f_2 = 2$$

$$f_2 = 2f$$

So, the focal length of part A of the lens is $2f$.

28. Correct option: (D) 20cm.

Given:

$$\mu = \frac{3}{2}$$

$$R_1 = 10\text{cm}$$

Using the lens maker formula,

$$\frac{1}{f}=(\mu-1)\left[\frac{1}{R_1}-\frac{1}{R_2}\right]$$

$$\frac{1}{f}=\left(\frac{3}{2}-1\right)\left[\frac{1}{10}-\frac{1}{\infty}\right]$$

$$f=20\text{cm}$$

The final image is generated at second focus or 20 cm from the concave lens, which is independent of b, if the object O is put at a distance of 20 cm from the plano-convex lens. The rays then become parallel.

So, Correct option: (D) 20c

LEVEL-II

1. Correct option: (B) $\frac{\mu R}{\mu-1}$.

The formula for refraction in spherical surfaces is given by as,

$$\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R}$$

Given:

$v=\infty$

$\mu_1=1$

$\mu_1=\mu$

Substitute all the value we get,

$$\frac{1}{\infty}-\frac{\mu}{u}=\frac{\mu-1}{R}$$

$$0-\frac{\mu}{u}=\frac{\mu-1}{R}$$

$$u=-\frac{\mu R}{\mu-1}$$

Here, negative sign due sign convention.

So, the distance of the objected is $\frac{\mu R}{\mu-1}$.

Correct option: (B) $\frac{\mu R}{\mu-1}$.

2. Correct option: (B) $n_2>n_1$.

The nature of lens changes, is $\mu_{medium}>\mu_{lens}$

Since $\mu_{medium}>\mu_{lens}$, so divergent lens and vice-versa.

So, Correct option: (B) $n_2>n_1$.

3. Correct option: (D) $\frac{f_1+f_2}{f_1 f_2}$.

Given: Focal length of lenses is f_1 and f_2.

We know that two lenses in contact have a combined effective power equal to the sum of their separate powers.

$P = P_1 + P_2$

So,

Power is given by as,

$P = \frac{1}{f}$

$P = \frac{1}{f_1} + \frac{1}{f_2}$

$P = \frac{f_1 + f_2}{f_1 f_2}$

So, Power of lens is $\frac{f_1 + f_2}{f_1 f_2}$.

Correct option: (D) $\frac{f_1 + f_2}{f_1 f_2}$.

4. Correct option: (B) 2.5D.

Given:

$\mu_1 = \frac{4}{3}$

$\mu_2 = 1.0$

$R = 10\text{cm}$

Let's examine the principal axis in order to observe where the parallel rays converge or diverge. Call it the focus and give the matching length f as the focal length.

Using the formula for refraction at curved surface,

$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

If we position our object at ∞, an image is created at focus.

Therefore,

$\frac{\frac{4}{3}}{f} - \frac{1.0}{\infty} = \frac{\frac{4}{3} - 1.0}{10}$

$f = 40\text{cm} = 0.4\text{m}$

Its power ought to be positive because the rays are convergent.

Hence, power is given by as,

$f = 40\text{cm} = 0.4\text{m}$

$P = \frac{1}{f} = \frac{1}{0.4} = 2.5\text{D}$

So, the power convex surface is 2.5D.

5. Correct option: (D) 2.

Given:

For given surface,

$u = -\infty$

$R = +R$

$v = +2R$

Using the formula for refraction at curved surface,

$$\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R}$$

$$\frac{\mu}{2R}-\frac{1}{(-\infty)}=\frac{\mu-1}{R}$$

$$\left(\mu_2=\mu,\mu_1=\mu_{air}=1\right)$$

$$\frac{\mu}{2R}=\frac{\mu-1}{R}$$

$$2\mu-2=\mu$$

$$\mu=2$$

The refractive index of the material of the cylinder is 2.

6. Correct option: (B) 15cm.

Given:

$$f=10\text{cm}$$

$$R_1=75\text{cm}$$

$$\mu=0.5$$

Using the lens maker formula we have,

$$\frac{1}{f}=\left(\mu-1\right)\left[\frac{1}{R_1}-\frac{1}{R_2}\right]$$

$$\frac{1}{10}=(0.5)\left[\frac{1}{75}-\frac{1}{R_2}\right]$$

$$\frac{1}{5}=\frac{1}{75}-\frac{1}{R_2}$$

$$\frac{1}{R_2}=-\frac{1}{5}+\frac{1}{75}$$

$$\frac{1}{R_2}=\frac{-3+2}{15}$$

$$R_2=15\text{cm}$$

So, the radius of curvature of the second surface is 15cm.

7. Correct option: (A) 1D.

Given: The Focal length of convex and concave length are given respectively:

$$f_1=0.5$$

$$f_2=-1$$

The following formula can be used to determine the power of a lens created by combining two lenses:

$$P=P_1+P_2$$

The power is given by as,

$$P=\frac{1}{f}$$

Therefore,

$$P = \frac{1}{f_1} + \frac{1}{f_2}$$

$$p = \frac{1}{0.5} + \frac{1}{-1}$$

$$P = 1D$$

So, the power of the resulting lens is 1D.

8. Correct option: (A,C)

9. Correct options: (A) and (C).

The magnification of refracting surface is given by as,

$$m = \frac{\mu_1 v}{\mu_2 u}$$

Therefore, Correct options: (A) and (C).

10. Correct options: (A) Actually meet at some point and (D) Meet (or appears to meet) at a distance of 60 cm from the spherical surface.

Given:

$\mu_1 = 1.0$

$\mu_2 = 1.5$

$R = 20\text{cm}$

$u = \infty$

The formula for refraction at curved surface is given by,

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

Substitute the all the value we get.

$$\frac{1.5}{v} - \frac{1.0}{\infty} = \frac{1.5-1}{20}$$

$$v = \pm 60\text{cm}$$

After the refraction from the spherical surface parallel rays is meet at distance 60 cm from the spherical surface.

Correct options: (A) Actually meet at some point and (D) Meet (or appears to meet) at a distance of 60 cm from the spherical surface.

11. Correct option: (B).

Speed of light in a medium is given by as,

$$f = \frac{f_0}{\mu}$$

Again, the frequency is constant.

$$f = f_0$$

We know that,

$v_0 = f_0 \lambda_0$

$v = f\lambda$

$$\Rightarrow \frac{v_0}{\mu} = f_0 \lambda$$

$$\Rightarrow \frac{v_0}{\mu} = \frac{v_0}{\lambda_0} \lambda$$

$$\Rightarrow \lambda = \frac{\lambda_0}{\mu}$$

12. Correct options: (A) the new focal length will become $4f$ and (C) New image will be virtual and magnified.

According to lens maker formula:

$$\frac{1}{f}=\left(\frac{\mu_L}{\mu_M}-1\right)\left(\frac{1}{R}-\frac{-1}{R}\right)$$

When lens was in air:

$$\frac{1}{f_1}=\left(\frac{3}{2}-1\right)\left(\frac{1}{2R}\right)$$

$$\frac{1}{f_1}=R$$

When lens was in medium, $\mu=\frac{4}{3}$

$$\frac{1}{f_2}=\left(\frac{\frac{3}{2}}{\frac{4}{3}}-1\right)\left(\frac{2}{R}\right)$$

$$\frac{1}{f_2}=\frac{1}{4R}$$

$$f_1=R,\ f_2=4R$$

In case1, object was between and $2f$.

However, in Case 2, the identical object will be inside the focus and pole,as $f_2=4f$.

Hence the image in case 2 will be virtual and magnified.

In the air, the image was magnified, real, and inverted. Consequently, the object was situated between f and $2f$. The focal length is currently $4f$. The object is now situated between the focus and the pole. As a result, the new image will be enhanced and virtual.

Correct options: (A) the new focal length will become $4f$ and (C) New image will be virtual and magnified.

13. Correct option: (C) $\frac{R}{2}$.

The formula for refraction at curved surface is given by,

$$\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R}$$

For first curved surface,

$$\frac{2}{v_1}+\frac{1}{x}=\frac{1}{R}$$

$$v_1=\frac{2Rx}{x-R}$$

For plane surface,

$$v=v_1-R$$

$$v=\frac{xR+R^2}{x-R}$$

$$\frac{1}{v}-\frac{2(x-R)}{R(x+R)}=0$$

$$\frac{1}{v}=\frac{2(x+R)}{R(x+R)}$$

For virtual image,

$\frac{1}{v} < 0 \Rightarrow \frac{2(x-R)}{R(x+R)} < 0$

$x < R$

So, the only possible option is $\frac{R}{2}$.

14. Correct option: (D) Inverted and same size.

For refraction from the first surface,

Using refraction formula we get,

$$\frac{2}{v_1} - \frac{1}{-2R} = \frac{1}{R}$$

Hence, $v_1 = 4R$

Magnification from the first refraction is given by as,

$$m = \frac{\mu_1 v_1}{\mu_r u_1}$$

$$m = -\frac{1}{2}\frac{4R}{-2R} = 1$$

For refraction from the second surface,

$$\frac{1}{v_2} - \frac{2}{3R} = \frac{-1}{\infty}$$

Hence, $v_2 = \frac{3R}{2}$

Magnification from second surface is given by as,

$$m = -\frac{2}{1}\frac{\frac{3R}{2}}{3R} = -1$$

Total magnification is given by as,

Net magnification= magnification from first surface × magnification from second surface

Net magnification=1 ×-1=-1

So, the image is inverted and same size.

Correct option: (D) Inverted and same size.

15. Correct option: (A) 90°**.**

A light beam follows the path tangent to the surface at the point of incidence when it encounters a spherical surface at grazing incidence, or when the angle of incidence is 90°. The ray will therefore leave the spherical surface parallel to the surface.

The angle of emergence from the planar surface will also be 90° if the ray emerges parallel to the surface.

16. Correct option: (D)

17. Correct option: (C)

18. Correct option: (D)

19. Correct Matching is: (a)-Q,(b)-Q,(c)-(R),(d)-(S).

Any lens dipped in a denser media than the lens exhibits convex lens behaviour, concave lens behaviour, convex lens convergence, and concave lens convergence.

(A) A convex lens in a denser media will act as a diverging lens, according to the theory.

(B) In a rarer medium, a concave lens will act like a diverging lens.

(C) A convex plano lens with silvering on the curved portion. It will therefore behave like a concave mirror.
(C) A convex plano lens with silvering on the curved portion. It will therefore behave like a concave mirror.
(D) The flat part of a plano concave lens then turned silvery. As a result, it will function like a convex mirror.

Prism and Human Eye

LEVEL-I

Prism and Dispersion of Light:

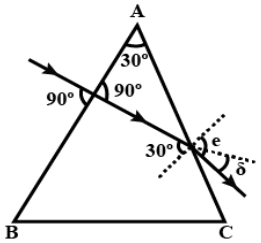
1 Correct option: (C) red ray.

More wavelength-diverse light will scatter less and have less dispersion.
Red colour deviates the least and violet deviates the most when white light is dispersed by a prism.
So, Correct option: (C) red ray.

2. Correct option: (A) $18^{\circ}36^{'}$**.**

Given: $\mu = 1.5$

Refracting angle$= 30^{\circ}$



For surface AC,

$$\frac{1}{\mu} = \frac{\sin 30^{\circ}}{\sin e}$$

$$\sin e = \mu \sin 30^{\circ}$$

$$\Rightarrow sin\, e = 1.5 \times \frac{1}{2} = 0.75$$

$$\Rightarrow e = sin^{-1}(0.75) = 48^{\circ}36^{'}$$

From figure,

$$\delta = e - 30^{\circ}$$

$$\Rightarrow 48^{\circ}36^{'} - 30^{\circ} = 18^{\circ}36^{'}$$

So, the deviation of monochromatic ray incident on its one surface is $18^{\circ}36^{'}$.

3. Correct option: (D) $\omega\delta$**.**

Dispersive power= ω
Angle of deviation= δ
Dispersive power is given by as,

$$\text{Dispersive power}(\omega) = \frac{\text{Angular dispersion}}{\text{mean deviation}(\delta_{mean})} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(1)$$

Angular dispersion is given by as,

$$\text{Angular dispersion} = \delta_v - \delta_R$$

if $\delta_{mean} = \delta$

Therefore, equation(1) become,

$$\omega = \frac{\delta_v - \delta_R}{\delta}$$

$$\delta_v - \delta_R = \omega\delta$$

Hence, Angular dispersion $= \omega\delta$

4. Correct option: (C) 82°**.**

Given: $\mu = 1.5$

Let's consider the angle of minimum deviation is δ_m.

Therefore, $A = \delta_m$

The refractive index of prism is given by as,

$$\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$1.5 = \frac{\sin\left(\frac{A+A}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$1.5 = \frac{\sin\left(\frac{A}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$1.5 = \frac{2\sin\left(\frac{A}{2}\right)\cos\left(\frac{A}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$1.5 = 2\cos\frac{A}{2}$$

$$0.75 = \cos\frac{A}{2}$$

$$\frac{A}{2} = \cos^{-1}(0.75)$$

$$\frac{A}{2} = 41^\circ$$

$$A = 82^\circ$$

So, the angle of the prism is 82°.

5. Correct option: (B) Material of prism.

The difference in refraction between the longest and shortest wavelengths that enter a prism is known as dispersive power. The refractive index of the prism determines refraction. This in turn depends on the prism's composition. Therefore, the prism's substance affects the dispersive power.

6. The correct option (A) Only deviation.

White light splits into a band of seven colours when it travels through a prism. However, when two prisms are merged so that the sum of their individual angular dispersion angles is zero, the result is an achromatic combination of prisms. When white light travels through an achromatic prism combination, the internally dispersed white light components from the Crown glass prism refract and collide at the edge of the Flint glass prism. As a result, the refracted light from the achromatic prism combination is once more white light with only slight deviation from the original colour.

7. Correct option: (B) 45° **.**

The refractive index of prism is given by as,

$$\mu = \frac{\sin\left(\frac{A+D_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$\sqrt{2} = \frac{\sin\left(\frac{60^\circ + D_m}{2}\right)}{\sin\left(\frac{60^\circ}{2}\right)}$$

$D_m = 30^\circ$

Now,

$D_m = 2i - A$

$30^\circ = 2i = A$

$i = 45^\circ$

So, the angle of incidence is 45°.

8. Correct option: (D) $\frac{1}{4}$ **.**

Given: Formula for deviation is given by as,

$\delta = (\mu - 1)A$

Where, $\delta =$ deviation, $A =$ angle of prism, $\mu =$ Refractive index of prism with medium.

Now,

$\delta = (\mu - 1)$

$$\frac{\delta_a}{\delta_w} = \frac{\frac{\mu_g}{\mu_a} - 1}{\frac{\mu_g}{\mu_w} - 1}$$

Where,

μ_a = Refractive index of air, $\mu_a = 1$

μ_g = refractive index of glass, $\mu_g = \frac{3}{2}$

μ_w =refractive index of water , $\mu_w = \frac{4}{3}$

Substitute the value we get,

$$\frac{\delta_a}{\delta_w} = \frac{\left(\frac{3}{2}-1\right)}{\left(\frac{\frac{3}{2}}{\frac{4}{3}}-1\right)}$$

$$\frac{\delta_a}{\delta_w} = \frac{\left(\frac{1}{2}\right)}{\left(\frac{9}{8}-1\right)} = 4$$

Hence , the ratio of angle of minimum deviation of a prism in air and when dipped in water is $\frac{1}{4}$.

9. Correct option: (A) Flint glass.

Compared to Refractive Index of Crown Glass ($\mu = 1.51$), Refractive Index of Flint Glass(1.65) is relatively higher. Since dispersive power depends on the refractive index, flint glass has the highest dispersive power.

10. Correct option: (C) Becomes distinct.

The path of the light ray through the prism is symmetrical in the minimum deviation position. Seven distinct colours of white light can be easily identified once white light has been distributed via a prism. As a result, the spectrum's visuals stand out.

11. Correct option: (C) 45° **.**

Given: Angle of prism, $A = 30^\circ$

Refractive index, $\mu = \sqrt{2}$

We know that,

$A = r + 0 = 30^\circ$

Using the snell's law at surface AB.

$$\mu = \frac{\sin i}{\sin r}$$

$$\sqrt{2} = \frac{\sin i}{\sin 30^\circ}$$

$$\sin i = \sqrt{2} \times \sin 30^\circ$$

$$i = 45^\circ$$

So, the angle of incidence is 45°.

12. Correct option: (C) 60° **.**

Given:

Angle of prism for equilateral prism, $A = 60^\circ$

Refractive index, $\mu = \sqrt{3}$

The refractive index of prism is given by as,

$$\mu = \frac{sin\left(\frac{60+\delta_m}{2}\right)}{sin\frac{A}{2}}$$

$$\sqrt{3} = \frac{sin\left(\frac{60+\delta_m}{2}\right)}{sin\frac{60}{2}}$$

$$\sqrt{3} = \frac{sin\left(\frac{60+\delta_m}{2}\right)}{sin\,3\,0}$$

$$\sqrt{3} = \frac{sin\left(\frac{60+\delta_m}{2}\right)}{\frac{1}{2}}$$

$$\frac{\sqrt{3}}{2} = sin\left(\frac{60+\delta_m}{2}\right)$$

$$sin\,6\,0 = sin\left(\frac{60+\delta_m}{2}\right)$$

$$60 = \frac{60+\delta_m}{2}$$

$$\delta_m = 120 - 60$$

$$\delta_m = 60^\circ$$

So, the minimum angle of deviation is 60°.

13. The correct option (A) Refraction.

Dispersion is the division of white light into its individual colours. Refraction or bending varies depending on the colour, which causes dispersion. When white light enters a different medium, its speed changes, causing it to split.

14. The correct option (B) Blue.

The deviation increases with decreasing wavelength. Blue deviates the greatest because, according to the question, it follows after violet.

15. Correct option: (A) 30°**.**

Given: Angle of prism, $A = 60^\circ$.

For minimum deviation,

$r = r^1$

We know that,

$r + r^1 = A$

$2r = A$

$r = \frac{A}{2}$

$r = \frac{60}{2}$

$r = 30^\circ$

So, the angle refraction is 30°.

16. Correct option: (C) 60°**.**

Given: the refractive index of prism, $\mu = 1.732$

Angle of minimum deviation is equal to the angle of the prism.

Therefore,

$u = \infty$

$v = 30cm$ $\quad A = \delta_m$

Let's consider the angle of minimum deviation represent by δ_m.

Using the prism formula,

$$\mu = \frac{\sin\frac{A+\delta_m}{2}}{\sin\frac{A}{2}}$$

Substitute all the value we get,

$$1.732 = \frac{sin\frac{A+A}{2}}{sin\frac{A}{2}}$$

$$1.732 = \frac{Sin\,A}{sin\frac{A}{2}}$$

$$1.732 = \frac{2\,sin\left(\frac{A}{2}\right)cos\left(\frac{A}{2}\right)}{sin\left(\frac{A}{2}\right)}$$

$$1.732 = 2\,cos\left(\frac{A}{2}\right)$$

$$0.866 = cos\left(\frac{A}{2}\right)$$

$$A = 60^\circ$$

So, the angle of prism is 60°.

17. Correct option: (A) 1.732.

Given: Angle of incident, $i = 60^\circ$.

Angle of prism, $A = 60^\circ$

Angle of deviation, $\delta = 60^\circ$

We know that,

$i_1 + i_2 = A + \delta$

The angle of emergence,

$i_2 = A + \delta - i_1$

$i_2 = 30 + 30 - 60$

And the emergent ray,

$r_1 + r_2 = A$

$i_2 = 0, r_2 = 0$

$r_1 = A = 30^\circ$

Using snell' law

$$\mu = \frac{\sin i_1}{\sin i_2} = \frac{\sin 60}{\sin 30} = 1.732$$

Since emergent ray is normal,

$\delta = \delta_m = 30^\circ$

$$\frac{\sin\left(30+\frac{30}{2}\right)}{\sin\frac{30}{2}} = 1.732$$

The value of refractive index of the prism is 1.732.

HUMAN EYE:

18. Correct option: (D) Accommodation.

The focal length of the eye lens allows the human eye to focus on objects at various distances. Accommodation refers to the eye's capacity to do so. The accommodation of an eye is the capacity of the eye to perceive objects at various distances. Our eye lens thins out during this process while we are attempting to see an object in the distant. This increases the focal length to about 2.5 cm. On the other hand, the eye lens takes on a more rounded shape when we are viewing an item that is closer to us. This shortens the focal length, allowing for sharp object focus.

So, Correct option: (D) Accommodation.

19. Correct option: (B) Yellow spot.

The central, spherical region of the retina known as the yellow spot or macula is located in the human eye. It's made to see objects as clearly as possible or with the most visual activity imaginable. It also goes by the name "blind spot." Our nervus opticus transmits electrical pulses to your brain when light hits our retina. Our brain transforms the impulses into an image. We are unable to see anything in the region where our nervus opticus connects to our retina since there are no light-sensitive cells there. Therein lies our blind spot.

20. Correct option: (A) 3D.

Given:

$u = 25\text{cm}$

$v = 100\text{cm}$

Power of lens is given by as,

$$P = \frac{1}{f} = \frac{1}{-v} + \frac{1}{u}$$

$$P = \frac{1}{-100} + \frac{1}{25}$$

$$P = +3D$$

So, the power of lens is +3D.

21. Correct option: (D) -15.8cm, -6.3D.

Given:

$v = -15\text{cm}$

$u = -300\text{cm}$

Using the formula of lens,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{-15} - \frac{1}{-300}$$

$$\frac{1}{f} = \frac{-19}{300}$$

$$f = \frac{-300}{19}$$

$f = -15.8\text{cm}$

The power of lens is given by as,

$$p = \frac{100}{f}\text{cm}$$

$$\text{P} = \frac{-100 \times 19}{300}$$

$$P = -6.33D$$

So, Focal length and power of lens is -15.8cm and -6.3D.

22. The correct option (B) Before retina.

Because the eye lens's focal length decreases in short-sightedness, the image forms before the retina.

In optics, a person with normal vision can clearly perceive both close-up and far-off objects that are within the range. This is due to the fact that on the retina, where the brain gets the picture information, the image of the objects is properly formed. What occurs to someone who has eye defects.

Myopia is a condition where a person can focus on close objects clearly, meaning that after refraction, the images of close things fall correctly on the retina whereas the images of distant objects do not exactly fall on the retina. This may be due to an excessively convex eye lens.

When the eyeball is overly tense in this situation, it will be impossible to focus on distant things. As a result, the image of the object will be produced in front of the retina, the area of the eye that is most sensitive to light. This results in hazy vision, which is treatable in a number of ways, one of which is the use of corrective lenses.

23. Correct option: (B) 30cm.

Given:

$u = \infty$

$v = -30\text{cm}$

Concave lens if necessary for increasing distant point and for this concave lens.

Using the lens formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{-30} - \frac{1}{\infty}$$

$$f = -30\text{cm}$$

For near point, $\frac{1}{-30} = \frac{1}{-15} - \frac{1}{u}$

$$u = -30\text{cm}$$

The near point is 30cm.

24. Correct option: (B) Double.

We know that, Focal length =-(Detected far point)

For curing myopia, a concave lens of focal length equal to the distance of the eye's own far point is used. i.e. d $f = -d$ Hence, if distance d is doubled, afterwards, the focus length will also double.

25. The correct option (D) optic nerve.

The optic nerve transmits the retina's sense of vision to the brain.

When light that has been reflected from the environment enters the retina through the eye lens, chemical and electrical impulses (sensory data) are produced. and the optic nerve sends electrical impulses carrying visual information to the brain.

26. Correct option: (C) Change in the convexity of the lens surface.

When ciliary muscles contract, the convexity of the lens changes. The ciliary muscles of the human eye hold a converging lens that is located behind the iris. Convexity of the lens varies when ciliary muscles exert force on it, focusing the picture on the retina. Focusing is accomplished by a change in the convexity of the ciliary tube, which is made up of these ciliary muscles.

27. Correct option: (A) Elongation of eyeball.

Myopia is an optical condition when light from an object is focussed just slightly in front of the retina rather than immediately hitting it. In this deficiency, a person can clearly see objects that are closer to him but cannot see objects that are farther away. Shortsightedness or near-sightedness are other names for it.

The difference in distance between the retina and eye lens is what causes myopia. When the eyeball lengthens along its axis, this distance grows. This causes the distant item to appear fuzzy.

Myopia, then, results from an) elongation of eye ball.

28. Correct option: (D) -0.5D.

Given: $f = 2.0\text{m}$

The power of lens is given by as,

$$P = \frac{1}{-(\text{defected far point})} = \frac{1}{-2} = -0.5\text{D}$$

So, the power of lens is -0.5D.

29. Correct option: (B) Long-sightedness.

When a person has trouble seeing close-up items (short sight) but not far-off objects (long sight), they are said to have hypermetropia. Images are created behind the retina because the eye lens is unable to focus light rays from adjacent objects onto the retina. Problems with near vision are caused by long sight. Hypermetropia is therefore a Long-side defect.

30. Correct option: (B) -1.0D.

Given: $v = -100\text{cm} = -1\text{m}$

Due to the person's inability to perceive objects beyond than 100 cm. Therefore, we must use a lens with a focal length f that can create an image of an object at infinity up to 100 cm.

Using lens formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{-1} - \frac{1}{\infty}$$

$$\frac{1}{f} = -1\text{cm}$$

The power of lens is given as,

$$P = \frac{1}{f} = \frac{1}{-1} = -1\text{D}$$

So, the power of the spectacle lens required to see a distant object is -1D.

31. Correct option: (C) varying refractive index.

The crystalline lens of the human eye. It is neither soft nor hard. Its refractive index varies. The crystalline lens that makes up the human eye has a variable refractive index. The refractive index gauges how much light is bent or refracted as it passes through the lens.

The refractive index of the human eye lens is highest in the centre and gradually drops towards the edges.
The human eye's lens has a variable refractive index, which aids in accommodation, allowing the eye to change its focal length and concentrate objects onto the retina, which is essential for sharp vision at various distances.

LEVEL-II

1. Correct option: (B) $(\mu-1)\alpha$.

Tha angle of minimum deviation is given by as,

$A=(\mu-1)\alpha$

Where, μ = refractive index of prism.

α = angle of incidence.

2. Correct option: (C) $\frac{3}{2}$.

Given:

$\delta_m = 40^\circ$

$A = 60^\circ$

The refractive index pf prism is given by as,

$$\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\frac{A}{2}}$$

$$\mu = \frac{\sin\left(\frac{60+40}{2}\right)}{\sin\frac{40}{2}}$$

$$\mu = \frac{\sin\left(\frac{100}{2}\right)}{\sin\frac{60}{2}}$$

$$\mu = \frac{\sin(50)}{\sin 30}$$

$\mu = 1.5$

So, the refractive index of material is $\frac{3}{2}$.

3. Correct option: (A) less than 41°.

We know that from the hypothesis,

$A+\delta = i_1 + i_2$

$\Rightarrow \delta = i_1 + i_2 - A$

$\Rightarrow \delta = 55^\circ + 46^\circ - 60^\circ = 41^\circ$

But we know that,

$\delta_m < \delta$

$\delta_m < 41^\circ$

So, the angle of minimum deviation will be (A) 41°.

4. Correct option: (B) $n = 2\cos A$**.**

Light rays should fall normally on the reflecting surface in order to retrace their route. So

$r_2 = 0$

$\Rightarrow r_1 = A - r_2$

$\Rightarrow r_1 = A$

Applying Snell's rule between incident ray and refracted ray,

$(1)\sin(2A) = n\sin(A)$

$2\sin A\cos A = n\sin A$

$n = 2\cos A$

5. Correct option: (A) 2:3.

We know that,

$\frac{\omega_1}{\omega_2} = -\frac{f_1}{f_2} = -\frac{2}{3}$

So, the ratio of dispersive power of glasses used is 2:3.

6. Correct option: (C) $2\cos^{-1}\left(\frac{\mu}{2}\right)$**.**

Using the formula of refractive index,

$$\mu = \frac{\sin\left(\frac{A+D}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

According to given condition, $A = D$.

$$\mu = \frac{sin\left(\frac{A+A}{2}\right)}{sin\left(\frac{A}{2}\right)}$$

$$\mu = \frac{sin(A)}{sin\left(\frac{A}{2}\right)}$$

$$\mu = \frac{2\,sin\left(\frac{A}{2}\right)cos\left(\frac{A}{2}\right)}{sin\left(\frac{A}{2}\right)}$$

$$\mu = 2\,cos\left(\frac{A}{2}\right)$$

$$A = 2\,cos^{-1}\left(\frac{\mu}{2}\right)$$

So, the value of A in terms of μ is $2\cos^{-1}\left(\frac{\mu}{2}\right)$.

7. Correct option: (C) -2.5D.

The focal length of the intended lens must match the distance to the distant object.

So, $f = -40\text{cm} = -0.40\text{m}$

Myopic person requires concave lenses, therefore f negative .

The power of lens is given by as,

$$P = \frac{1}{f} = -\frac{1}{0.40} = -2.5\text{D}$$

So, the power of lens is -2.5D.

8. Correct option: (B) 2D and -0.33D.

The required focal length for correcting the near point,

$$f = \frac{50-50}{(50-25)} = 50cm$$

The power of lens is given by as,

$$P = \frac{1}{f} = \frac{100}{50} = +2\text{D}$$

The required focal length for correcting the far point,

$$f = -(\text{defected the far point}) = -3\text{m}$$

$$P = -\frac{1}{3}\text{D} = -0.33\text{D}$$

So, Powers of lens are +2D and -0.33D.

9. Correct option: (B) $\left(1+\frac{D}{f}\right)$**.**

Given: Final image formed at, $\text{D} = 25\text{cm}$.

In the case, $v = -D$ from lens formula,

Now using lens formula,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{-D} - \frac{1}{-u} = \frac{1}{f}$$

$$i.e \frac{D}{u} = 1 + \frac{D}{f}$$

So, magnifying power is given by as,

$$\frac{D}{u} = \left(1 + \frac{D}{f}\right)$$

So, the magnifying power od simple microscope is given by as, $\left(1 + \frac{D}{f}\right)$.

10. Correct option: (A) 1.8cm.

Length of compound microscope, $L_\infty = 14\text{cm}$.

The magnifying power of relaxed eye , $m = 25$.

The focal length of the eye, $f_e = 5\text{cm}$

The length compound microscope,

$$L_\infty = v_0 + f_e$$

$$14 = v_0 + 5$$

$$v_0 = 9\text{cm}$$

The magnifying power of relaxed eye is given by as,

$$m = \frac{v_0}{u_0} . \frac{D}{f_e}$$

$$25 = \frac{9}{u_0} . \frac{25}{5}$$

$$u_0 = \frac{9}{5}$$

$$u_0 = 1.8\text{cm}$$

So, the distance for objective lens is 1.8 cm.

11. Correct option: (C) The deviation of the prism is given by $\delta = i - A$.

The angle by which a light beam is bent while passing through a prism is referred to as the deviation of the prism. Both the material's refractive index and the light beam's angle of incidence play a role.

The correct equation for a prism's deviation is $\delta = i - A$, where stands for the deviation angle, i for the angle of incidence, and A for the prism's angle. The absence of a positive sign implies that light is bending towards the prism's base.

So, Correct option: (C) The deviation of the prism is given by $\delta = i - A$.

12. Correct option: (B) and (C).

Solution.

A rainbow is a naturally occurring spectrum that forms after a rainstorm (each raindrop functions as a tiny prism) as a result of the dispersion, refraction, and internal reflection of various hues. Refraction, dispersion, and total internal reflection are all factors in the production of a rainbow. A water droplet is illuminated by the sun. The light bends or refracts as it enters the raindrop. White light is divided into seven different colours as a result of the slowing of light. On the other end of the plunge, complete internal reflection takes place.

Total internal reflection only occurs once during the development of the primary rainbow, but total internal reflection occurs twice during the formation of the secondary rainbow, changing the order of colours.

13. Correct options: (A), (C) and (D).

Explanation: (A). The red colour has the shortest frequency and the longest wavelength. The red colour has a wavelength of about two nanometers. The highest frequency and shortest wavelength are found in the colour violet.

(C) Inverse relationship exists between wavelength and refractive index. Violet, green, yellow, and red are the colours listed in ascending order of wavelength for the options. As a result, the violet wavelength is the shortest. It therefore has the greatest refractive index.

(D) Because the refractive index of the prism is higher for violet light, it distorts violet light more than red light in the dispersion of white light from a prism.

Due to its shortest wavelength among all the hues, violet light deviates the most when white light is scattered by a prism.

14. Correct options: (A),(B) and (C).

It is true that Option A) "The image of a nearby object is formed behind the retina" depicts a hypermetropia symptom. This issue causes a fuzzy image because the light's focus point is behind the retina.

Since "longsightedness" is another word for hypermetropia, Option B) is equally accurate. It means that while distant items may be viewed more clearly, it is harder to see nearby objects clearly.

Option C) "Corrected by a convex lens of suitable focal length" is likewise accurate. Hypermetropia can be treated with a convex lens, which is thicker in the centre and thinner on the edges. The converging property of the convex lens aids in moving the focal point forward so that light rays properly focus on the retina, enabling the formation of a sharp image.

15. Correct options: (B),(C) and (D).

Explanation. a) There are two angles of incidence for every angle of deviation. 9f is inaccurate since there is a separate angle of departure for each incidence angle.

16. The correct matching is (a)-(S), (b)-(Q), (c)-(P), (d)-(R)

Age-related presbyopia is a medical disease that explains how eyes gradually lose their capacity to concentrate on close objects. For adapting the eye to concentrate at close objects (often the bottom lens) and distant objects (typically the top lens), bifocal lenses (consisting of top and bottom lenses) with an appropriate focal length are prescribed. So (a)-(S)

When the eye is unable to direct light from close objects at the retina, hypermetropia develops. Therefore, a convex lens with the appropriate power is placed close to the eye to focus light rays on the retina. So (b)-(Q)

Sphero-cylindrical lenses are utilised to ensure that the picture formed at the retina is formed properly since astigmatism is caused by uneven curves in the eyeball. thus (c) -(P)

When the eye's lens focuses light from far-off objects in front of the retina, myopia develops. Therefore, an external concave lens with a proper focal length is used to enable precise focus of light beams from distant objects on the retina. So (d)-(R).

17. Correct Answer is: 2

3. Electrostatics

Solutions

LEVEL-I

1.Correct option: (C)

Electric field lines are continuous curves. They start from a positively charged body and end at a negatively charged body. Electric field lines are continuous but do not form closed loops.

2.Correct option:(B)

The electrostatic force between two charges is given by

$$F = \frac{Kq_1q_2}{R^2}$$

Here, K =electrostatic force constant ($K=9x10^9 Nm^2/C^2$)

q_1 = charge on the object 1, q_2 = charge on object 2

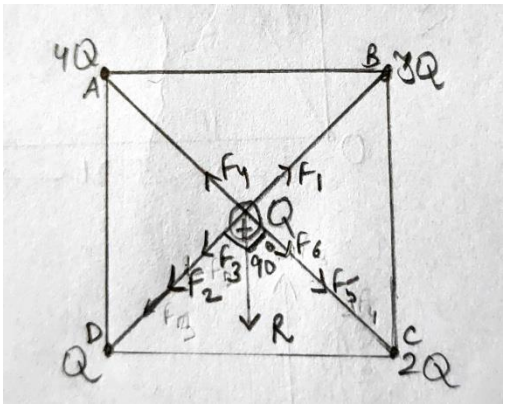
R = separation between the charges

As the force is directly proportional to the product of the charges so it will be maximum when the two charges are equal.

That is

x:y = 1:1

3.Correct option(D)



The electrostatic force between two charges is given by

$$F = \frac{q_1q_2}{4\pi\epsilon_o r^2}$$

q_1 = charge on the object 1, q_2 = charge on object 2

r= separation between the charges

Separation between charges is same w. r. t centre = ½ of the diagonal of square $= \frac{\mathbf{1}}{\mathbf{2}}\sqrt{\mathbf{2}}\boldsymbol{l} = \frac{\boldsymbol{l}}{\sqrt{\mathbf{2}}}$

Now repulsive force between charge Q and Q =

$$F_1 = \frac{2Q^2}{4\pi\epsilon_o l^2} \text{ towards B}$$

Repulsive force between charge Q and 3Q =

$$F_2 = \frac{6Q^2}{4\pi\epsilon_o l^2} \text{ towards D}$$

As magnitude of 3Q is greater so net force is towards corner D = $F_3 = \frac{6Q^2}{4\pi\epsilon_o l^2} - \frac{2Q^2}{4\pi\epsilon_o l^2} = \frac{4Q^2}{4\pi\epsilon_o l^2}$ *towards D*

Repulsive force between charge Q and 2Q =

$$F_4 = \frac{4Q^2}{4\pi\epsilon_0 l^2} \text{towards A}$$

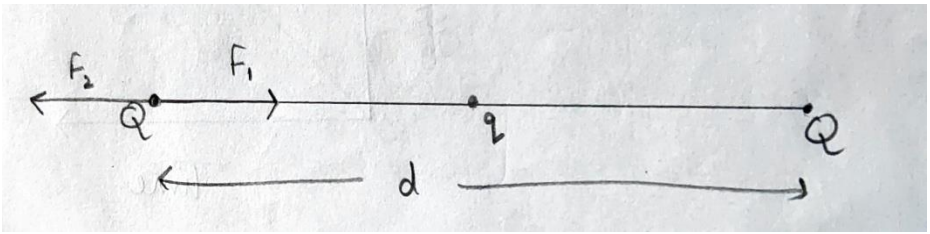
Repulsive force between charge Q and 4Q =

$$F_5 = \frac{8Q^2}{4\pi\epsilon_0 l^2} \text{towards } C$$

As magnitude of 4Q is greater so net force is towards corner C = $F_6 = \frac{8Q^2}{4\pi\epsilon_o l^2} - \frac{4Q^2}{4\pi\epsilon_o l^2} = \frac{4Q^2}{4\pi\epsilon_o l^2}\ towards\ C$

Resultant force R = $\sqrt{F_3^2 + F_6^2 + 2F_3F_6 cos90°} = \sqrt{2F_3^2} = \frac{\sqrt{2}\,x4Q^2}{4\pi\epsilon_o l^2} - j = -\frac{\sqrt{2}Q^2}{\pi\epsilon_o l^2} j$

4.Correct option(B)



For the equilibrium of charge Q, the force applied by both the charges i.e. another Q and q should be equal in magnitude and opposite in direction. This is only possible when q has negative charge.
If d is the distance between both the pairs of q and Q then
The electrostatic force between two charges q and Q is given by

$$F_1 = \frac{qQ}{4\pi\epsilon_o \left(\frac{d}{2}\right)^2} = \frac{4qQ}{4\pi\epsilon_o d^2}$$

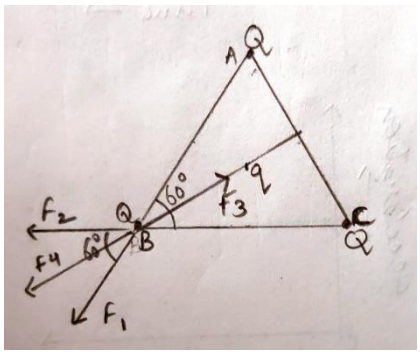
The electrostatic force between two charges Q and Q is given by

$$F_2 = \frac{QQ}{4\pi\epsilon_o d^2}$$

For equilibrium

$$F_1 = -F_2 = 4qQ = -Q^2\ or\quad q = -\frac{Q}{4}$$

5.Correct option(B)



Net force on q will be zero for any values of q and Q. Net force on each Q should also be zero. For this q should be negative of Q. By symmetry, total force on charge q will be zero.
Now to calculate total force on any on charge Q
The electrostatic force between two charges is given by

$$F = \frac{q_1 q_2}{4\pi\epsilon_o r^2}$$

q_1 = charge on the object 1, q_2 = charge on object 2
r= separation between the charges
Let side of the triangle =l
Separation between charges is same w. r. t centre = distance of centroid of the triangle from centre =$\frac{l}{\sqrt{3}}$
Now repulsive force between charge Qand Q =

$$F_1 = \frac{Q^2}{4\pi\epsilon_0 l^2} \text{ away from A}$$

Repulsive force between charge Q and Q =

$$F_2 = \frac{Q^2}{4\pi\epsilon_0 l^2} \text{ away from C}$$

Attractive force between charge Q and q =

$$F_3 = \frac{3qQ}{4\pi\epsilon_0 l^2} \text{ away from B}$$

As it is an equilateral triangle so each angle is 60°

Resultant of force 1 and 2 = $F_4 = \sqrt{F_1^2 + F_2^2 + 2F_1F_2cos60°} = \sqrt{2F_1^2 + \frac{2F_1^2 1}{2}} = \sqrt{3}F_1 = \sqrt{3}\frac{Q^2}{4\pi\epsilon_o l^2}$

For the equilibrium of charge Q, the force applied by all the charges i.e.
another Q and q should be equal in magnitude and opposite in direction.
For equilibrium

$$F_3 = -F_4 = \frac{3qQ}{4\pi\epsilon_o l^2} = -\sqrt{3}\frac{Q^2}{4\pi\epsilon_o l^2} \quad or \quad \frac{q}{Q} = -\frac{1}{\sqrt{3}} = 1:-\sqrt{3}$$

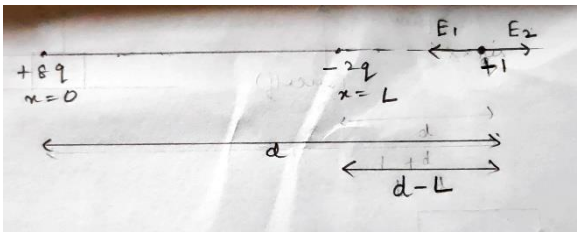
6.Correct option(D)

When the conducting sphere of charge Q is touched with the first sphere then charges will be distributed so that their potential is equal

$$V_1 = V_2 = \frac{q_1}{4\pi\epsilon r} = \frac{q_2}{4\pi\epsilon r} \text{ or } q_1 = q_2 = q$$

According to conservation of charge net charge must be same
Means q+q = Q Or q=$\frac{Q}{2}$
Similarly when the sphere having charge Q/2 is touched with the second sphere then charge on the sphere will be Q/4
And after touching 3rd sphere charge on the sphere will be Q/8

7.Correct option(A)



Let the unit positive test charge is placed at a distance d from the charge 8q
The electric field due to the charge 8q =

$$E_1 = \frac{8q}{4\pi\epsilon_o (d)^2}$$

The electric field due to the charge - 2q =

$$E_2 = \frac{-2q}{4\pi\epsilon_o(d-L)^2}$$

Net electric field will be zero when magnitude of

$$E_1 = E_2$$

$$\frac{8q}{4\pi\epsilon_o(d)^2} = \frac{2q}{4\pi\epsilon_o(d-L)^2}$$

$$4(d-L)^2 = d^2$$

$$4d^2 + 4L^2 - 8Ld = d^2$$

$$3d^2 - 8Ld + 4L^2 = 0$$

By splitting the middle term we get

$3d^2 - 6Ld - 2Ld + 4L^2 = 0$

3d(d-2L) - 2L(d-2L) =0

Or

3d-2L =0

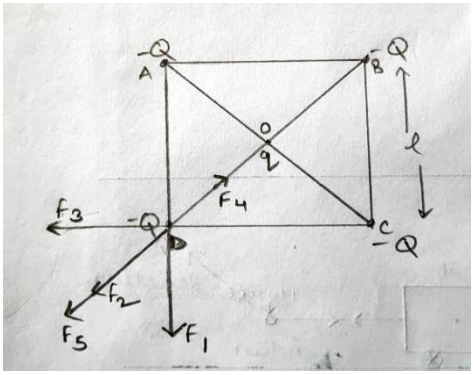
Gives d=2L/3 which is not possible

Or d-2L=0

Gives d= 2L

Means charge is at d=2L

8.Correct option(A)



By symmetry, Force on central charge will be equal and opposite due to the diagonally opposite charges, which will cancel each other.

Hence, Net force on central charge will always be zero, irrespective of value of charge q.

Therefore, to find value of q we have to check equilibrium of any one charge at the corner.

The electrostatic force between two charges is given by

$$F = \frac{q_1 q_2}{4\pi\epsilon_o r^2}$$

q_1 = charge on the object 1, q_2 = charge on object 2

r= separation between the charges

Separation between charges is same w. r. t centre = ½ of the diagonal of square $= \frac{1}{2}\sqrt{2}l = \frac{l}{\sqrt{2}}$

Now repulsive force between charge - Q and- Q =

$$F_1 = \frac{Q^2}{4\pi\epsilon_o l^2} away\ from\ D$$

Similarly

$$F_3 = \frac{Q^2}{4\pi\epsilon_o l^2} away\ from\ C$$

$$F_2 = \frac{Q^2}{4\pi\epsilon_o 2l^2} away\ from\ D$$

Resultant force $\sqrt{F_1^2 + F_3^2 + 2F_1F_3cos90°} = F_5 = \sqrt{2F_1^2} = \frac{\sqrt{2}\,Q^2}{4\pi\epsilon_o l^2}$

Force of attraction between q and - Q

$$F_4 = \frac{2qQ}{4\pi\epsilon_o l^2} towards\ B$$

For equilibrium

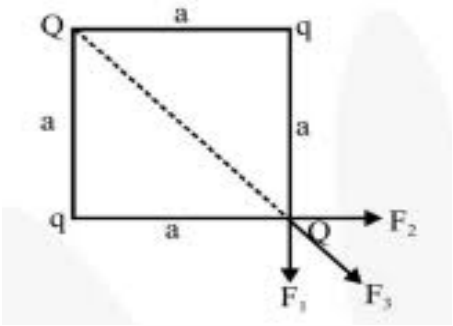
$F_2 + F_5 = -F_4$

$$\frac{Q^2}{4\pi\epsilon_o 2l^2} + \frac{\sqrt{2}\,Q^2}{4\pi\epsilon_o l^2} = -\frac{2qQ}{4\pi\epsilon_o l^2}$$

$$\frac{Q}{2} + \frac{\sqrt{2}\,Q}{1} = -2q$$

$$-\frac{Q(1+2\sqrt{2}\,)}{4} = q$$

9.Correct option(B)



The electrostatic force between two charges is given by

$$F = \frac{q_1 q_2}{4\pi\epsilon_o r^2}$$

q_1 = charge on the object 1, q_2 = charge on object 2

r= separation between the charges

diagonal of square =$\sqrt{2}a$

Now repulsive force between charge Q and Q =

$$F_3 = \frac{Q^2}{4\pi\epsilon_o 2a^2}$$

Repulsive force between charge q and Q

$$F_1 = F_2 = \frac{qQ}{4\pi\epsilon_o a^2}$$

Resultant of forces 1 and 2 $= \sqrt{F_1^2 + F_2^2 + 2F_1F_2cos90°} = F_4 = \sqrt{2F_1^2} = \frac{\sqrt{2}\,qQ}{4\pi\epsilon_o a^2}$

Now net force on Q is zero

$F_3 + F_4 = 0$

$$\frac{Q^2}{4\pi\epsilon_o 2a^2} + \frac{\sqrt{2}\,qQ}{4\pi\epsilon_o a^2} = 0$$

$$\frac{Q}{2} + \frac{\sqrt{2}\,q}{1} = 0$$

$$\frac{Q}{q} = -2\sqrt{2}$$

10.Correct option(C)

Electric field due to one positive charge at point R = $\frac{q}{4\pi\epsilon_o r^2}$

Now in order to double this field

Charges have to placed such that all the pairs cancel out and only one pair is left to give net electric field

For this same sign charge has to be placed at pairs (P, S) (Q,T)

So only pair left will be of U and R whose electric field will get added up and give double the previous electric field.

11.Correct option(A)

The ratio of number of electric field lines emerging from charge 1 and 2 is proportional to the ratio of charges

Field lines emerging from 1 =4

Field lines entering charge 2 = - 8

$$\frac{q_1}{q_2} = \frac{-4}{8} = -\frac{1}{2}$$

12.Correct option(C)

Potential at any point is the work done in bringing a unit positive test charge from infinity to that point.

$$\text{potential V} = \frac{\text{work done}}{\text{test charge}} = \frac{100}{10^{-2}} = 10^4 \text{volt}$$

13.Correct option(B)

Work done by fixed charge is electrostatic force which will push the negative charge away that is possible only when fixed charge is negative.

Hence, potential due to charge q is also negative

14.Correct option(A)

An electron moves away from the nucleus, its potential energy increases. The potential energy of the electron is minimum at equilibrium distance from the nucleus. As this distance increases / decreases, its potential energy increases.

So, work done will be positive.

15.Correct option(B)

$$V_1 = \frac{q_1}{4\pi\epsilon_o r} = 10 \ldots\ldots (1)$$

$$V_2 = \frac{q_1 + q_2}{4\pi\epsilon_o r} = 4 \ldots\ldots (2)$$

$$4 = \frac{q_1 + q_2}{4\pi\epsilon_o r} = \frac{q_1}{4\pi\epsilon_o r} + \frac{q_2}{4\pi\epsilon_o r}$$

From eqn 1

$4 = 10 + \frac{q_2}{4\pi\epsilon_o r}$

$-6 = \frac{q_2}{4\pi\epsilon_o r} \ldots\ldots (3)$

Dividing eq 1 and 3

$-\frac{6}{10} = \frac{q_2}{q_1}$

16.Correct option(D)

$work\ done\ W = q(V_2 - V_1)$

Charge on electron = q=-1.6x10^{-19}C

$V_1 = -500\,V, \quad V_2 = 1500V$

$work\ done\ W = -1.6x10^{-19}\big(1500 - (-500)\big) = -3.2x\ 10^{-16}J$

17.Correct option: (D)

$$\text{work done } W = q(V_2 - V_1)$$

As the charge – Q is shifted from C to D we have to calculate potential at C and D due to other charges

diagonal of square =$\sqrt{2}l$

Potential at point C due to charge at point A = $\frac{-Q}{4\pi\epsilon_o\sqrt{2}l}$

Potential at point C due to charge at point B= $\frac{Q}{4\pi\epsilon_o l}$

Net potential at point C = $\frac{-Q}{4\pi\epsilon_o\sqrt{2}l} + \frac{Q}{4\pi\epsilon_o l} = \frac{Q}{4\pi\epsilon_o l\sqrt{2}}(-1+\sqrt{2})$

Potential at point D due to charge at point B = $\frac{Q}{4\pi\epsilon_o\sqrt{2}l}$

Potential at point D due to charge at point A= $\frac{-Q}{4\pi\epsilon_o l}$

Potential at point D due to charge at point C= $\frac{-Q}{4\pi\epsilon_o l}$

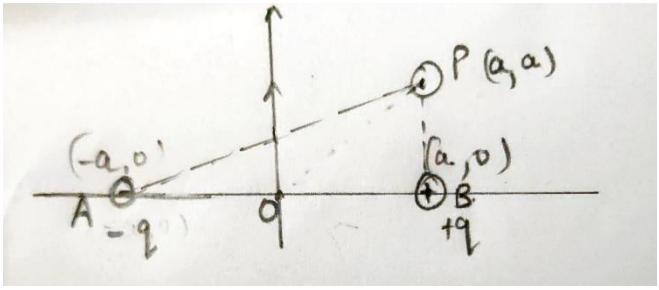
Net potential at point D = $\frac{Q}{4\pi\epsilon_o\sqrt{2}l} + \frac{-Q}{4\pi\epsilon_o l} + \frac{-Q}{4\pi\epsilon_o l} = \frac{Q}{4\pi\epsilon_o l\sqrt{2}}(1-2\sqrt{2})$

$$\text{work done } W = -Q\left(\frac{Q}{4\pi\epsilon_o l\sqrt{2}}\left(1-2\sqrt{2}+1-\sqrt{2}\right)\right) = -\frac{kQ^2}{l}\left(\frac{2-3\sqrt{2}}{\sqrt{2}}\right) = -\frac{kQ^2}{l}\left(\sqrt{2}-3\right)$$

18.Correct option(A)

$$\text{work done } W = q(V_2 - V_1)$$

As the charge Q is shifted from origin to P we have to calculate potential at Origin and P due to other charges



Distance of point P from charge – q is calculated by pythagoras theorem
In triangle PAB
Base =2a
Perpendicular = a
$H^2 = P^2 + B^2$

$H^2 = a^2 + 4a^2$
$H = \sqrt{5}a$

Potential at point O due to charge at point A = $\frac{-q}{4\pi\epsilon_o a}$
Potential at point O due to charge at point B= $\frac{q}{4\pi\epsilon_o a}$
Net potential at point O=0

Potential at point P due to charge at point A= $\frac{-q}{4\pi\epsilon_o \sqrt{5}a}$
Potential at point P due to charge at point B= $\frac{q}{4\pi\epsilon_o a}$
Net potential at point P= $\frac{-q}{4\pi\epsilon_o \sqrt{5}a} + \frac{q}{4\pi\epsilon_o a} = \frac{kq}{a}(1 - \frac{1}{\sqrt{5}})$

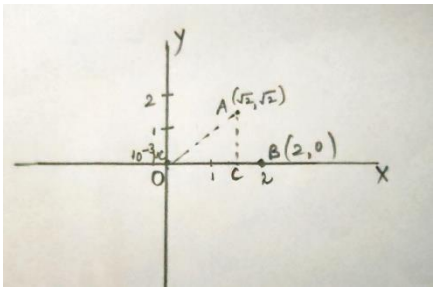
$$\text{work done } W = Q\left(\frac{kq}{a}\left(1 - \frac{1}{\sqrt{5}}\right) - 0\right) = \frac{kqQ}{a}(1 - \frac{1}{\sqrt{5}})$$

19.Correct option(B)
As the electron brings from infinity to the orbit near the nucleus, the pull of nucleus on the electron increases which decreases the potential energy .
20.Correct option(B)
Charge on proton =e
Charge on alpha particle =2e
Mass of proton = m_p
Mass of alpha particle = $4m_p$
We know that kinetic energy $K_p = \frac{1}{2}m_p v_p^2 = eV$ …….(1)

kinetic energy $K_\alpha = \frac{1}{2}m_\alpha v_\alpha^2 = 2eV = \frac{1}{2}4m_p v_\alpha^2$ ……(2)
Dividing eq 1 by 2
$\frac{v_p^2}{4v_\alpha^2} = \frac{1}{2}$
$\frac{v_p^2}{v_\alpha^2} = 2$

$\frac{v_p}{v_\alpha} = \sqrt{2}:1$

21.Correct option(B)



Distance of point A from origin is calculated by pythagoras theorem
In triangle AOC
Base =$\sqrt{2}$
Perpendicular = $\sqrt{2}$
$H^2 = P^2 + B^2$

$H^2 = \sqrt{2}^2 + \sqrt{2}^2$
$H = 2$

Potential at point A due to charge at point O = $\frac{10^{-3}\mu C}{4\pi\epsilon_o\, 2}$
Potential at point B due to charge at point O= $\frac{10^{-3}\mu C}{4\pi\epsilon_o\, 2}$
Potential difference between A and B= 0

22.Correct option(A)
$work\ done\ W = q(V_2 - V_1)$
$2 = 20\ (V_2 - V_1)$
$potential\ difference(V_2 - V_1) = 0.1\ V$

23.Correct option(B)
Distance between Q and +q is calculated by pythagoras theorem
Base =a
Perpendicular = a
$H^2 = P^2 + B^2$

$H^2 = a^2 + a^2$
$H = \sqrt{2}a$

Net potential energy of the system U = $\frac{Qq}{4\pi\epsilon_o a} + \frac{q^2}{4\pi\epsilon_o a} + \frac{Qq}{4\pi\epsilon_o\sqrt{2}a} = 0$

$Q + q + \frac{Q}{\sqrt{2}} = 0$

$Q\left(1 + \frac{1}{\sqrt{2}}\right) = -q \Rightarrow Q = -\frac{q\sqrt{2}}{\sqrt{2}+1}$

24.Correct option(C)

Since the surface is not closed so the net flux of electric field is zero.

25.Correct option(C)

$\boldsymbol{\phi = \frac{q_{enc}}{\epsilon_o}}$

Charge enclosed by the **surface** $\boldsymbol{S_1 = +Q}$

Charge enclosed by the surface $\boldsymbol{S_2 = +Q - \frac{4Q}{3} = -\frac{Q}{3}}$

$$\frac{\phi_1}{\phi_2} = \frac{Q}{\frac{-Q}{3}} = -3:1$$

26.Correct option(A)

$$\phi = \frac{q_{enc}}{\epsilon_o}$$

Charge enclosed by the surface $S = q_1 + q_2 = q - q = 0$

So net flux passing through surface S = 0

27.Correct option(A)

We know that, electric flux $\boldsymbol{\phi_1}$ (or electric field lines) entering in a closed surface is -ve and electric flux $\boldsymbol{\phi_2}$ (or electric field lines) leaving a closed surface is +ve.

Hence, net electric flux through the closed surface,= $\phi_2 - \phi_1$

$$\phi = \frac{q_{enc}}{\epsilon_o}$$

$$(\phi_2 - \phi_1)\epsilon_o = q$$

28.Correct option(D)

The electric field at any point is the resultant effect of all charges. So the electric field on surface will be due to all charges in this system.

29.Correct option(A)

$$\text{electric flux } \phi = \frac{q_{enc}}{\epsilon_o}$$

As the charge enclosed by the cube is q

So,

$$\text{electric flux } \phi = \frac{q}{\epsilon_o}$$

30.Correct option(B)

According to the gauss's law

$$\text{electric flux } \phi = \frac{q_{enc}}{\epsilon_o}$$

As the charge enclosed by the surface is $= \boldsymbol{q_3 + q_4}$

So,

$$\text{electric flux due to all the charges } (\phi) = \frac{q_3 + q_4}{\epsilon_o}$$

31.Correct option(C)

- Gauss law is valid also other medium like dielectric medium

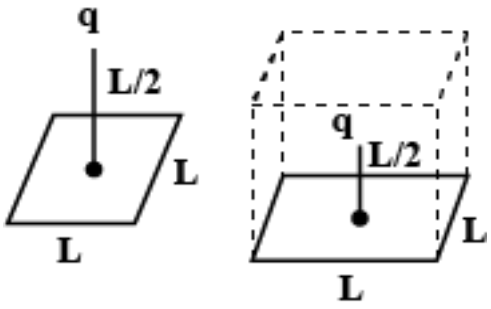
According to the gauss law, net electric flux through any closed surface.

$$\boldsymbol{net\ electric\ flux\ \phi = \frac{q_{enc}}{\epsilon_o}}$$

So, the net flux through the closed surface is equal to the flux due to charges enclosed by the surface in any medium.

The Gauss's law is valid most commonly for the charge distribution of symmetric body like spherical, cylindrical, plane symmetry. The charge distribution on the symmetry body may be uniform or non-uniform. It depends only on the charge enclosed by the closed surface.

32.Correct option(C)



The given square of side L may be considered as one of the faces of a cube with edge L. Then given charge q will be considered to be placed at the centre of the cube. Then according to Gauss's theorem, the magnitude of the electric flux through the faces (six) of the cube is given by

$$\text{electric flux } \phi = \frac{q_{enc}}{\epsilon_o}$$

Hence, electric flux through one face of the cube for the given square will be

Hence, electric flux through one face of the cube for the given square will be

$$\text{electric flux } \phi' = \frac{1}{6}\phi = \frac{q}{6\epsilon_o}$$

LEVEL-II

SINGLE OPTION CORRECT TYPE

1.Correct option(B)

The electrostatic force between two charges is given by

$$F = \frac{q_1 q_2}{4\pi\epsilon_o r^2}$$

$$5 = \frac{q_1 q_2}{4\pi\epsilon_o (0.06)^2} \ldots\ldots \{1\}$$

When the two charges move towards each other then new distance becomes 0.06-0.01-0.01=0.04

$$F_2 = \frac{q_1 q_2}{4\pi\epsilon_o (0.04)^2} \ldots\ldots \{2\}$$

Dividing equation 2 by 1

$$\frac{F_2}{5} = \frac{(0.06)^2}{(0.04)^2} = \frac{0.0036}{0.0016} = 2.25$$

$$F_2 = 2.25x\ 5 = \ 11.25N$$

2.Correct option(A)

The electrostatic force between two charges is given by

$$F = \frac{q_1 q_2}{4\pi\epsilon_o r^2}$$

$$40 = \frac{3\ x\ 8}{4\pi\epsilon_o (r)^2} \ldots\ldots \{1\}$$

When $charge\ of\ -5\mu C\ is\ added\ \ then\ q_1 = \ 3-5 = -2\ \mu C, \quad q_2 = 8-5 = 3\mu c$

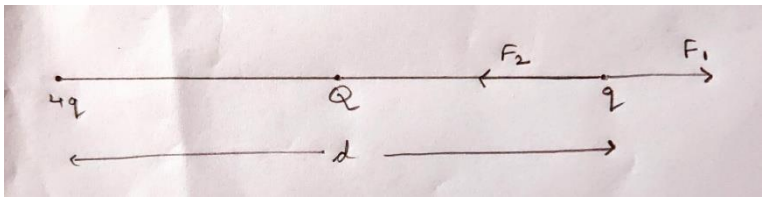
$$F_2 = \frac{-2\ x\ 3}{4\pi\epsilon_o (r)^2} \ldots\ldots \{2\}$$

Dividing equation 2 by 1

$$\frac{F_2}{40} = \frac{-2\ x\ 3}{3\ x\ 8} = -\frac{1}{4}$$

$$F_2 = -\frac{40}{4} = \ -10N$$

3.Correct option(A)



Let d is the distance between both the pairs of 4q and q then

The electrostatic force between two charges q and Q is given by

$$F_1 = \frac{qQ}{4\pi\epsilon_o \left(\frac{d}{2}\right)^2} = \frac{4qQ}{4\pi\epsilon_o d^2}$$

The electrostatic force between two charges 4q and q is given by

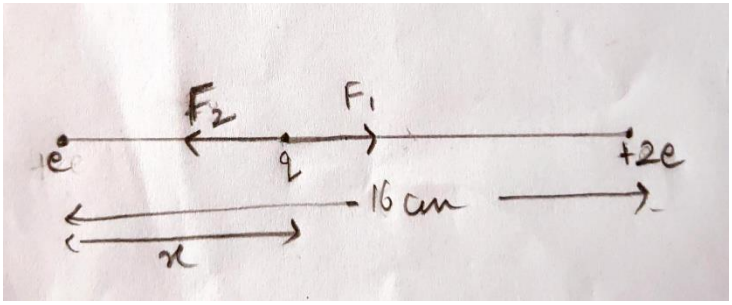
$$F_2 = \frac{4q^2}{4\pi\epsilon_o d^2}$$

For the force on charge q to be zero the sign of Q must be negative then only forces on q will act in opposite direction

$$F_1 = -F_2$$

$$4q^2 = -4qQ \quad or \quad Q = -q$$

4.Correct option(D)



For equilibrium of the whole system the sign of q must be negative
Let the distance between the charge q and e is x
The electrostatic force between two charges q and e is given by

$$F_1 = \frac{eq}{4\pi\epsilon_o(x)^2}$$

The electrostatic force between two charges q and 2e is given by

$$F_2 = \frac{2eq}{4\pi\epsilon_o(16-x)^2}$$

For equilibrium $F_1 = F_2$

$$\frac{eq}{4\pi\epsilon_o(x)^2} = \frac{2eq}{4\pi\epsilon_o(16-x)^2}$$

$$\frac{1}{(x)^2} = \frac{2}{(16-x)^2}$$

$$\frac{1}{x} = \frac{\sqrt{2}}{16-x}$$

$$16 - x = \sqrt{2}\,x$$

Incorrect values.

5.Correct option(C)

The charge is kept at a corner of a cube so it can be visualised as being at the centre of a cube of double its side i.e.,comprising of 8 such cubes kept besides and top of each other. Thus contribution to each smaller cube will be $= \frac{q}{8\epsilon_o}$

6.Correct option(C)

A charge q is placed at the center of open end Q a cylindrical vessel, we have to find the flux through the surface of the vessel.
so, when charge Q is placed at the center of open end of a cylindrical vessel then only half of the charge will contribute to the flux, because half will lie inside the surface and half will lie outside the surface.
so, flux through the surface of vessel is $\frac{q}{2\epsilon_o}$

7.Correct option(B)

$$\text{electric flux } \phi = \frac{q_{enc}}{\epsilon_o}$$

For unit positive charge q= 1

$$\text{electric flux } \phi = \frac{1}{\epsilon_o} = \epsilon_o^{-1}$$

8.Correct option(B)

We know that, electric flux $\boldsymbol{\phi_1}$ (or electric field lines) entering in a closed surface is -ve and electric flux $\boldsymbol{\phi_2}$ (or electric field lines) leaving a closed surface is +ve.
Hence, net electric flux through the closed surface,$= \phi_2 - \phi_1$

$$\phi = \frac{q_{enc}}{\epsilon_o}$$

$$(\phi_2 - \phi_1)\epsilon_o = q$$

9.Correct option(C)

Potential at any point is the work done in bringing a unit positive test charge from infinity to that point.

$$\text{potential } V = \frac{\text{work done}}{\text{test charge}} = \frac{50}{-10} = -5 \text{ volt}$$

10.Correct option(C)

$$\text{potential V} = \frac{\text{potential energy at the point}}{\text{charge}} = \frac{50}{5} = 10 \text{ volt}$$

11.Correct option(D)

Negative sign shows that the direction of electric field is the direction of decreasing potential.

12.Correct option(D)

We know that if W is the work done in bringing a unit positive charge q from infinity to a point in the field, the electric potential V at this point is given by:

$$\boldsymbol{potential\ V = \frac{work\ done}{test\ charge} = \frac{W}{q}}$$

We know that electric potential is $\boldsymbol{V = \frac{kQ}{r}}$

Combining both the equation, $W=\frac{kqQ}{r}$

Which is nothing but electric potential energy. So, we can say that electric potential energy is related to work done. So, we can say if potential energy will be zero then work done also will be zero.

Electric field is conservative. So, work done in taking a charge on the closed path is zero or two small spheres, each carrying a charge q are placed at a distance of r metre apart and they interact with forceF . If one of the spheres is taken around the other once in a circular path, the work done will be equal to Work done in equipotential surface is zero.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

13.Correct options(A), (B), (C)

E depends upon both Q1 and Q2 but ϕ depends only on Q1 because according to Gauss law flux from any surface depends only on the charge enclosed by that surface.

so if Q1 changes, both E and ϕ will change.

Q2 changes, then E will change and ϕ will remain same.

If Q1 =0 , E will be non-zero and ϕ will be 0.

14.Correct options(A), (C)

The electric field at any point and any plane on the equipotential surface is always perpendicular to S.

As the electric field is directed away from the surface so it has only radial component and there is no tangential component.

15Correcr options(C), (D)

Electric field is a vector quantity. The electric field at a point due to a number of point charges is the vector sum of electric field due to individual charges. So, when a positive charge is brought into an electric field, the electric field due to the positive charge is added to the electric field already present. Therefore, the electric field increases.

When a negative charge is brought into an electric field, the electric field due to the negative charge is subtracted from the electric field already present. Therefore, the electric field decreases.

16.Correct options(B), (D)

The potential difference between two points and the change in potential energy are relative terms and do not depend on the definition of an absolute reference like zero potential or zero potential energy.

17.Correct options(A), (C), (D)

we know that two equipotential surfaces can not intersect each other and lines of force are always perpendicular to the equipotential surface.

When two equipotential lines are close together, the slope will be steep. So, close equipotential surfaces indicate a maximum electric field

$E=-\frac{dv}{dr}$

Both the surfaces can be plane surfaces

18.Correct options(A), (B), (C), (D)

Uniform field means a constant field at every point and it is directed in a fixed direction for all points. Thus equipotential surfaces must be plane surface. The electric field is always perpendicular to equipotential surface because $\rightarrow E=-\rightarrow \nabla V$. As surfaces are equipotential so it must be spaced such that surfaces having equal differences in potential are separated by equal distances. As $E=-dv/dr$ so it must have decreasing potentials in the direction of the field.

COMPREHENSION TYPE

PASSAGE-1

19.Correct option(D)

In order to overcome his or her weight, the electrostatic force must balance his or her weight i.e., $qE=mg$

$\Rightarrow q=mg/E = 60\times10/\ 150=4C$

As the electric field is directed towards the earth's center, so the sign of charge should be negative. Thus, $q=-4C$

20.Correct option(D)

The electrostatic force between two charges is given by

$$F=\frac{q_1q_2}{4\pi\epsilon_o r^2}$$

$$F=\frac{-4\ \mathrm{x}-4}{4\pi\epsilon_o(100)^2}=16\mathrm{x}\ 9\ \mathrm{x}\ 10^9\mathrm{x}\ 10^{-4}=144\mathrm{x}10^5=1.44\mathrm{x}\ 10^7\mathrm{N}$$

PASSAGE-II

21.Correct option(A)

According to question the force must be qE = 10mg

$$\frac{q}{m}=\frac{10g}{E}=\frac{10\mathrm{x}9.8}{1.4\mathrm{x}\ 10^5}=7.0\ \mathrm{x}10^{-4}\mathrm{Ckg}^{-1}$$

22.Correct option(D)

According to question the force must be qE = 10mg

Mass of carbon atom m= 12 g = $12\ \mathrm{x}10^{-3}$kg

$$q=\frac{10mg}{E}=\frac{10x12\ \mathrm{x}10^{-3}\mathrm{x}9.8}{1.4\mathrm{x}\ 10^5}=8.4x\ 10^{-6}C$$

$$\text{no. of electrons } n = \frac{q}{e} = \frac{8.4x10^{-6}}{1.6x10^{-19}} = 5.25x10^{13}$$

12 g of carbon has avagadro no. of carbon atoms

Number of carbon atoms $= \frac{N_A}{n} = \frac{6.023x10^{23}}{5.25x10^{13}} = 1.15x10^{10}\ per\ excess\ electron$

ASSERTION AND REASON TYPE

23.Correct option(B)

Force on a proton and an electron in a uniform electric field is same because magnitude of charge on both is exactly the same.

Electric force on a test charge depends only on the source charge and the distance between source and test charge.

24.Correct option(C)

According to the gauss's law

$$\text{electric flux } \phi = \frac{q_{enc}}{\epsilon_o}$$

As the charge enclosed by the surface is $= -\boldsymbol{q} + \boldsymbol{2q} - \boldsymbol{q} = \mathbf{0}$

So,

$$\text{electric flux due to all the charges } (\phi) = 0$$

MATRIX MATCHING TYPE

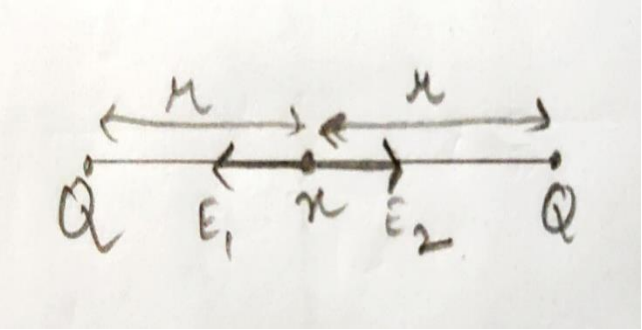
25 a…… P, S

b…….R

c…… T

d…….. Q, R

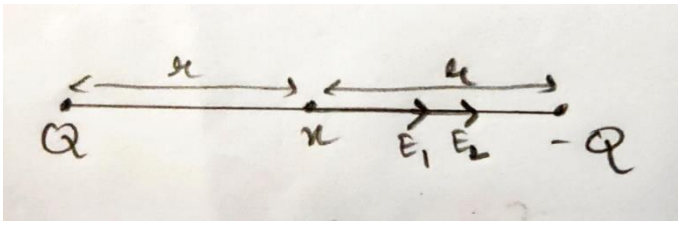
a)



Electric field at the point x will be equal in magnitude and opposite in direction hence it will cancel out to 0

Electric potential at point x = $V = \frac{kQ}{r} + \frac{kQ}{r} = \frac{\boldsymbol{2kQ}}{\boldsymbol{r}}$

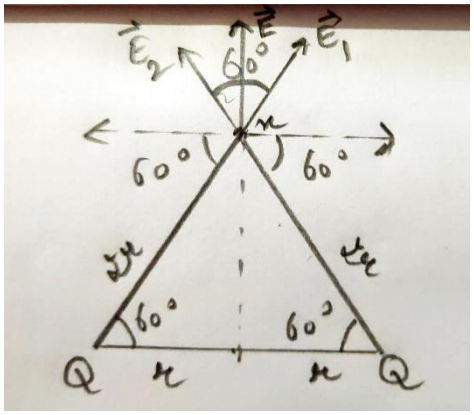
b)



Electric potential at point x = V = $\frac{kQ}{r} - \frac{kQ}{r} = 0$

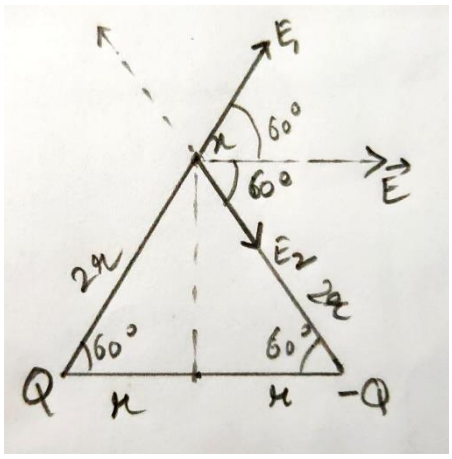
Electric field at point x will add up = E = $\frac{kQ}{r^2} + \frac{kQ}{r^2} = \frac{2kQ}{r^2}$

c)



Electric potential at point x = $V = \frac{kQ}{2r} + \frac{kQ}{2r} = \frac{kQ}{r}$

Resultant of electric field 1 and 2 = $\sqrt{E_1^2 + E_2^2 + 2E_1E_2cos60°} = \sqrt{2E_1^2 + E_1^2} = \sqrt{3}\,E_1 = \sqrt{3}\,\frac{\boldsymbol{kQ}}{\boldsymbol{4r^2}}$

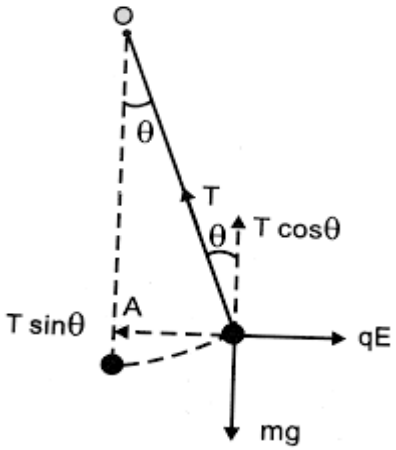


d) Electric potential at point x = $V = \frac{kQ}{2r} - \frac{kQ}{2r} = 0$

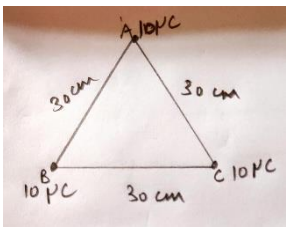
Resultant of electric field 1 and 2 = $\sqrt{E_1^2 + E_2^2 + 2E_1E_2cos120°} = \sqrt{2E_1^2 - E_1^2} = E_1 = \frac{kQ}{4r^2}$

INTEGER TYPE

26. 1



Mass =80mg
Charge =2x10^{-8}C
Electric field =40,000 Vm^{-1}
Given, mass =80mg=80x10^{-6}kg
Let T be the tension in string and θ is the angle if string with vertical.
In equilibrium
Tsinθ=qE…….. (1)
Tcosθ=mg …… (2)
Divide (1) by (2) we get
tanθ=$\frac{qE}{mg}$
⇒tanθ=$\frac{2x10^{-8}x40000}{80x10^{-6}x10} = \frac{10^{-4}}{10^{-4}} = 1$
Tanθ=1
From question
tanθ=x
We get x= 1
27. 9



Work done in arranging this configuration = potential energy of the system

$$potential\ energy = \frac{q_1 q_2}{4\pi\epsilon_o r}$$

Net potential energy of the system U =$U_{AB} + U_{AC} + U_{BC}$
$= \frac{9x\ 10^9 x10x10x10^{-12}}{30\ x10^{-2}} + \frac{9x\ 10^9 x10x10x10^{-12}}{30\ x10^{-2}} + \frac{9x\ 10^9 x10x10x10^{-12}}{30\ x10^{-2}} = 9\ Joules$
Work done = 9 J

4. Current Electricity

Solutions

LEVEL-I

Electric Current and Drift Velocity:

1. Correct option: (A)

Solution: We know, the drift velocity v_d of the charge carriers is given by:

$$v_d = \frac{J}{ne}$$

where J is the current density and ne is the charge carrier density.

We also know, the current density when the current is uniform across the surface and parallel to area vector, is defined as:

$$J = \frac{i}{A}$$

where A is the total area of the surface.

Given, a steady current is set up in a metallic wire of non-uniform cross-section.

Now, consider one such non-uniform cross-section of the metallic wire with area A. Then the drift velocity of the charge carrier is:

$$v_d = \frac{J}{ne}$$

$$v_d = \frac{i}{Ane}$$

Since the current is steady, it will be the same along any cross section.

Therefore, if a steady current is set up in a metallic wire of non-uniform cross-section, then the

drift velocity is related the area of cross-section A as $v_d \propto A^{-1}$.

2. Correct option: (A)

Solution: We know the expression for resistivity of a metal is given by:

$$\rho = \frac{m}{ne^2\tau}$$

where n is the number of free electrons per unit volume and τ is the mean time between the collisions of an electron with the atoms of the metal, m is the mass of electron and e is the elementary charge.

We also know, the resistance R of a conducting wire of length L and uniform cross section is

$$R = \rho\frac{L}{A}$$

where A is the cross-sectional area.

Given, n, e, τ, m represents electrons density charge, relaxation time and mass of an electron respectively.

Hence, the resistance of wire of length l and cross-sectional area A is given by:

$$R = \rho \frac{l}{A}$$

Substituting $\rho = \frac{m}{ne^2\tau}$ for resistivity, we get:

$$R = \frac{m}{ne^2\tau}\frac{l}{A}$$

Therefore, if n, e, τ, m represents electrons density charge, relaxation time and mass of an electron respectively, then , the resistance of wire of length l and cross-sectional area A is $\frac{ml}{ne^2\tau A}$.

3. Correct option: (C)

Solution: We know, an electric current i in a conductor is defined by

$$i = \frac{dq}{dt}$$

where dq is the amount of positive charge that passes in time dt.

Also, the drift velocity in a metallic wire carrying steady current i is:

$$v_d = \frac{i}{Ane}$$

where ne is the charge carrier density, and A is the area of cross section.

Given, a steady current pass through a wire of non-uniform cross-section.

We know, free electron density n is defined as the number of charge carriers per unit volume. It is the characteristic of the material, and is thus independent of the area of cross section.

Similarly, resistivity is also the characteristic of a material. It does not depend on the area of cross-section.

Now, the charge crossing in a given time interval is:

$$dq = idt$$

It is also thus independent of the area of cross section.

Since the drift velocity is $v_d = \frac{i}{Ane}$, hence it does depend on the cross-section.

4. Correct option: (A)

Solution: We know, for a material obeying Ohm's law:

$$E = \rho J$$

where ρ is the resistivity of the material and J is the current density.

We also know, the drift velocity v_d of the charge carriers is given by:

$$v_d = \frac{J}{ne}$$

where J is the current density and ne is the charge carrier density.

Now, using $E = \rho J$, the expression for drift velocity $v_d = \frac{J}{ne}$ can be rewritten as:

$$v_d = \frac{E}{ne\rho}$$

Since n, ρ are constant for a material, we can conclude $v_d \propto E$.

Therefore, if Ohm's law is obeyed, then $v_d \propto E$.

5. Correct option: (A)

Solution: We know, an electric current i in a conductor is defined by

$$i = \frac{dq}{dt}$$

where dq is the amount of positive charge that passes in time dt.

Given, a steady current is passing through a linear conductor of non-uniform cross-section.

Therefore, the net quantity of charge crossing any cross-section per second is just the current i. Since, the current is steady, hence the current is same in any cross section.

Therefore, if a steady current is passing through a linear conductor of non-uniform cross-section, then the net quantity of charge crossing any cross-section per second is independent of area of cross-section.

6. Correct option: (B)

Solution: We know the expression for resistivity of a metal is given by:

$$\rho = \frac{m}{ne^2\tau}$$

where n is the number of free electrons per unit volume and τ is the mean time between the collisions of an electron with the atoms of the metal, m is the mass of electron and e is the elementary charge.

Now, as we increase the temperature, the collision rate of the charge carriers increases. Thus the mean time between the collision or the relaxation time decreases. This ultimately leads to the increase in resistivity of the metal as n is very large and nearly independent of the temperature.

Therefore, the relaxation time in conductors decreases with the increase of temperature.

7. Correct option: (B)

Solution: We know, the drift velocity in a metallic wire carrying steady current i is:

$$v_d = \frac{i}{Ane}$$

where ne is the charge carrier density, and A is the area of cross section.

Given, a wire is carrying a current of 4A has cross – sectional area $10^{-8}\, m^2$. The free electron density in wire is $10^{28}\, m^{-3}$.

Therefore, the drift velocity is:

$$v_d = \frac{4}{10^{-8}10^{28}(1.6\times10^{-19})}$$

$$v_d = 0.25$$

Therefore, if a wire is carrying a current of 4A has cross – sectional area $10^{-8}\ m^2$ and the free electron density in wire is $10^{28}\ m^{-3}$, then the drift velocity is 0. 25 m/s.

8. Correct option: (D)

Solution: We know, the drift velocity in a metallic wire carrying steady current i is:

$$v_d = \frac{i}{Ane}$$

where ne is the charge carrier density, and A is the area of cross section.

Given, a steady current flow in a metallic conductor of non-uniform cross-section.

Since drift velocity depends on the area of cross section, and since the conductor has non-uniform cross section, so the cross section varies along the length of the conductor, therefore the drift velocity will also vary along the length of the conductor.

Since steady current flows in a metallic conductor, hence the current through any surface or along any length of the conductor will be the same.

We know, for a material obeying Ohm's law:

$$E = \rho J$$

where ρ is the resistivity of the material and J is the current density.

Now, the current density when the current is uniform across the surface and parallel to area vector, is:

$$J = \frac{i}{A}$$

where A is the total area of the surface.

Since the cross section along the length of the conductor is varying, hence the current density J and hence the electric field E too will vary along the length of the conductor. Therefore, if a steady current flow in a metallic conductor of non-uniform cross-section, then quantity which is constant along the length of the conductor is current only.

9. Correct option: (C)

Solution: We know, the drift velocity in a metallic wire carrying steady current i is:

$$v_d = \frac{i}{Ane}$$

where ne is the charge carrier density, and A is the area of cross section.

We also know, the resistance R of a conducting wire of length L and uniform cross section is $R = \rho\frac{L}{A}$

where A is the cross-sectional area.

Given, a potential difference of V is applied at the ends of copper wire of length l and diameter d.

Now, if we only double the diameter d, then the resistance becomes:

$$R' = \rho\frac{L}{\pi(2d/2)^2}$$

$$R' = \frac{R}{4}$$

Since current depends on the voltage V and resistance R by $V = iR$, therefore the new current is:

$$i' = \frac{4V}{R}$$

$$i' = 4i$$

Therefore, the new drift velocity is:

$$v_d' = \frac{4i}{(\pi(2d/2)^2)ne}$$

$$v_d' = \frac{4i}{4Ane}$$

$$v_d' = \frac{i}{Ane}$$

$$v_d' = v_d$$

Therefore, if a potential difference of V is applied at the ends of copper wire of length l and diameter d, and on doubling only d, drift velocity does not change.

10. Correct option: (D)

Solution: We know, an electric current i in a conductor is defined by

$$i = \frac{dq}{dt}$$

where dq is the amount of positive charge that passes in time dt.

Given, the current passing through a conductor is 5 Ampere.

Therefore, the charge that passes through that conductor in 5 minutes is:

$$q = it$$

$$q = 5(5)(60)$$

$$q = 1500\,C$$

Therefore, if the current passing through a conductor is 5 Ampere, then the charge that passes through that conductor in 5 minute is $1500\,C$.

11. Correct option: (B)

Solution: We know, an electric current i in a conductor is defined by

$$i = \frac{dq}{dt}$$

where dq is the amount of positive charge that passes in time dt.

Further, the time period T is related to the angular frequency ω by:

$$T = \frac{2\pi}{\omega}$$

Given, in a Hydrogen atom, an electron is revolving with an angular frequency 6. 28 rad/s around the nucleus.

Since the charge carried by an electron is e, therefore the equivalent electric current is:

$$i = \frac{e}{T}$$

$$i = \frac{e\omega}{2\pi}$$

$$i = \frac{e(6.28)}{2(3.14)}$$

$$i = e$$

$$i = 1.6 \times 10^{-19}$$

Therefore, if in a Hydrogen atom, an electron is revolving with an angular frequency 6.28 rad/s around the nucleus, then the equivalent electric current is $1.6*10^{-19}$ A.

Ohm's law:

12. Correct option: (D)

Solution: We know, Ohm's law states that the current through a device is always directly proportional to the potential difference applied to the device. Mathematically,

$V = iR$

where i current flows through the device when V voltage is applied across the device, R is its resistance. A given device obeys Ohm's law if its resistance is independent of the applied potential difference V.

We therefore see $i \propto V$ and $V \propto i$.

Therefore, Ohm's law states that $i \propto V, V \propto i$ and $V = iR$.

13. Correct option: (B)

Solution: We know, for a material obeying Ohm's law:

$E = \rho J$

where ρ is the resistivity of the material and J is the current density.

Since ρ is a constant for any material at a particular temperature, therefore $E \propto J$.

Therefore, electric field E and current density J have the relation $E \propto J$.

14. Correct option: (A)

Solution: We know, Ohm's law states that the current through a device is always directly proportional to the potential difference applied to the device. Mathematically,

$V = iR$

where i current flows through the device when V voltage is applied across the device, R is its resistance.

A given device obeys Ohm's law if its resistance is independent of the applied potential difference V. This is true for metals at low temperature.

As we increase the temperature, the resistance no longer remains constant - It also increases for the metals. Hence Ohm's law remains no longer true for metals at high temperature.

Diodes are non-linear devices, the current across the diode is not proportional to the voltage across the diode. It doesn't follow Ohm's law.

Electrolytes also don't obey the Ohm's law unless the potential is very high. This is because, as the voltage increase, the number of conduction ions also increases, and the overall current increases non-linearly.

Therefore, Ohm's law is true for metallic conductors at low temperature.

15. Correct option: (A)

Solution: When there is a temperature difference, heat flows from a body at higher temperature to a body at low temperature.

Similarly, if there is potential difference between two bodies connected by an electric circuit, charge flows from the body at higher potential low to low potential.

Therefore, the quantity which is analogous to temperature in electricity is Potential.

16. Correct option: (A)

Solution: In a metallic conductor, there are lots of free conduction electron. These electrons constitute the current flow in a conductor.

Therefore, the flow of the electric current through a metallic conductor is only due to electrons.

17. Correct option: (A)

Solution: We know, Ohm's law states that the current through a device is always directly proportional to the potential difference applied to the device. Mathematically,

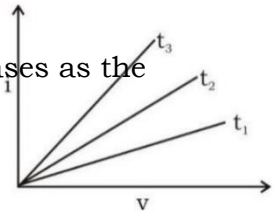
$V = iR$

where i current flows through the device when V voltage is applied across the device, and R is its resistance.

Therefore, the slope of $i - V$ curve gives the inverse of the resistance.

We know, as temperature increases, the resistance of the metal also increases as the electrons suffer more collusion due to increase in thermal motion.

Given, $i - V$ graph for a metal at temperatures t_1, t_2, t_3 are shown.



Since t_1 has the lowest slope, so it has the highest resistance and as resistance increases with increasing temperature, so t_1 should be the highest temperature.

Terminal Voltage:

18. Correct option: (A)

Solution: We know, during discharging, a battery acts like a pump, such that the current flows in the circuit from the positive terminal of the battery to the negative terminal. This means, inside the battery the direction of current in the cell is (-) ve pole to (+) ve pole during discharging.

19. Correct option: (A)

Solution: We know, the terminal-to-terminal potential difference V across the discharging battery is related to its emf E by the equation:

$V = E - ir$

where i is the current through the battery and r is its internal resistance.

Further, the terminal-to-terminal potential difference V across the charging battery is related to its emf E by the equation:

$V = E + ir$

where i is the current through the battery and r is its internal resistance.

Therefore, when the battery is being charged, its terminal voltage is greater than its emf.

20. Correct option: (A)

Solution: We know, an idea cell is defined to have zero internal resistance. Therefore, the value of internal resistance of an ideal cell is zero.

21. Correct option: (A)

Solution: We know, Ohm's law states that the current through a device is always directly proportional to the potential difference applied to the device. Mathematically,

$V = iR$

where i current flows through the device when V voltage is applied across the device, and R is its resistance.

In an open circuit, no current flows for any value of voltage V. This implies infinite resistance.

Therefore, the resistance of an open circuit is infinity.

22. Correct option: (C)

Solution: We know, the terminal-to-terminal potential difference V across the discharging battery is related to its emf E by the equation:

$V = E - ir$

where i is the current through the battery and r is its internal resistance.

Given, a $50\,V$ battery is connected across a 10 ohm resistor. The current is 4. 5 A flowing through the resistor.

Now, the voltage across the resistor from Ohm's law is $V = iR$ where R is the resistance of the resistor. Hence, the internal resistance of the battery is given by:

$V = E - ir$

$iR = E - ir$

$$r = \frac{E - iR}{i}$$

Substituting the values we get:

$$r = \frac{E - iR}{i}$$

$$r = \frac{50 - 4.5(10)}{4.5}$$

$$r = \frac{5}{4.5}$$

$$r = 1.1\Omega$$

Therefore, if a $50\,V$ battery is connected across a 10 ohm resistor. The current is 4. 5 A flowing through the resistor, then the internal resistance of the battery is 1.1Ω.

23. Correct option: (A)

Solution: We know, the terminal-to-terminal potential difference V across the discharging battery is related to its emf E by the equation:

$V = E - ir$

where i is the current through the battery and r is its internal resistance.

Further, the terminal-to-terminal potential difference V across the charging battery, when the current flows in opposite direction than that of discharging battery, is related to its emf E by the equation:

$V = E + ir$

Therefore, when the battery is being charged, its terminal voltage is greater than its emf.

24. Correct option: (D)

Solution: We know, the terminal-to-terminal potential difference V across the discharging battery is related to its emf E by the equation:

$V = E - ir$

where i is the current through the battery and r is its internal resistance.

Therefore, when a battery is discharging, its terminal voltage is always less than the emf of the battery: $V < E$

Further, the terminal-to-terminal potential difference V across the charging battery is related to its emf E by the equation:

$V = E + ir$

where i is the current through the battery and r is its internal resistance.

Therefore, when a battery is charging, its terminal voltage is always greater than the emf of the battery: $V > E$

Now, when a cell is short-circuited, such that there is no external resistance of the load, the terminal voltage, which is the voltage across the external load is zero.

When a cell is in open circuit, the current across the battery is zero, thus the terminal voltage is:

$V = E - 0r$

$V = E$

Therefore, the correct match is:

a -h, b-g, c -e, d-f

25. Correct option: (B)

Solution: We know, the terminal-to-terminal potential difference V across the charging battery is related to its emf E by the equation:

$V = E + ir$

where i is the current through the battery and r is its internal resistance.

Given, a cell of emf 6V is being charged by 1A current and the internal resistance of the cell is 1 ohm.

Therefore, the potential difference across the terminals of the cell is:

$V = 6 + (1)(1)$

$V = 7V$

Therefore, if a cell of emf 6V is being charged by 1A current and the internal resistance of the cell is 1 ohm, then the potential difference across the terminals of the cell is $7V$.

26. Correct option: (Doubt (A)

Solution: We know, the terminal-to-terminal potential difference V across the discharging battery is related to its emf E by the equation:

$V = E - ir$

where i is the current through the battery and r is its internal resistance.

Now, if the external resistance is R, we can rewrite the above expression as:

$iR = E - ir$

$$i = \frac{E}{R + r}$$

Therefore, current provided by a battery is maximum when the denominator in above expression is minimum when internal resistance r is less than external resistance R.

Joules Law And Electric Power:

27. Correct option: (B)

Solution: We know, the heat generated in a resistor R when a current i flows through it is:

$$Q = i^2 Rt = \frac{V^2 t}{R}$$

where t is time for which the current is passed and V is the voltage across the resistor.

We also know, the resistance R of a conducting wire of length L and uniform cross section is

$$R = \rho \frac{L}{A}$$

where A is the cross-sectional area.

Given, a heater coil is cut into two equal parts and only one part is used in the heater.

Now, when the heater coil is cut into two equal parts, the length is halved but the area remain unchanged, thus the new resistance becomes:

$$R' = \rho \frac{(L/2)}{A}$$

$$R' = \frac{R}{2}$$

Hence the heat generated becomes:

$$Q' = i^2 \frac{R}{2} t$$

$$Q' = \frac{Q}{2}$$

Therefore, if a heater coil is cut into two equal parts and only one part is used in the heater, then the heat generated becomes halved.

28. Correct option: (D)

Solution: We know, the heat generated in a resistor R when a current i flows through it is:

$$Q = i^2 Rt = \frac{V^2 t}{R}$$

where t is time for which the current is passed and V is the voltage across the resistor.

We also know, the resistance R of a conducting wire of length L and uniform cross section is

$$R = \rho \frac{L}{A}$$

where A is the cross-sectional area.

Given, A heater of 220 volt boils a volume of water in 5 minutes time.

Therefore, the amount of heat Q required to boil the water is:

$$Q = \frac{V^2 t}{R}$$

$$Q = \frac{220^2 (5)(60)}{R}$$

where R is the resistance of the heater.

Now, If the heater is operated at 110 volt, we will require the same amount of heat to boil the water. Let the time required now to boil the water be t minutes, then:

$$\frac{110^2 (t)(60)}{R} = \frac{220^2 (5)(60)}{R}$$

$$t = 5(2^2)$$

$$t = 20 \text{ min}$$

Therefore, if a heater of 220 volt boils a volume of water in 5 minutes time, then If the heater is operated at 110 volt the same volume of water will boil in 20 minutes.

29. Correct option: (D)

Solution: We know, the heat generated in a resistor R when a current i flows through it is:

$$Q = i^2 Rt = \frac{V^2 t}{R}$$

where t is time for which the current is passed and V is the voltage across the resistor.

We see, the heat generated in a conductor carrying current depends upon the time t for which the current is passed, resistance R of the conductor and strength of the current i.

30. Correct option: (A)

Solution: We know, the power P or rate of energy transfer, in a resistor R across which a potential difference V is maintained is

$$P = i^2 R = \frac{V^2}{R}$$

Further, when two resistors R_1 and R_2 are connected in series, the equivalent resistance is the sum of the individual resistance:

$$R_{eq} = R_1 + R_2$$

Given, A 25 Watt, 220 Volt bulb and a 100 Watt, 220 Volt bulb are connected in series across 440 volt line.

Now, the bulb will fuse when its power exceed the power rating of the bulb.

Here, for the first bulb with power rating of 25 Watt, at 220 Volt, its resistance is:

$$P = \frac{V^2}{R_1}$$

$$R_1 = \frac{V^2}{P}$$

$$R_1 = \frac{220^2}{25}$$

$$R_1 = 1936\,\Omega$$

Similarly for the second bulb with power rating of 100 Watt at 220 Volt, its resistance is:

$$P = \frac{V^2}{R_2}$$

$$R_2 = \frac{V^2}{P}$$

$$R_2 = \frac{220^2}{100}$$

$$R_2 = 484\,\Omega$$

Now, when the two bulbs are connected in series across 440 volt line, the current across each bulb will be the same:

$$i = \frac{V}{R_{eq}}$$

$$i = \frac{V}{R_1 + R_2}$$

$$i = \frac{440}{1936 + 484}$$

$$i = 0.181A$$

Therefore, the power of the first bulb is:

$$P = i^2 R_1$$

$$P = (0.181)^2 (1936)$$

$$P = 64W$$

Since this power exceeds the power rating of $25W$, the first bulb will fuse.

Now, the power of the second bulb is:

$$P = i^2 R_2$$

$$P = (0.181)^2 (484)$$

$$P = 16W$$

Since this power is less than the power rating of $100W$, the second bulb will not fuse.

Therefore, if a 25 Watt, 220 Volt bulb and a 100 Watt, 220 Volt bulb are connected in series across 440 volt line, then only 25 Watt bulb will fuse.

31. Correct option: (D)

Solution: We know, the power P or rate of energy transfer, in a resistor R across which a potential difference V is maintained is

$$P = i^2 R = \frac{V^2}{R}$$

Given, an electric bulb is rated 220 volt and 100 watt.

Therefore, the resistance of the bulb is:

$$P = \frac{V^2}{R}$$
$$R = \frac{V^2}{P}$$
$$R = \frac{220^2}{100}$$
$$R = 484\Omega$$

Now, power consumed by it when it is operated on 110 volt is:

$$P = \frac{V^2}{R}$$
$$P = \frac{110^2}{484}$$
$$P = 25W$$

Therefore, if an electric bulb is rated 220 volt and 100 watt, then power consumed by it when it is operated on 110 volt is 25 watt.

32. Correct option: (Question incomplete (B)

Solution: We know, the heat generated in a resistor R when a current i flows through it is:

$$Q = i^2 Rt = \frac{V^2 t}{R}$$

where t is time for which the current is passed and V is the voltage across the resistor.

Given, a current of 2A passing through a conductor produces 80J of heat in t seconds.

Therefore, the resistance of a conductor in ohm is:

$$Q = i^2 Rt$$
$$R = \frac{Q}{i^2 t}$$
$$R = \frac{80}{2^2 t}$$
$$R = \frac{20}{t}$$

We thus need the value of t to solve this question.

33. Correct option: (C)

Solution: We know, the power P or rate of energy transfer, in a resistor R across which a potential difference V is maintained is

$$P = i^2R = \frac{V^2}{R}$$

Given, two electric bulbs whose resistances are in the ratio of 1:2 are connected in parallel to a constant voltage source.

Hence, the powers dissipated in them have the ratio:

$$\frac{P_1}{P_2} = \frac{\frac{V^2}{R_1}}{\frac{V^2}{R_2}}$$

$$\frac{P_1}{P_2} = \frac{R_2}{R_1}$$

$$\frac{P_1}{P_2} = \frac{2}{1}$$

Therefore, if two electric bulbs whose resistances are in the ratio of 1:2 are connected in parallel to a constant voltage source, then the powers dissipated in them have the ratio 2:1.

LEVEL-II

1. Correct option: (B)

Solution: We know the force on a charged particle q in an electric field $\vec{E}$ is given by:

$$\vec{F} = q\vec{E}$$

Given, an electric field $\vec{E}$ is applied across the ends of a conductor. Since the electrons carry the charge $-e$, therefore the force on the electrons is:

$$\vec{F} = -e\vec{E}$$

Therefore, the force acts in the opposite direction of the electric field.

Hence, if an electric field $\vec{E}$ is applied across the ends of a conductor, the direction of motion of free electrons is opposite to the direction of $\vec{E}$.

2. Correct option: (B)

Solution: We also know, the resistance R of a conducting wire of length L and uniform cross section is

$$R = \rho\frac{L}{A}$$

where A is the cross-sectional area.

We also know, Ohm's law states that the current through a device is always directly proportional to the potential difference applied to the device. Mathematically,

$V = iR$

where i current flows through the device when V voltage is applied across the device, R is its resistance.

Given, a steady current is set up in a metallic wire of non-uniform cross-section.

Now, the rate of flow of electrons is the current i in the metallic wire. It is given by:

$V = iR$

$$i = \frac{V}{R}$$
$$i = \frac{V}{\rho \frac{L}{A}}$$
$$i = \frac{VA}{\rho L}$$
$$i \propto A$$

Therefore, if a steady current is set up in a metallic wire of non-uniform cross-section, then the rate

of flow of electrons r is related to the area of cross-section A as $r \propto A$

3. Correct option: (D)

Solution: We know, the terminal-to-terminal potential difference V across the discharging battery is related to its emf E by the equation:

$V = E - ir$

where i is the current through the battery and r is its internal resistance.

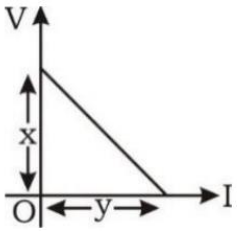
We also know the equation of a straight line with x intercept a and y intercept b is given by:

$$\frac{x}{a} + \frac{y}{b} = 1$$

Transforming the above equation in this form, we get:

$$V + ir = E$$
$$\frac{V}{E} + \frac{ir}{E} = 1$$

Therefore, we see plotting $V - i$ graph, the y intercept is the emf E and the x-intercept is E / r.



Given, a graph plotted between potential difference V between the terminals of a cell against current between the terminals a cell against current i in a circuit by varying as load rheostat is as shown.

From the figure, we see the emf of the cell is x and E / r is y.

Therefore the internal resistance of the cell is:

$$\frac{E}{r} = y$$
$$r = \frac{E}{y}$$
$$r = \frac{x}{y}$$

Since the conductance of the cell is the reciprocal of the resistance, thus the internal conductance of the cell is $\frac{y}{x}$.

4. Correct option: (C)

Solution: We know when many cells are connected in series, the net emf of the combination is the sum of the emf of each of the individual cell.

Further, when two resistors R_1 and R_2 are connected in series, the equivalent resistance is the sum of the individual resistance:

$R_{eq} = R_1 + R_2$

Given, n cells each of emf E and internal resistance r are connected in series.

Therefore, using the Ohm's law $V = iR$, current through an external resistance R is given by:

$$i = \frac{V}{R_{eq}}$$

$$i = \frac{nE}{R + nr}$$

Therefore, if n cells each of emf E and internal resistance r are connected in series, then current through an external resistance R is $\frac{nE}{R + nr}$.

5. Correct option: (D)

Solution: We know, the heat generated in a resistor R when a current i flows through it is:

$$Q = i^2Rt = \frac{V^2t}{R}$$

where t is time for which the current is passed and V is the voltage across the resistor.

We also know, the resistance R of a conducting wire of length L and uniform cross section is

$$R = \rho\frac{L}{A}$$

where A is the cross-sectional area.

Given, A heater of 220 volt boils a volume of water in 5 minutes time.

Therefore, the amount of heat Q required to boil the water is:

$$Q = \frac{V^2t}{R}$$

$$Q = \frac{220^2(5)(60)}{R}$$

where R is the resistance of the heater.

Now, If the heater is operated at 110 volt, we will require the same amount of heat to boil the water. Let the time required now to boil the water be t minutes, then:

$$\frac{110^2(t)(60)}{R} = \frac{220^2(5)(60)}{R}$$

$$t = 5(2^2)$$

$$t = 20\,\text{min}$$

Therefore, if a heater of 220 volt boils a volume of water in 5 minutes time, then If the heater is operated at 110 volt the same volume of water will boil in 20 minutes.

6. Correct option: (B)

Solution: We know, the power P or rate of energy transfer, in a resistor R across which a potential difference V is maintained is

$$P = i^2 R = \frac{V^2}{R}$$

Given, of the two bulbs in a house one glows brighter than the other.

Now, if the bulb glows brighter, then it has more power.

Since power P is directly proportional to the resistance R when the two bulbs are connected to the same main voltage supple of the house, therefore the bulbs that glows dimly has the higher resistance than the other.

7. Correct option: (A)

Solution: We know, the terminal-to-terminal potential difference V across the discharging battery is related to its emf E by the equation:

$$V = E - ir$$

where i is the current through the battery and r is its internal resistance.

Given, the terminal potential difference of a cell is equal to emf of the cell.

Then,

$$E = E - ir$$

$$i = 0$$

Since $i = \frac{V}{R}$, therefore the external resistance is infinity.

Therefore, if the terminal potential difference of a cell is equal to emf of the cell, then the external resistance is infinity.

8. Correct option: (A)

Solution: We know, the terminal-to-terminal potential difference V across the discharging battery is related to its emf E by the equation:

$$V = E - ir$$

where i is the current through the battery and r is its internal resistance.

Given, the external resistance is equal to the internal resistance of a cell of emf E

Then, using the Ohm's law $V = iR$ where R is the external resistance, we get

$$iR = E - ir$$

$$iR = E - iR$$

$$i = \frac{E}{2R}$$

Therefore, the potential difference across its terminal is:

$$V = E - \frac{E}{2R} R$$

$$V = \frac{E}{2}$$

Therefore, if the external resistance is equal to the internal resistance of a cell of emf E, then the potential difference across its terminal is $\frac{E}{2}$.

9. Correct options: B), C), D)

Solution: We know, the drift velocity in a metallic wire carrying steady current i is:

$$v_d = \frac{i}{Ane}$$

where ne is the charge carrier density, and A is the area of cross section.

Given, a steady current flow in a metallic conductor of non-uniform cross-section.

Since drift velocity depends on the area of cross section, and since the conductor has non-uniform cross section, so the cross section varies along the length of the conductor, therefore the drift velocity will also vary along the length of the conductor.

Since steady current flows in a metallic conductor, hence the current through any surface or along any length of the conductor will be the same.

We know, for a material obeying Ohm's law:

$$E = \rho J$$

where ρ is the resistivity of the material and J is the current density.

Now, the current density when the current is uniform across the surface and parallel to area vector, is:

$$J = \frac{i}{A}$$

where A is the total area of the surface.

Since the cross section along the length of the conductor is varying, hence the current density J and hence the electric field E too will vary along the length of the conductor. Therefore, if a steady current flow in a metallic conductor of non-uniform cross-section, then quantities which are not constant along the length of the conductor are current density, electric field and drift speed.

10. Correct options: A), C), D)

Solution: We know, the drift velocity in a metallic wire carrying steady current i is:

$$v_d = \frac{i}{Ane}$$

where ne is the charge carrier density, and A is the area of cross section.

From this expression, we see, drift velocity depends upon cross-section of the wire A, number of free electrons n and magnitude of the current i.

11. Correct option: (C)

Solution: A material is called as non-ohmic resistance if it doesn't obey Ohm's law, that is, if its resistance is not independent of the applied potential difference V.

All metals at low temperature obey Ohm's law. Therefore, copper wire and tungsten wire are both ohmic resistance.

Diodes are non-linear devices, the current across the diode is not proportional to the voltage across the diode. It doesn't follow Ohm's law. Therefore, diode is an example of a non – ohmic resistance. Therefore, example of non ohmic resistance is diode.

12. Correct options: A), B), C)

Solution: We know, the heat generated in a resistor R when a current i flows through it is:

$$Q = i^2 Rt = \frac{V^2 t}{R}$$

where t is time for which the current is passed and V is the voltage across the resistor.

We see, the heat generated in a conductor carrying current depends upon the time t for which the current is passed, resistance R of the conductor and strength of the current i.

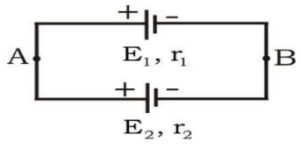
13. Correct options: A), B), C), D)

Solution: We know, for a move through a resistance in the direction of the current, the change in potential is $-iR$; in the opposite direction it is iR.

Also, for a move through an ideal emf device in the direction of from positive terminal to negative terminal, the change in potential is E; in the opposite direction it is $-E$

We know, according to the junction rule, algebraic sum of the changes in potential encountered in a complete traversal of any loop of a circuit must be zero.

Given, two cells of unequal e. m. fs E_1 and E_2 and internal resistances r_1 and r_2. V_A and V_B are the potentials at A and B respectively. Let's assume E_1 is greater than E_1.



Now, traversing the loop counterclockwise, we get:

$$-E_2 - ir_2 + ir_1 + E_1 = 0$$

$$i = \frac{E_2 - E_1}{r_1 - r_2}$$

Since the current in flowing in opposite direction through the batteries, therefore one cell will continuously supply energy to the other. Hence, one cell is continuously charging another cell is continuously discharging

The potential difference across both the cells will be equal. This follows from the fact that algebraic sum of the changes in potential encountered in a complete traversal of any loop of a circuit must be zero.

Since for a charging battery the potential across its is greater than its emf, thus here the potential difference across one cell will be greater than its e. m. f .

14. Correct options: A)

Solution: We know, the current density J is defined as the amount current i passing through unit area:

$$J = \frac{i}{A}$$

Given, a current of 1. 8 A flows through a wire of area of cross section 0. 5 mm^2.

Hence the current density in the wire is:

$$J = \frac{1.8}{0.5 \times 10^{-6}}$$

$$J = 3600000$$

Therefore, if a current of 1. 8 A flows through a wire of area of cross section 0. 5 mm^2, then the current density in the wire is $3.6 \times 10^{-6}\ Am^{-2}$

15. Correct option: (C)

Solution: We know, the current density J is defined as the amount current i passing through unit area:

$$J = \frac{i}{A}$$

We also know, an electric current i in a conductor is defined by

$$i = \frac{dq}{dt}$$

where dq is the amount of positive charge that passes in time dt.

Given, a silver wire 1 mm in diameter carries a charge of 90 coulombs in one hour and 15 minutes. Silver contains 5.8×10^{22} free electrons cm^{-3}.

Now, the current in the wire is:

$$i = \frac{90}{(60+15)(60)}$$

$$i = 0.02\,A$$

Hence the current density in the wire is:

$$J = \frac{0.02}{\pi(0.5\times10^{-3})^2}$$

$$J = 25464.79 Am^{-2}$$

$$J = 2.55\times10^{4}\,Am^{-2}$$

Therefore, if a silver wire 1 mm in diameter carries a charge of 90 coulombs in one hour and 15 minutes, then the current density is $2.55\times10^{4}\,Am^{-2}$.

16. Correct option: (A)

Solution: We know, the current density J is defined as the amount current i passing through unit area: $J = \frac{i}{A}$

Given, two rods A, B of radii 2 cm and 4 cm connected one by one respectively and the current flowing through them is same.

Consequently, the ratio of current densities is:

$$\frac{J_1}{J_2} = \frac{\frac{i}{A_1}}{\frac{i}{A_2}}$$

$$\frac{J_1}{J_2} = \frac{A_2}{A_1}$$

$$\frac{J_1}{J_2} = \frac{\pi R_2^2}{\pi R_1^2}$$

$$\frac{J_1}{J_2} = \frac{\pi 4^2}{\pi 2^2}$$

$$\frac{J_1}{J_2} = 4$$

Therefore, if two rods A, B of radii 2 cm and 4 cm connected one by one respectively and the current flowing through them is same, then the ratio of current densities is $4:1$.

17. Correct option: (B)

Solution: We know, the power P or rate of energy transfer of a device across which a potential difference V is maintained is

$P = iV$

where i is the current through the device.

Given, a house has 3 tube lights, two fans and a Television. Each tube light draws 40W. The fan draws 80W and the Television draws 60W. On the average, all the tube lights are kept on for five hours, two fans for 12 hours and the television for five hours every day.

Therefore, the total electrical energy consumed per day in the house is:

$E = Pt$

$E = 3(40)(5)(60)(60) + 2(80)(12)(60)(60) + 1(60)(5)(60)(60)$

$E = 2160000 + 6912000 + 1080000$

$E = 10152000J$

Since $1kWh$ contains $3.6\times10^6\ J$, therefore the total energy consumed per day in the house in kWh is:

$$E = \frac{10152000}{3.6\times10^6} kWh$$

$E = 2.82\,kWh$

Therefore, if a house has 3 tube lights, two fans and a Television. Each tube light draws 40W. The fan draws 80W and the Television draws 60W. On the average, all the tube lights are kept on for five hours, two fans for 12 hours and the television for five hours every day, then the total electrical energy consumed per day is $2.82\,kWh$.

18. Correct option: (C)

Solution: We know, the power P or rate of energy transfer of a device across which a potential difference V is maintained is

$P = iV$

where i is the current through the device.

Given, a house has 3 tube lights, two fans and a Television. Each tube light draws 40W. The fan draws 80W and the Television draws 60W. On the average, all the tube lights are kept on for five hours, two fans for 12 hours and the television for five hours every day.

Therefore, the total electrical energy consumed per day in the house is:

$E = Pt$

$E = 3(40)(5)(60)(60) + 2(80)(12)(60)(60) + 1(60)(5)(60)(60)$

$E = 2160000 + 6912000 + 1080000$

$E = 10152000\,J$

Since $1kWh$ contains $3.6\times10^6\ J$, therefore the total energy consumed per day in the house in kWh is:

$$E = \frac{10152000}{3.6\times10^{6}} kWh$$
$$E = 2.82\,kWh$$

Therefore, the number of units consumed in a month is:

$$E = 2.82\times30$$
$$E = 84.6\,kWh$$

Therefore, if a house has 3 tube lights, two fans and a Television. Each tube light draws 40W. The fan draws 80W and the Television draws 60W. On the average, all the tube lights are kept on for five hours, two fans for 12 hours and the television for five hours every day, then the no. of units consumed in 30 days is $84.6\,kWh$.

19. Correct option: (A)

Solution: We know, the power P or rate of energy transfer of a device across which a potential difference V is maintained is

$$P = iV$$

where i is the current through the device.

Given, a house has 3 tube lights, two fans and a Television. Each tube light draws 40W. The fan draws 80W and the Television draws 60W. On the average, all the tube lights are kept on for five hours, two fans for 12 hours and the television for five hours every day.

Therefore, the total electrical energy consumed per day in the house is:

$$E = Pt$$
$$E = 3(40)(5)(60)(60) + 2(80)(12)(60)(60) + 1(60)(5)(60)(60)$$
$$E = 2160000 + 6912000 + 1080000$$
$$E = 10152000\,J$$

Since $1kWh$ contains $3.6\times10^{6}\,J$, therefore the total energy consumed per day in the house in kWh is:

$$E = \frac{10152000}{3.6\times10^{6}} kWh$$
$$E = 2.82\,kWh$$

Therefore, the number of units consumed in a month is:

$$E = 2.82\times30$$
$$E = 84.6\,kWh$$

Since the rate per kWh of energy consumed is Rs 3, therefore the cost of electrical energy used in 30 days is:

$$84.6\times3 = 253.8$$

Therefore, if a house has 3 tube lights, two fans and a Television. Each tube light draws 40W. The fan draws 80W and the Television draws 60W. On the average, all the tube lights are kept on for five hours, two fans for 12 hours and the television for five hours every day, then the cost of electrical energy used in 30 days is Rs 253.8.

20. Correct option: (A)

Solution: We know, if an electric current i passes through a conductor of resistance R for a duration of t sec. Then the heat energy or thermal energy produced in the conductor is $H = i^2Rt$.

Given, a resistor develops $400\,J$ of thermal energy in $10\,s$, when a current of 2A is passed through it.

Therefore, the resistance of the resistor is:

$$H = i^2Rt$$

$$400 = 2^2R(10)$$

$$R = \frac{400}{40}$$

$$R = 10\,\Omega$$

Thus, if a resistor develops $400\,J$ of thermal energy in $10\,s$, when a current of 2A is passed through it, then the resistance of the resistor is 10 ohm.

21. Correct option: (C)

Solution: We know, if an electric current i passes through a conductor of resistance R for a duration of t sec. Then the heat energy or thermal energy produced in the conductor is $H = i^2Rt$.

Given, a current of 4 A flows through a resistance of 10 ohm.

Thus, the thermal energy developed in 10 sec is:

$$H = i^2Rt$$

$$H = 4^2(10)(10)$$

$$H = 1600\,J$$

Therefore, if a current of 4 A flows through a resistance of 10 ohm, then the thermal energy developed in 10 sec is 1600 J.

22. Correct option: (D)

Solution: We know, the power P or rate of energy transfer, in a resistor R across which a potential difference V is maintained is

$$P = i^2R = \frac{V^2}{R}$$

We also know, the maximum power is transferred to a load by the battery when the internal resistance of the battery is equal to the resistance of the load.

Given, a battery of emf 2V and internal resistance is 0. 5 ohm, and the maximum power transferred to the load is 0. 05 W.

Since the maximum power is transferred to the load, its resistance should be equal to that of the internal resistance of the battery.

Hence, the current passing through the battery is:

$P = i^2 R$

$0.05 = i^2 (0.5)$

$i = \sqrt{10}$

$i = 3.16\,A$

Therefore, if a battery of emf 2V and internal resistance is 0. 5 ohm, and the maximum power transferred to the load is 0. 05 W, then the current passing through the battery is 3 A approximately.

23. Correct match: a) ->P), b) ->P), c) -> R), S),d) -> Q)

Solution: We know, electric charge and electric current are scalar quantities.

Also, electric current is defined as charge flowing per unit time, $i = \frac{Q}{t}$.

We also know, the current density when the current is uniform across the surface and parallel to area vector, is defined as:

$J = \frac{i}{A}$

where A is the total area of the surface.

Consequently, the unit of current density is Ampere/m^2

Also, current density is a vector.

Therefore, correct match is: a) ->P), b) ->P), c) -> R), S),d) -> Q)

24. Correct match: a) ->P),Q), b) ->P),Q), c) -> R),d) -> S)

Solution: We know, the power P or rate of energy transfer, in a resistor R across which a potential difference V is maintained is

$P = i^2 R = \frac{V^2}{R}$

We know, the terminal-to-terminal potential difference V across the discharging battery is related to its emf E by the equation:

$V = E - ir$

where i is the current through the battery and r is its internal resistance.

Further, the terminal-to-terminal potential difference V across the charging battery is related to its emf E by the equation:

$V = E + ir$

where i is the current through the battery and r is its internal resistance.

Therefore, correct match is: a) ->P),Q), b) ->P),Q), c) -> R),d) -> S)

Resistance and Resistivity

LEVEL-I

1. Correct option: (B)

Solution: We know, the resistance R of a conducting wire of length L and uniform cross section is

$R = \rho \frac{L}{A}$

where A is the cross-sectional area.

Given, a piece of aluminium wire of finite length is drawn through a series of dies to reduce its diameter to half its original value.

Therefore, the new resistance of the wire is:

$$R' = \rho \frac{L}{(\pi(d/(2(2))^2))}$$

$$R' = \rho \frac{4}{(\pi(d/2)^2)}$$

$$R' = 4\rho \frac{L}{A}$$

$$R' = 4R$$

Therefore, if a piece of aluminium wire of finite length is drawn through a series of dies to reduce its diameter to half its original value, its resistance will become four times.

2. Correct option: (A)

Solution: We know, the resistance R of a conducting wire of length L and uniform cross section is

$$R = \rho \frac{L}{A}$$

where A is the cross-sectional area.

Given, a wire of resistance R is elongated n – fold to make a new uniform wire.

Therefore, the new resistance of the wire is:

$$R' = \rho \frac{nL}{A}$$

$$R' = nR$$

Therefore, a wire of resistance R is elongated n – fold to make a new uniform wire, then the new resistance of the wire is nR.

3. Correct option: (B)

Solution: We know, for a material obeying Ohm's law:

$$E = \rho J$$

where ρ is the resistivity of the material and J is the current density.

Since specific resistance is also called as resistivity, therefore electric intensity, current density and specific resistance are related to each other by the relation:

$E = kJ$.

4. Correct option: (B)

Solution: We know, Ohm's law states that the current through a device is always directly proportional to the potential difference applied to the device. Mathematically,

$$V = iR$$

where i current flows through the device when V voltage is applied across the device, R is its resistance. A given device obeys Ohm's law if its resistance is independent of the applied potential difference V .

Given, an electric wire is connected across a cell of e. m. f. E . The current i is measured by an ammeter of resistance R . Therefore, according to Ohm's law: $E = iR$

5. Correct option: (A)

Solution: We know, the resistance R of a wire a temperature T is related to the resistance at some reference temperature T_0 by:

$$R = R_0(1+\alpha(T-T_0))$$

where α is the coefficient of expansion of resistance.

Therefore, the resistance of a wire a temperature $t^\circ C$ and $0^\circ C$ are related by:

$$R = R_0(1+\alpha(t-0))$$

$$R = R_0(1+\alpha t)$$

6. Correct option: (B)

Solution: We know, the resistance R of a conducting wire of length L and uniform cross section is

$$R = \rho\frac{L}{A}$$

where A is the cross-sectional area.

Given, an electric wire of length L and area of cross section A has a resistance R ohm.

Therefore, another wire of the same material having same length and area of cross – section $4A$ has a resistance of:

$$R' = \rho\frac{L}{4A}$$

$$R' = \frac{R}{4}$$

Therefore, if an electric wire of length L and area of cross section A has a resistance R ohm, then another wire of the same material having same length and area of cross – section $4A$ has a resistance of $\frac{R}{4}$.

7. Correct option: (D)

Solution: We know, the resistance R of a conducting wire of length L and uniform cross section is $R = \rho\frac{L}{A}$

where A is the cross-sectional area.

Given, the resistance of a wire is 10 ohms, Its length is increased by 10% by stretching.

Therefore, the new length of the wire is:

$$L' = L + L\times\frac{10}{100}$$

$$L' = L + 0.1L$$

$$L' = 1.1L$$

Therefore, the new resistance of the wire is:

$$R' = \rho \frac{1.1L}{A}$$

$$R' = 1.1R$$

$$R' = 1.1(10)$$

$$R' = 11\Omega$$

8. Correct option: (A)

Solution: We know, the resistance R is related to conductance K by:

$$K = \frac{1}{R}$$

Therefore, the reciprocal of resistance is conductance.

9. Correct option: (D)

Solution: We know, the resistance R of a conducting wire of length L and uniform cross section is

$$R = \rho \frac{L}{A}$$

where A is the cross-sectional area.

The resistivity of a wire depends on its material. It does not depend on length, shape and area of cross-section of the material.

10. Correct option: (A)

Solution: We know, the resistance R is related to conductance K by:

$$K = \frac{1}{R}$$

Now, a superconductor is defined to have zero resistance.

Therefore, the conductivity of a superconductor is theoretically infinite.

11. Correct option: (B)

Solution: We know, the resistance R of a conducting wire of length L and uniform cross section is

$$R = \rho \frac{L}{A}$$

where A is the cross-sectional area.

Given, a uniform wire of resistance R is uniformly compressed along is length, until its radius becomes n times the original radius.

Therefore, the new resistance of the wire is:

$$R' = \rho \frac{L}{(\pi (nr)^2))}$$

$$R' = \rho \frac{4}{n^2 (\pi r^2)}$$

$$R' = \frac{R}{n^2}$$

Therefore, a uniform wire of resistance R is uniformly compressed along is length, until its radius becomes n times the original radius, then new resistance of the wire is $\frac{R}{n^2}$.

12. Correct option: (D)

Solution: We know, the resistance R of a conducting wire of length L and uniform cross section is

$$R = \rho \frac{L}{A}$$

where A is the cross-sectional area.

We know, the resistance R of a wire a temperature T is related to the resistance at some reference temperature T_0 by:

$$R = R_0(1 + \alpha(T - T_0))$$

where α is the coefficient of expansion of resistance.

Therefore, the resistance of a conductor increases with increase in length, decrease in cross – sectional area and increase in temperature.

13. Correct option: (A)

Solution: We know, the resistance R of a conducting wire of length L and uniform cross section is

$$R = \rho \frac{L}{A}$$

where A is the cross-sectional area.

Given, the length of a given cylindrical wire is increased by 100%.

Since the volume of the cylinder should remain unchanged, therefore the new radius if given by:

$$\pi r^2 h = \pi r'^2 (2h)$$

$$r' = \frac{r}{\sqrt{2}}$$

Therefore, the new resistance of the wire is:

$$R' = \rho \frac{2L}{\pi r'^2}$$

$$R' = \rho \frac{4L}{\pi r^2}$$

$$R' = 4R$$

Therefore, the increase in percentage of the resistance of the wire is:

$$\frac{R' - R}{R} \times 100 = 300$$

Therefore, if the length of a given cylindrical wire is increased by 100%, then the change in the resistance of the wire will be 300 %.

Grouping Of Resistances

14. Correct option: (D)

Solution: We know, when two resistances are connected in parallel, the equivalent resistance is always less than the each of the two resistances of the combination.

Given, two resistances of resistance R_1 and R_2 having $R_1 > R_2$ are connected in parallel.

Therefore, for equivalent resistance R, we have:

$R < R_1$

15. Correct option: (B)

Solution: When two resistors R_1 and R_2 are connected in series, the equivalent resistance is the sum of the individual resistance:

$R_{eq} = R_1 + R_2$

Also, when two resistors R_1 and R_2 are connected in parallel, the equivalent resistance is given by:

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

Using these relations, we can say, when n resistors are connected in series, the net resistance is:

$R_{eq} = nR$

And when n resistors are connected in parallel, the net resistance is:

$$R_{eq} = \frac{R}{n}$$

Given, there are n similar conductors each of resistance R and the resultant resistance comes out to be x when connected in parallel. Therefore:

$$x = \frac{R}{n}$$

$R = nx$

Now, if they are connected in series, the resistance comes out to be:

$R_{eq} = nR$

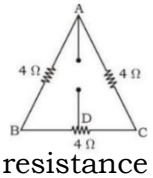
$R_{eq} = n^2 x$

16. Correct option: (C)

Solution: We know, the resistance R of a conducting wire of length L and uniform cross section is

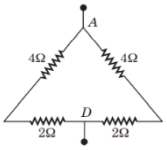
$$R = \rho \frac{L}{A}$$

where A is the cross-sectional area.



Also, when two resistors R_1 and R_2 are connected in parallel, the equivalent resistance is given by: $R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$

Given, three resistance of 4 ohm each are connected as shown in figure. The point D divides the resistance into two equal halves.



Since the resistance is directly proportional to the length of the wire, the equivalent circuit will look like as shown on right.

We see, with reference to AD, 2 ohm resistor is in series with 4 ohm, and the whole combination

is in parallel with identical combination.

The resistance between point A and D will be therefore:

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

$$R_{eq} = \frac{(4+2)(4+2)}{(4+2)+(4+2)}$$

$$R_{eq} = \frac{6}{2}$$

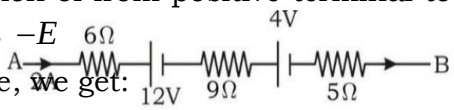
$$R_{eq} = 3\Omega$$

Therefore, if three resistances of 4 ohm each are connected as shown in figure and the point D divides the resistance into two equal halves, then the resistance between point A and D will be 3 ohm.

17. Correct option: (C)

Solution: We know, for a move through a resistance in the direction of the current, the change in potential is $-iR$; in the opposite direction it is iR.

Also, for a move through an ideal emf device in the direction of from positive terminal to. The change in potential is E; in the opposite direction it is $-E$



Applying the above mentioned loop rules in the given figure, we get:

$$V_A = 2(6) - 12 - 9(2) - 4 - 5(2) - V_B$$

$$V_B - V_A = 32$$

18. Correct option: (D)

Solution: When two resistors R_1 and R_2 are connected in series, the equivalent resistance is the sum of the individual resistance:

$$R_{eq} = R_1 + R_2$$

Therefore, the equivalent resistance of resistors connected in series is always equal to sum of component resistors.

19. Correct option: (C)

Solution: When two resistors R_1 and R_2 are connected in series, the equivalent resistance is the sum of the individual resistance:

$$R_{eq} = R_1 + R_2$$

Also, when two resistors R_1 and R_2 are connected in parallel, the equivalent resistance is given by:

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

Hence, when n resistors are connected in series, the net resistance is:

$$R_{eq} = nR$$

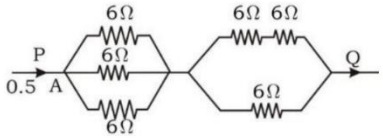
And when n resistors are connected in parallel, the net resistance is:

$$R_{eq} = \frac{R}{n}$$

Given, resistance of 6 ohm each are connected in the manner shown in adjoining figure.

Now, the net resitance of first three resistors connected in parallel is:

$$R_1 = \frac{6}{3}$$
$$R_1 = 2\Omega$$



Next, we have two 6 ohms resistors connected in series, which then is connected in parallel to another 6 ohms resistor.

The combination of two 6 ohms resistors connected in series is $6+6=12\Omega$, hence the net resistance of the second loop of the circuit, using $R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$ is:

$$R_2 = \frac{12(6)}{12+6}$$
$$R_2 = \frac{72}{18}$$
$$R_2 = 4\Omega$$

Finally, we have two resistors of resistance $R_1 = 2\Omega$ and $R_2 = 4\Omega$ connected in series. The net resistance of the whole circuit is therefore:

$$R_{eq} = R_1 + R_2$$
$$R_{eq} = 2+4$$
$$R_{eq} = 6\Omega$$

Since the current flowing through the circuit is 0. 5 A, therefore the potential difference is:

$$V_P - V_Q = iR_{eq}$$
$$V_P - V_Q = 6(0.5)$$
$$V_P - V_Q = 3V$$

Therefore, if resistance of 6 ohm each are connected in the manner shown in adjoining figure, with the current 0. 5 ampere, the potential difference $V_P - V_Q$ is 3V.

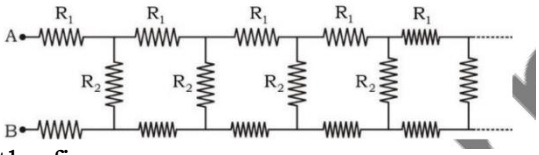
20. Correct option: (C)

Solution: When two resistors R_1 and R_2 are connected in series, the equivalent resistance is the sum of the individual resistance:

$$R_{eq} = R_1 + R_2$$

Also, when two resistors R_1 and R_2 are connected in parallel, the equivalent resistance is given by:

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

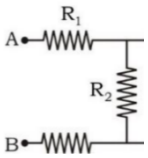


Given, an infinite sequence of resistance is shown in the figure

where $R_1 = 1\Omega, R_2 = 2\Omega$.

Let the net resistance of infinite sequence of resistance be R.

Since adding or subtracting something from infinity is still infinity, the above circuit can be simplified as consisting of the combination shown in right, with the whole sequence of infinite resistance connected in parallel to R_2 resistance.



Sine this simplified combination will also have the same resistance R, we have:

$$R = R_1 + \frac{RR_2}{R + R_2}$$

$$R = 1 + \frac{R(2)}{R + 2}$$

$$R = \left(\frac{3R + 2}{R + 2}\right)$$

$$R^2 + 2R = 3R + 2$$

$$R^2 - R - 2 = 0$$

Using the quadratic root formula, the resistance is:

$$R = \frac{1 \pm \sqrt{1 - 4(1)(-2)}}{2}$$

$$R = \frac{1 \pm \sqrt{9}}{2}$$

$$R = \frac{1 \pm 3}{2}$$

$$R = 2, -1$$

Since resistance is a positive quantity, therefore

$$R = 2\Omega$$

Therefore, if an infinite sequence of resistance is shown in the figure where $R_1 = 1\Omega, R_2 = 2\Omega$, then the resultant resistance between A and B will be 2 ohm.

21. Correct option: (C)

Solution: When two resistors R_1 and R_2 are connected in series, the equivalent resistance is the sum of the individual resistance:

$$R_{eq} = R_1 + R_2$$

Also, when two resistors R_1 and R_2 are connected in parallel, the equivalent resistance is given by:

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

Using these relations, se see, when n resistors are connected in series, the net resistance is:

$$R_{eq} = nR$$

And when n resistors are connected in parallel, the net resistance is:

$$R_{eq} = \frac{R}{n}$$

Given, n equal resistors are first connected in series and then connected in parallel.

Therefore, the ratio of the maximum to the minimum resistance is:

$$\frac{nR}{R/n} = n^2$$

22. Correct option: (D)

Solution: When two resistors R_1 and R_2 are connected in series, the equivalent resistance is the sum of the individual resistance:

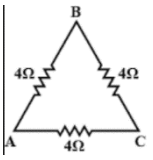
$$R_{eq} = R_1 + R_2$$

Also, when two resistors R_1 and R_2 are connected in parallel, the equivalent resistance is given by:

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

Given, three resistances 4 ohm each are connected in the form of an equilateral triangle.

We are interested in effective resistance between two corners, say AB.



We see, with reference to AB, two 4 ohms resistance are connected in series and then this combination is in parallel to another 4 ohm resistor.

Therefore, the effective resistance between two corners is:

$$R_{eq} = \frac{4(4+4)}{4+(4+4)}$$

$$R_{eq} = \frac{32}{12}$$

$$R_{eq} = \frac{8}{3}$$

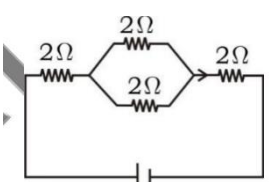
Therefore, if three resistances 4 ohm each are connected in the form of an equilateral triangle, then the effective resistance between two corners $\frac{8}{3}\Omega$.

23. Correct option: (C)

Solution: When two resistors R_1 and R_2 are connected in series, the equivalent resistance is the sum of the individual resistance:

$$R_{eq} = R_1 + R_2$$

Also, when two resistors R_1 and R_2 are connected in parallel, the equivalent resistance is given by:



$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

Also, when n resistors are connected in parallel, the net resistance is:

$$R_{eq} = \frac{R}{n}$$

We want to calculate the net resistance in the given circuit in the figure on right.

Now, the net resitance of two resistors connected in parallel in the middle is:

$$R_2 = \frac{2}{2}$$

$$R_2 = 1\Omega$$

Finally, we have three resistors of resistance $R_1 = 2\Omega, R_2 = 1\Omega, R_3 = 2\Omega$ connected in series. The net resistance of the whole circuit is therefore:

$$R_{eq} = R_1 + R_2 + R_3$$

$$R_{eq} = 2 + 1 + 2$$

$$R_{eq} = 5\Omega$$

Therefore, the equivalent resistance of the circuit shown in the figure is 5 ohms.

24. Correct option: (D)

Solution: When two resistors R_1 and R_2 are connected in series, the equivalent resistance is the sum of the individual resistance:

$$R_{eq} = R_1 + R_2$$

Also, when two resistors R_1 and R_2 are connected in parallel, the equivalent resistance is given by:

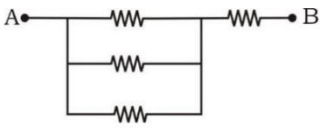
$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

Also, when n resistors are connected in parallel, the net resistance is:

$$R_{eq} = \frac{R}{n}$$

We want to calculate the net resistance in the given circuit in the figure on right.

Given, all the resistors shown have the value 2 ohm each.



Now, the net resitance of the first three resistors connected in parallel is:

$$R_1 = \frac{2}{3}\Omega$$

Finally, we have two resistors of resistance $R_1 = \frac{2}{3}\Omega, R_2 = 2\Omega$ connected in series. The net resistance of the whole circuit is therefore:

$$R_{eq} = R_1 + R_2$$

$$R_{eq} = \frac{2}{3} + 2$$

$R_{eq} = \frac{8}{3}\Omega$

$R_{eq} = 2\frac{2}{3}\Omega$

Therefore, the equivalent resistance of the circuit shown in the figure is $2\frac{2}{3}\Omega$.

25. Correct option: (B)

Solution: When two resistors R_1 and R_2 are connected in series, the equivalent resistance is the sum of the individual resistance:

$R_{eq} = R_1 + R_2$

We also know, the resistance R of a conducting wire of length L and uniform cross section is

$R = \rho\frac{L}{A}$

where A is the cross-sectional area.

Given, two wires of the same dimensions but resistivities ρ_1 and ρ_2 are connected in series.

Let the equivalent resistivity of the combination is ρ.

Now, since the combination has twice the length of the wire but the same area of cross-section, the net resistance of the circuit is:

$R_{eq} = R_1 + R_2$

$\rho\frac{2L}{A} = \rho_1\frac{L}{A} + \rho_2\frac{L}{A}$

$\rho = \frac{(\rho_1 + \rho_2)}{2}$

Therefore, if two wires of the same dimensions but resistivities ρ_1 and ρ_2 are connected in series, then equivalent resistivity of the combination is $\frac{(\rho_1 + \rho_2)}{2}$.

26. Correct option: (A)

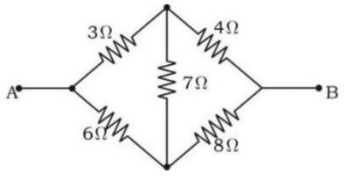
Solution: When two resistors R_1 and R_2 are connected in series, the equivalent resistance is the sum of the individual resistance:

$R_{eq} = R_1 + R_2$

Also, when two resistors R_1 and R_2 are connected in parallel, the equivalent resistance is given by:

$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$

Given, resistance of 6 ohm each are connected in the manner shown in adjoining figure.



Now, we see the given cicruit is like a wheatstone's bridge which is balanced:

$$\frac{P}{Q}=\frac{R}{S}$$

$$\frac{3}{4}=\frac{6}{8}$$

Therefore, no current flows through the 7 ohm resistor, and it can be dropped from the circuit.

Finally, we have two resistors of resistance 3 and 4 ohms, connected in parallel to another series set of resistors of resistances 6 and 8 ohm. The net resistance of the whole circuit is therefore:

$$R_{eq}=\frac{(3+4)(6+8)}{3+4+6+8}$$

$$R_{eq}=\frac{7(14)}{21}$$

$$R_{eq}=\frac{14}{3}$$

Therefore, if resistance of 6 ohm each are connected in the manner shown in adjoining figure, then equivalent resistance between A and B will be 14/3 ohm.

27. Correct option: (B)

Solution: When two resistors R_1 and R_2 are connected in series, the equivalent resistance is the sum of the individual resistance:

$$R_{eq}=R_1+R_2$$

Also, when two resistors R_1 and R_2 are connected in parallel, the equivalent resistance is given by:

$$\frac{1}{R_{eq}}=\frac{1}{R_1}+\frac{1}{R_2}$$

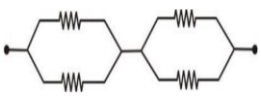
Using these relations, se see, when n resistors are connected in series, the net resistance is:

$$R_{eq}=nR$$

And when n resistors are connected in parallel, the net resistance is:

$$R_{eq}=\frac{R}{n}$$

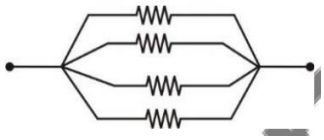
We want to know which arrangement of four identical resistance should be used to maximum energy from a cell of voltage V from the given option.

a) 

Here we have two parallel resistors connected in series with another identical combination of two parallel resistors. The equivalent resistance is therefore:

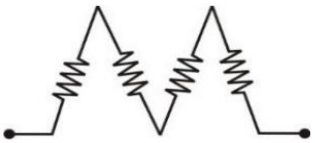
$$R_{eq} = \frac{R}{2} + \frac{R}{2}$$

$$R_{eq} = R$$

b) 

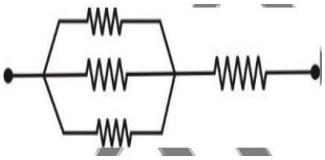
Here we have four identical resitors connected in parallel. The equivalent resitance is therefore:

$$R_{eq} = \frac{R}{4}$$

c) 

Here, we have four identical resitors connected in series. The equivalent resitance is therefore:

$$R_{eq} = 4R$$

d) 

Here we have three identical resitors connected in parallel which is then again connected to another similar resitance in series. The net resitance is therefore:

$$R_{eq} = \frac{R}{3} + R$$

$$R_{eq} = \frac{4R}{3}$$

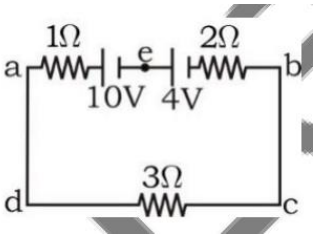
Now, we know, the power P or rate of energy transfer, in a resistor R across which a potential difference V is maintained is

$$P = \frac{V^2}{R}$$

Therefore, for maximum power to be transferred, the resistance should be minimum.

Therefore, the configuration in B) gives the maximum energy transfer from the cell.

28. Correct option: (D)



Solution: We know, for a move through a resistance in the direction of the current, the change in potential is $-iR$; in the opposite direction it is iR.

Also, for a move through an ideal emf device in the direction of from positive terminal to. The change in potential is E; in the opposite direction it is $-E$

We know, the algebraic sum of the changes in potential encountered in a complete traversal of any loop of a circuit must be zero.

Let us initially assume the current is in the direction from b to a.

Since the algebraic sum of the changes in potential encountered in a complete traversal of any loop of a circuit must be zero, thus:

$-2i+4-10-i-3i=0$

$-6i-6=0$

$i=-1A$

Since we get the current negative, this means the current originally flows in the direction opposite to what we assumed earlier. Therefore, the magnitude and direction of the current in the circuit shown will be 1 A from a to b through e.

inco

29. Correct option: (B)

Solution: We know, the terminal-to-terminal potential difference V across the discharging battery is related to its emf E by the equation:

$V=E-ir$

where i is the current through the battery and r is its internal resistance.

Given, a cell of e. m. f. E is connected with an external resistance R then potential difference across cell is V.

Now using Ohm's law, the voltage across the resistance is $V=iR$, which is also equal to the potential difference across the cell, thus:

$$iR=E-ir$$

$$i=\frac{E}{R+r}$$

Substituting this for current in $V=E-ir$, we get:

$$V=E-\frac{E}{R+r}r$$

$$\frac{E}{R+r}r=E-V$$

$$Er=(E-V)(R+r)$$

$$Er=r(E-V)+R(E-V)$$

$$r(E-E+V)=R(E-V)$$

$$r=\frac{R(E-V)}{V}$$

Therefore, if a cell of e. m. f. E is connected with an external resistance R then potential difference across cell is V, the internal resistance of cell will be $\frac{R(E-V)}{V}$.

30. Correct option: (A)

Solution: We know, according to the Kirchhoff's law of meshes, the sum of the currents entering any junction must be equal to the sum of the currents leaving that junction.

Since the Kirchhoff's law of meshes is effectively the statement that charge cannot pile up in a junction, hence the Kirchhoff's law of meshes is in accordance with law of conservation of charge.

31. Correct option: (A)

Solution: We know, the algebraic sum of the changes in potential encountered in a complete traversal of any loop of a circuit must be zero.

Therefore, according to the Kirchhoff's law the algebraic sum of the products of current and resistance as well as e. m. f in a closed loop is zero.

32. Correct option: (B)

Solution: Wheatstone bridge is used to measure the value of unknown resistance.

Whetstone's bridge cannot be used for measurement of very low resistance. This is because measuring very low resistance can give wrong results due to resistance offered by the wires, the devices like galvanometer etc.

Hence, Whetstone's bridge cannot be used for measurement of very low resistance.

33. Correct option: (A)

Solution: We know, the maximum power is transferred to a load by the battery when the internal resistance of the battery is equal to the resistance of the load.

Also, when two resistors R_1 and R_2 are connected in parallel, the equivalent resistance is given by:

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

Given, two cells, each of e. m. f. E and internal resistance r are connected in parallel between the resistance R.

Since the two cells are connected in parallel with the resistance, as looked from the end points of the load, the two internal resistances of the cell are in parallel. Their equivalent resistance is:

$$R_{eq} = \frac{rr}{r + r}$$

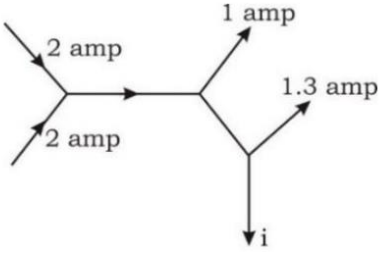
$$R_{eq} = \frac{r}{2}$$

Therefore, the maximum energy given to the resistor will be, only when $R = \frac{r}{2}$.

34. Correct option: (A)

Solution: We know, according to junction rule, the sum of the currents entering any junction must be equal to the sum of the currents leaving that junction.

Given, the figure below shows currents in a part of electric circuit.



Applying the junction rule to the first junction, the current flowing to the right is:

$2 + 2 = 4\,A$

Since 1 A current flows away from the next junction, the current flowing down that junction is:

$4-1=3A$

In the last junction, 1. 3 A current flows away, hence the current i flowing downward is:

$i=3-1.3$

$i=1.7A$

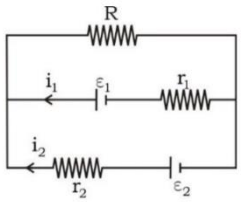
Therefore, the current i in the figure is 1. 7 A.

35. Correct option: (A)

Solution: We know, for a move through a resistance in the direction of the current, the change in potential is $-iR$; in the opposite direction it is iR.

Also, for a move through an ideal emf device in the direction of from positive terminal to negative terminal, the change in potential is E; in the opposite direction it is $-E$

We know, according to the junction rule, algebraic sum of the changes in potential encountered in a complete traversal of any loop of a circuit must be zero.



Given, electrical circuit shown in the figure on right.

Applying the junction rule to the outer loop, traversing it clockwise, we get:

$-i_2r_2-(i_2+i_1)R+\varepsilon_2=0$

Now, applying the junction rule to the inner lower loop, traversing it clockwise, we get:

$-i_2r_2-\varepsilon_1+i_1r_1+\varepsilon_2=0$

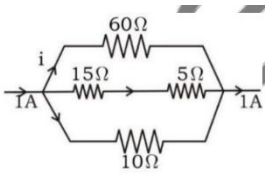
Now, applying the junction rule to the inner upper loop, traversing it clockwise, we get:

$\varepsilon_1-(i_2+i_1)R-i_1r_1=0$

Therefore, if electrical circuit shown in the figure on right, then the correct equation for it is $\varepsilon_1-(i_2+i_1)R-i_1r_1=0$.

36. Correct option: (D)

Solution: We know, according to the junction rule, algebraic sum of the changes in potential encountered in a complete traversal of any loop of a circuit must be zero.



Given, an electric circuit as shown in the figure on right.

Let a current i flows through 60 ohm resistor, a current i_1 through 15 ohm resistor.

Then from the junction rule, the current flowing through the 10 ohm resistor is:

$1-i-i_1$

Since the voltage in parallel connection remain the same, we have:

$60i=(15+5)i_1$

$60i=10(1-i-i_1)$

From the first equation, we get:

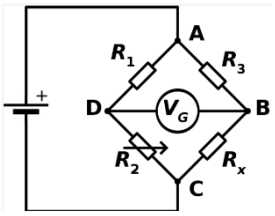
$3i=i_1$

Substituting this in the second equation, we get:

$60i = 10(1 - i - 3i)$
$60i = 10(1 - 4i)$
$100i = 10$
$i = 10A$

Therefore, the magnitude of the current i is 10 A.

37. Correct option: (C)



Solution: We know, for a balanced Wheatstone's bridge:

$\frac{R_1}{R_x} = \frac{R_3}{R_2}$

where R_x is the unknown resistance.

Given, in a balanced Wheatstone's network, the resistances in the arms Q and S are interchanged.

In terms of arms Q and S, the balanced condition can be written as:

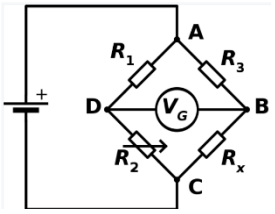
$\frac{P}{S} = \frac{R}{Q}$

Now, if the arms Q and S are interchanged, the above ration changes to:

$\frac{P}{Q} = \frac{R}{S}$

Since this is not the same ratio as earlier, the network is not balanced now.

38. Correct option: (B)



Solution: We know, for a balanced Wheatstone's bridge:

$\frac{R_1}{R_x} = \frac{R_3}{R_2}$

where R_x is the unknown resistance.

Given, galvanometer and battery are interchanged in balanced Wheatstone bridge.

Now, when galvanometer and battery are interchanged, the new configuration is still a Wheatstone bridge. Since the value of resistances are the same, the bridge still balances.

39. Correct option: (D)

Solution: We know, for a balanced Wheatstone's bridge:

$\frac{R_1}{R_x} = \frac{R_3}{R_2}$

where R_x is the unknown resistance.

Hence, if we know the value of two resistors, we can find compare the value of two unknown resistors.

The working principle of meter bridge is also based on the Wheatstone's bridge.

Also, Wheatstone's bridge can measure small strains produced in hard metals.

LEVEL-II

1. Correct option: (C)

Solution: We know, the resistance R of a conducting wire of length L and uniform cross section is

$$R = \rho \frac{L}{A}$$

where A is the cross-sectional area.

Given, two wires made of same material have lengths in the ratio 1:2 and their volumes in the same ratio.

Since we can write for the area of the wire $A = \frac{V}{L}$ where V is the volume of the wire, the formula for resistance becomes:

$$R = \rho \frac{L^2}{V}$$

Hence, the ratio of their resistances is:

$$\frac{R_1}{R_2} = \frac{\rho \frac{L_1^2}{V_1}}{\rho \frac{L_2^2}{V_2}}$$
$$\frac{R_1}{R_2} = \frac{L_1^2 V_2}{V_1 L_2^2}$$
$$\frac{R_1}{R_2} = \frac{2}{2^2}$$
$$\frac{R_1}{R_2} = \frac{1}{2}$$

Therefore, if two wires made of same material have lengths in the ratio 1:2 and their volumes in the same ratio, then the ratio of their resistances is 1:2.

2. Correct option: (D)

Solution: We know, the resistance R of a conducting wire of length L and uniform cross section is $R = \rho \frac{L}{A}$

where A is the cross-sectional area.

Given, the resistance of a wire is 10 ohm. Another wire is made of same material, whose length is twice and the radius is half.

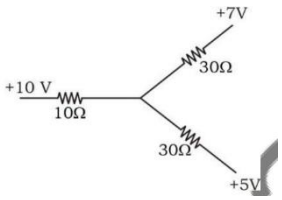
Therefore, the resistance of another wire is:

$$R' = \rho \frac{L'}{A'}$$
$$R' = \rho \frac{2L}{\pi(r/2)^2}$$
$$R' = \rho \frac{4L}{\pi r^2}$$
$$R' = \rho \frac{4L}{A}$$
$$R' = 4R$$
$$R' = 4(10)$$
$$R' = 40\Omega$$

Therefore, if the resistance of a wire is 10 ohm and another wire is made of same material, whose length is twice and the radius is half, then its resistance is 40 ohms.

3. Correct option: (B)

Solution: We know, for a move through a resistance in the direction of the current, the change in potential is $-iR$; in the opposite direction it is iR.



We also know, according to the Kirchhoff's law of meshes, the sum of the currents entering any junction must be equal to the sum of the currents leaving that junction.

Given, an electric circuit as shown in the right.

Let the current through the 10 ohm resistor be i and current through upward 30 ohm resistor be i_1.

Then in accordance to the Kirchhoff's law of meshes, the current through the downward 30 ohm resistor is:

$i - i_1$

Now, accounting for all the changes in the potential in the top branch of the circuit we get:

$10 - 10i - 30i_1 - 7 = 0$

$3 - 10i - 30i_1 = 0$

Similarly, accounting for all the changes in the potential in the lower branch of the circuit we get:

$10 - 10i - 30(i - i_1) - 5 = 0$

$5 - 40i + 30i_1 = 0$

Adding these two equations, we get:

$8 - 50i = 0$

$i = \frac{8}{50}$

$i = 0.16\,A$

Therefore, the current through the 10 ohm resistor is 0. 172 A approximately.

4. Correct option: (C)

Solution: We know, the resistance R of a conducting wire of length L and uniform cross section is $R = \rho\frac{L}{A}$

where A is the cross-sectional area.

Given, an aluminium wire of resistivity $\rho = 2.2\times10^{-8}\Omega m$ and diameter 1. 4 mm is used to make a 4 ohm resistor.

The length of the wire is therefore:

$R = \rho\frac{L}{A}$

$L = \frac{RA}{\rho}$

$L = \frac{4(\pi(1.4/2\times10^{-3})^2)}{2.2\times10^{-8}}$

$L = 280\,m$

Hence, if an aluminium wire of resistivity $\rho = 2.2 \times 10^{-8} \Omega m$ and diameter 1. 4 mm is used to make a 4 ohm resistor, then its length is 280 m.

5. Correct option: (Incomplete figure(A)

Solution: Points P and Q are not properly labelled in the figure. Since the resistance of the network depends on the exact location of the terminal points, we cannot solve this question unless proper location of points P and Q are made clear.

6. Correct option: (A)

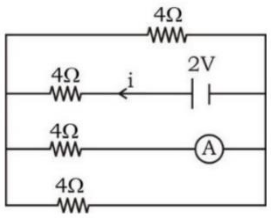
Solution: When two resistors R_1 and R_2 are connected in series, the equivalent resistance is the sum of the individual resistance:

$$R_{eq} = R_1 + R_2$$

Also, when two resistors R_1 and R_2 are connected in parallel, the equivalent resistance is given by:

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

Given, an electric circuit as shown in the figure on the right.



The three 4 ohms resistance are in parallel, therefore the net resistance is:

$$R_{eq} = 4/3 + 4$$

$$R_{eq} = \frac{16}{3}$$

Therefore, current coming from the battery is:

$$i = V / R_{eq}$$

$$i = 2 / (16/3)$$

$$i = \frac{3}{8}$$

Since equal current flows from each 4 ohm resistor from the current division rule, the ammeter reading is therefore:

$$i_a = i / 3$$

$$i_a = 1/8\, A$$

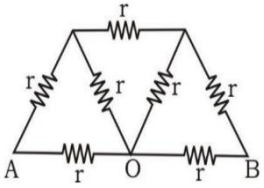
7. Correct option: (B)

Solution: When two resistors R_1 and R_2 are connected in series, the equivalent resistance is the sum of the individual resistance:

$$R_{eq} = R_1 + R_2$$

Also, when two resistors R_1 and R_2 are connected in parallel, the equivalent resistance is given by:

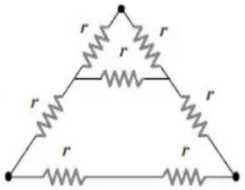
$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$



Given, an electric circuit as shown in the figure on the right.

Now, here from symmetry the potential at the point O is zero.

Hence, the given circuit can be simplified to as shown in the figure below:



The equivalent resistance of the topmost triangular part which consists of 2 r ohm resistors connected in parallel to one r ohm resistor is:

$$R_1 = \frac{(2r)(r)}{2r + r}$$

$$R_1 = \frac{2r}{3}$$

Next, we have this R_1 resistance connected in series with 2 r ohms resistors an all these combination connected in parallel to another series combination of two r ohm resistors. The net resistance is thus:

$$R_1 = \frac{\left(\frac{2r}{3} + 2r\right)(2r)}{\frac{2r}{3} + 2r + 2r}$$

$$R_1 = \frac{\left(\frac{8r}{3}\right)(2r)}{\frac{14r}{3}}$$

$$R_1 = \frac{8r}{7}$$

Therefore, equivalent resistance across A and B in the given circuit is 8r/7.

8. Correct option: (D)

Solution: When two resistors R_1 and R_2 are connected in series, the equivalent resistance is the sum of the individual resistance:

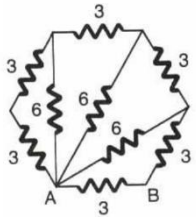
$$R_{eq} = R_1 + R_2$$

Also, when two resistors R_1 and R_2 are connected in parallel, the equivalent resistance is given by:

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

Given, an electric as shown in the figure on the right:

Now, at the leftmost of the figure, we see two 3 ohm resistor in series is connected to another 6 ohm resistor in parallel. Their net resistance is therefore:



$$R_1 = \frac{6(6)}{6 + 6}$$

$$R_1 = 3$$

Next, we again have two 3 ohm resistor in series is connected to another 6 ohm resistor in parallel. Their net resistance is therefore:

$$R_2 = \frac{6(6)}{6+6}$$
$$R_2 = 3$$

Next, we again have two 3 ohm resistor in series is connected to another 6 ohm resistor in parallel. Their net resistance is therefore:

$$R_3 = \frac{6(6)}{6+6}$$
$$R_3 = 3$$

Finally we have two 3 ohm resistor in series is connected to a 3 ohm resistor in parallel. The net resistance is therefore:

$$R_{eq} = \frac{6(3)}{6+3}$$
$$R_{eq} = 2$$

Therefore, the effective resistance between points A and B in the figure is 2 ohm.

9. Correct option: (B)

Solution: When two resistors R_1 and R_2 are connected in series, the equivalent resistance is the sum of the individual resistance:

$$R_{eq} = R_1 + R_2$$

We also know, the resistance R of a conducting wire of length L and uniform cross section is

$$R = \rho\frac{L}{A}$$

where A is the cross-sectional area.

Given, n conducting wires of same dimensions but having resistivities 1,2,3,4 … N are connected in series.

Let the equivalent resistivity of the combination be ρ.

Now, since all the wire have same dimensions, the net resistance of the circuit is:

$$R_{eq} = R_1 + R_2 + ... + R_n$$
$$\rho\frac{L}{A} = \rho_1\frac{L}{A} + \rho_2\frac{L}{A} + ... + \rho_n\frac{L}{A}$$
$$\rho\frac{L}{A} = \frac{L}{A} + 2\frac{L}{A} + ... + n\frac{L}{A}$$
$$\rho = 1 + 2 + ... + n$$

Using the summation formula for the sum of first n natural number, $\frac{n(n+1)}{2}$, we get:

$$\rho = \frac{n(n+1)}{2}$$

Therefore n conducting wires of same dimensions but having resistivities 1,2,3,4 … N are connected in series, then equivalent resistivity of the combination is $\frac{n(n+1)}{2}$.

10. Correct options: A), B), C), D)

Solution: We know, the resistance R of a conducting wire of length L and uniform cross section is $R = \rho\frac{L}{A}$

where A is the cross-sectional area.

We also know, the resistance R of a wire a temperature T is related to the resistance at some reference temperature T_0 by:

$R = R_0(1+\alpha(T-T_0))$

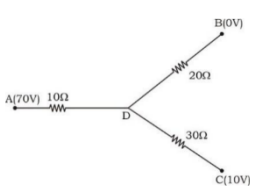
where α is the coefficient of expansion of resistance.

From these expressions we see, the resistance of a conductor depends upon length of the conductor, area of cross section of the conductor, nature of the material and temperature of the conductor.

11. Correct options: A),B),D)

Solution: We know, for a move through a resistance in the direction of the current, the change in potential is $-iR$; in the opposite direction it is iR.

We also know, according to the Kirchhoff's law of meshes, the sum of the currents entering any junction must be equal to the sum of the currents leaving that junction.



We know, the power P or rate of energy transfer, in a resistor R across which a potential

difference V is maintained is

$P = i^2R = \frac{V^2}{R}$

Given, an electric circuit as shown in the right.

Let the current through the 10 ohm resistor be i and current through 20 ohm resistor be i_1

.

Then in accordance to the Kirchhoff's law of meshes, the current through the 30 ohm resistor is:

$i - i_1$

Now, accounting for all the changes in the potential in the top branch of the circuit we get:

$70 - 10i - 20i_1 = 0$

Similarly, accounting for all the changes in the potential in the lower branch of the circuit we get:

$70 - 10i - 30(i - i_1) - 10 = 0$

$60 - 40i + 30i_1 = 0$

Now, multiplying $70 - 10i - 20i_1 = 0$ by -4 and adding to $60 - 40i + 30i_1 = 0$, we get:

$-4(70-10i-20i_1)+60-40i+30i_1=0$

$-280+40i+80i_1+60-40i+30i_1=0$

$-220+110i_1=0$

$i_1=2$

Substituting $i_1=2$ in $70-10i-20i_1=0$, we get:

$70-10i-20(2)=0$

$30-10i=0$

$i=3$

Therefore, the current in the section AD is $i=3$, in section DB is $i_1=2$ and in section DC is $i-i_1=1$

Now, accounting for all the changes in the potential from A to D, we get:

$70-10i=V_D$

$70-10(3)=V_D$

$40=V_D$

Therefore, point D is at a potential of 40V.

Now, the total power draws by the network is:

$P=i^2 10+i_1^2 20+(i-i_1)^2 30$

$P=3^2 10+2^2 20+1^2 30$

$P=90+80+30$

$P=200W$

Therefore, in the given figure point D is at a potential of 40V, the currents in the sections AD, DB, DC are in the ratio 3: 2:1, and the network draws a total power of 200 W.

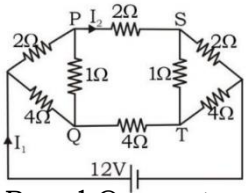
12. Correct option: (A), B), C), D)

Solution: When two resistors R_1 and R_2 are connected in series, the equivalent resistance is the sum of the individual resistance:

$R_{eq}=R_1+R_2$

Also, when two resistors R_1 and R_2 are connected in parallel, the equivalent resistance is given by:

$R_{eq}=\dfrac{R_1R_2}{R_1+R_2}$



Given, a resistance network as shown in the figure on the right.

Now, due to the formation of balanced Wheatstone's bridge, the point P and Q are at same potential and the points S and T are also at the same potential.

Therefore, no current flows through PQ and ST. Consequently, the resistance in PQ and ST can be dropped from the figure.

Now, the net resistance of the circuit is:

$R_{eq}=\dfrac{6(12)}{6+12}$

$R_{eq}=4$

Hence from the Ohm's law, the current i_1 in the circuit is:

$$i_1 = \frac{V}{R}$$
$$i_1 = \frac{12}{4}$$
$$i_1 = 3A$$

Let the current through PS be i_2. Then from the junction rule, the current through QT is $i_1 - i_2$. Since the voltage in parallel combination is the same, we get:

$$6i_2 = 12(i_1 - i_2)$$
$$i_2 = \frac{12}{12+6} i_1$$
$$i_2 = \frac{12}{18} 3$$
$$i_2 = 2A$$

Since the current i_2 flows in the direction from P to S, therefore, the potential at S is less than that at Q.

Therefore, if a resistance network as shown in the figure on the right, then the current through PQ is zero, $i_1 = 3A$, $i_2 = 2A$ and the potential at S is less than that at Q.

13. Correct option: (C)

Solution: We know, the resistance R of a conducting wire of length L and uniform cross section is

$$R = \rho \frac{L}{A}$$

where A is the cross-sectional area.

Also, when two resistors R_1 and R_2 are connected in parallel, the equivalent resistance is given by:

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

Given, a regular n - sided polygon is made from a uniform wire of resistance R where n is even.

We are interested in the resistance of the polygon between opposite corners.

Now, length of each side of the polygon is:

$$\frac{L}{n}$$

where L is the total length of the wire.

Hence, the resistance of each side of polygon is:

$$R' = \rho \frac{L/n}{A}$$
$$R' = \frac{R}{n}$$

Now, since the opposite corner of the polygon divides either side of the polygon in equal parts, each such part has resistance given by:

$$\frac{n}{2}\frac{R}{n}=\frac{R}{2}$$

Since these resistances are connected in parallel for the opposite point, the net resistance is:

$$R_{eq}=\frac{(R/2)(R/2)}{R/2+R/2}$$

$$R_{eq}=\frac{R}{4}$$

Now, since the adjacent corners of the polygon divides the polygon such that there are $n-1$ sides on one side with resistance given by:

$$n-1\frac{R}{n}=\frac{n-1}{n}R$$

Since these resistances are connected in parallel for the adjacent point with one side, the net resistance is:

$$R_{eq}=\frac{(n-1/n)R)(R/n)}{R/n+(n-1/n)R}$$

$$R_{eq}=\frac{(n-1)R}{n+n^2-n}$$

$$R_{eq}=\frac{(n-1)R}{n^2}$$

Therefore, if a regular n - sided polygon is made from a uniform wire of resistance R where n is even, resistance of each side of polygon is $\frac{R}{n}$, the resistance of the polygon between opposite corners is $\frac{R}{4}$ and the resistance of the polygon between any two adjacent side is $\frac{(n-1)R}{n^2}$.

Hence, the equivalent resistance between any two corners can have the maximum value $\frac{R}{4}$ and the minimum value $\frac{(n-1)R}{n^2}$.

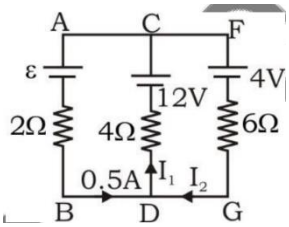
14. Correct option: (A), B)

Solution: We know, for a move through a resistance in the direction of the current, the change in potential is $-iR$; in the opposite direction it is iR.

Also, for a move through an ideal emf device in the direction of from positive terminal to negative terminal, the change in potential is E; in the opposite direction it is $-E$

We know, according to the junction rule, algebraic sum of the changes in potential encountered in a complete traversal of any loop of a circuit must be zero.

Given, an electric circuit as shown in the figure on the right.



Now, traversing the inner left loop in the counterclockwise direction, we get:

$-\varepsilon - 2(0.5) - 4i_1 + 12 = 0$

$-\varepsilon - 4i_1 + 11 = 0$

Now, traversing the inner right loop in the clockwise direction, we get:

$-4 - 4i_1 - 6i_2 + 12 = 0$

$-4i_1 - 6i_2 + 8 = 0$

Now, traversing the outer loop in the clockwise direction, we get:

$\varepsilon - 4 - 6i_2 + 2(0.5) = 0$

$\varepsilon - 3 - 6i_2 = 0$

Adding $\varepsilon - 3 - 6i_2 = 0$ to $-\varepsilon - 4i_1 + 11 = 0$, we get:

$-4i_1 - 6i_2 + 8 = 0$

$i_1 = 2 - \frac{3}{2} i_2$

From the junction rule, we also have:

$i_1 = 0.5 + i_2$

$2 - \frac{3}{2} i_2 = 0.5 + i_2$

$1.5 = 2.5 i_2$

$i_2 = 0.6$

Hence,

$i_1 = 2 - \frac{3}{2} i_2$

$i_1 = 2 - \frac{3}{2}(0.6)$

$i_1 = 2 - 0.9$

$i_1 = 1.1A$

Substituting the values in $-\varepsilon - 4i_1 + 11 = 0$, we get:

$\varepsilon = 4i_1 - 11$

$\varepsilon = 4(1.1) - 11$

$\varepsilon = -6.6$

The negative sign shows that the current flows through the battery in a direction opposite to that of what we have assumed.

Therefore, if an electric circuit as shown in the figure on the right, then $i_1 = 1.1A$ and $\varepsilon = 6.6V$ in magnitude.

15. Correct option: (B)

Solution: We know, the resistance R of a conducting wire of length L and uniform cross section is

$$R = \rho \frac{L}{A}$$

where A is the cross-sectional area.

Given, the dimensions of a block are $1cm \times 1cm \times 100cm$. The specific resistance of its material is $3 \times 10^{-7} \Omega m$

We are interested in the resistance between the opposite rectangular faces.

Since for the rectangular faces, the area of cross section is $1cm \times 100cm = 100cm^2$ and the length between the faces is $1cm$, the resistance is:

$$R = \rho \frac{L}{A}$$

$$R = 3 \times 10^{-7} \frac{1 \times 10^{-2}}{100 \times 10^{-4}}$$

$$R = 3 \times 10^{-7} \Omega$$

Therefore, if the dimensions of a block are $1cm \times 1cm \times 100cm$ and the specific resistance of its material is $3 \times 10^{-7} \Omega m$ then the resistance between the opposite rectangular faces is $3 \times 10^{-7} \Omega$.

16. Correct option: (D)

Solution: We know, the resistance R of a conducting wire of length L and uniform cross section is

$$R = \rho \frac{L}{A}$$

where A is the cross-sectional area.

Given, the dimensions of a block are $1cm \times 1cm \times 100cm$. The specific resistance of its material is $3 \times 10^{-7} \Omega m$

We are interested in the resistance between the opposite square faces.

Since for the square faces, the area of cross section is $1cm \times 1cm = 1cm^2$ and the length between the faces is $100cm$, the resistance is:

$$R = \rho \frac{L}{A}$$

$$R = 3 \times 10^{-7} \frac{100 \times 10^{-2}}{1 \times 10^{-4}}$$

$$R = 3 \times 10^{-3} \Omega$$

Therefore, if the dimensions of a block are $1cm \times 1cm \times 100cm$ and the specific resistance of its material is $3 \times 10^{-7} \Omega m$ then the resistance between the square faces is $3 \times 10^{-3} \Omega$.

17. Correct option: (A)

Solution: We know, the resistance R of a conducting wire of length L and uniform cross section is

$$R = \rho \frac{L}{A}$$

where A is the cross-sectional area.

Given, a wire has a resistance R.

Now, the wire is stretched to double its length.

Therefore, if wire has a resistance R then its resistance if it is stretched to double its length is $2R$.

18. Correct option: (A)

Solution: We know, the resistance R of a conducting wire of length L and uniform cross section is

$$R = \rho \frac{L}{A}$$

where A is the cross-sectional area.

Given, a coil is formed by winding 250 turns of insulated 16 –gauge copper wire (diameter = 1. 3 mm) in a single layer on a cylindrical form of radius 12 cm. Further, resistivity of copper is $1.69 \times 10^{-8}\ \Omega m$

Since the coil has 250 turns where each turn is winded on a cylindrical form of radius 12 cm, the total length of the coil is:

$$L = 250(2\pi(12))$$

$$L = 6000\pi\, cm$$

Therefore, the resistance of the coil is:

$$R = \rho \frac{L}{A}$$

$$R = 1.69 \times 10^{-8} \frac{6000\pi \times 10^{-2}}{\pi(1.3/2 \times 10^{-3})^2}$$

$$R = 2.4\Omega$$

Therefore, if a coil is formed by winding 250 turns of insulated 16 –gauge copper wire (diameter = 1. 3 mm) in a single layer on a cylindrical form of radius 12 cm and resistivity of copper is $1.69 \times 10^{-8}\ \Omega m$, then the resistance of the coil is 2. 4 ohm.

19. Correct option: (A)

Solution: We know, the resistance R of a conducting wire of length L and uniform cross section is

$$R = \rho \frac{L}{A}$$

where A is the cross-sectional area.

We know, Ohm's law states that the current through a device is always directly proportional to the potential difference applied to the device. Mathematically,

$$V = iR$$

where i current flows through the device when V voltage is applied across the device, R is its resistance.

We also know, the current density when the current is uniform across the surface and parallel to area vector, is defined as:

$$J = \frac{i}{A}$$

where A is the total area of the surface.

Given, when 115V is applied across a wire that is 10 m long and has a 0. 30 mm radius, the magnitude of the current density is $1.40 \times 10^4 \, Am^{-2}$.

Now, the current flowing through the wire is:

$$J = \frac{i}{A}$$

$$i = JA$$

Sine 115V is applied across the wire, from ohm's law, the resistance of the wire is:

$$V = iR$$

$$R = \frac{V}{i}$$

$$R = \frac{V}{JA}$$

Hence, the resistivity of the wire is:

$$R = \rho \frac{L}{A}$$

$$\rho = \frac{RA}{L}$$

$$\rho = \frac{VA}{JAL}$$

$$\rho = \frac{V}{JL}$$

Substituting the values, we get:

$$\rho = \frac{115}{1.40 \times 10^4 (10)}$$

$$\rho = 8.2 \times 10^{-4} \, \Omega m$$

Therefore, when 115V is applied across a wire that is 10 m long and has a 0. 30 mm radius, the magnitude of the current density is $1.40 \times 10^4 \, Am^{-2}$, then resistivity is $8.2 \times 10^{-4} \, \Omega m$.

20. Correct option: (A)

Solution: We know, the resistance R of a conducting wire of length L and uniform cross section is

$$R = \rho \frac{L}{A}$$

where A is the cross-sectional area.

Given, a regular n - sided polygon is made from a uniform wire of resistance R where n is even.

Now, length of each side of the polygon is:

$\frac{L}{n}$

where L is the total length of the wire.

Hence, the resistance of each side of polygon is:

$$R' = \rho \frac{L/n}{A}$$

$$R' = \frac{R}{n}$$

Therefore, if a regular n - sided polygon is made from a uniform wire of resistance R where n is even, then the resistance of each side of polygon is $\frac{R}{n}$.

21. Correct option: (D)

Solution: We know, the resistance R of a conducting wire of length L and uniform cross section is

$$R = \rho \frac{L}{A}$$

where A is the cross-sectional area.

Also, when two resistors R_1 and R_2 are connected in parallel, the equivalent resistance is given by:

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

Given, a regular n - sided polygon is made from a uniform wire of resistance R where n is even.

We are interested in the resistance of the polygon between opposite corners.

Now, length of each side of the polygon is:

$\frac{L}{n}$

where L is the total length of the wire.

Hence, the resistance of each side of polygon is:

$$R' = \rho \frac{L/n}{A}$$

$$R' = \frac{R}{n}$$

Now, since the opposite corner of the polygon divides either side of the polygon in equal parts, each such part has resistance given by:

$$\frac{n}{2}\frac{R}{n} = \frac{R}{2}$$

Since these resistances are connected in parallel for the opposite point, the net resistance is:

$$R_{eq} = \frac{(R/2)(R/2)}{R/2 + R/2}$$

$$R_{eq} = \frac{R}{4}$$

Therefore, if if a regular n - sided polygon is made from a uniform wire of resistance R where n is even, then the resistance of the polygon between opposite corners $\frac{R}{4}$.

22. Correct option: (None of the given option is correct. (D)

Solution: We know, the resistance R of a conducting wire of length L and uniform cross section is

$$R = \rho \frac{L}{A}$$

where A is the cross-sectional area.

Also, when two resistors R_1 and R_2 are connected in parallel, the equivalent resistance is given by:

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

Given, a regular n - sided polygon is made from a uniform wire of resistance R where n is even.

We are interested in the resistance of the polygon between opposite corners.

Now, length of each side of the polygon is:

$$\frac{L}{n}$$

where L is the total length of the wire.

Hence, the resistance of each side of polygon is:

$$R' = \rho \frac{L/n}{A}$$

$$R' = \frac{R}{n}$$

Now, since the adjacent corners of the polygon divides the polygon such that there are $n-1$ sides on one side with resistance given by:

$$n-1\frac{R}{n} = \frac{n-1}{n}R$$

Since these resistances are connected in parallel for the adjacent point with one side, the net resistance is:

$$R_{eq} = \frac{((n-1/n)R)(R/n)}{R/n + (n-1/n)R}$$

$$R_{eq} = \frac{(n-1)R}{n + n^2 - n}$$

$$R_{eq} = \frac{(n-1)R}{n^2}$$

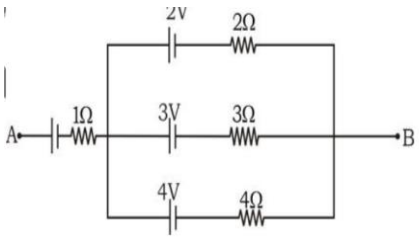
Therefore, if a regular n - sided polygon is made from a uniform wire of resistance R where n is even, then the resistance of the polygon between adjacent corners is $\frac{(n-1)R}{n^2}$.

23. Correct option: (Incomplete data (A)

Solution: We know, for n number of cells connected in parallel with emf $E_1, E_2, ..., E_n$ and internal resistance $r_1, r_2, ..., r_n$, the equivalent emf E_{eq} and equivalent resistance r_{eq} is given by:

$$\frac{1}{r_{eq}} = \frac{1}{r_1} + \frac{1}{r_2} + ... + \frac{1}{r_n}$$

$$\frac{r_{eq}}{E_{eq}} = \frac{E_1}{r_1} + \frac{E_2}{r_2} + ... + \frac{E_n}{r_n}$$



Here, the equivalent resistance of the cell is:

$$\frac{1}{r_{eq}} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}$$

$$\frac{1}{r_{eq}} = \frac{1}{2} + \frac{1}{3} + \frac{1}{4}$$

$$\frac{1}{r_{eq}} = \frac{6+4+3}{12}$$

$$r_{eq} = \frac{12}{13}\Omega$$

Now, the equivalent emf of the cell is:

$$\frac{r_{eq}}{E_{eq}} = \frac{E_1}{r_1} + \frac{E_2}{r_2} + \frac{E_3}{r_3}$$

$$\frac{r_{eq}}{E_{eq}} = \frac{2}{2} + \frac{3}{3} + \frac{4}{4}$$

$$\frac{r_{eq}}{E_{eq}} = 3$$

$$E_{eq} = \frac{r_{eq}}{3}$$

$$E_{eq} = \frac{12/13}{3}$$

$$E_{eq} = \frac{4}{13}$$

Now, the equivalent emf between the terminals A and B cannot be found unless we have the value of the battery before 1 ohm resistor.

24. Correct option: (B)

Solution: We know, for n number of cells connected in parallel with emf $E_1, E_2, ..., E_n$ and internal resistance $r_1, r_2, ..., r_n$, the equivalent emf E_{eq} and equivalent resistance r_{eq} is given by:

$$\frac{1}{r_{eq}} = \frac{1}{r_1} + \frac{1}{r_2} + ... + \frac{1}{r_n}$$

$$\frac{r_{eq}}{E_{eq}} = \frac{E_1}{r_1} + \frac{E_2}{r_2} + ... + \frac{E_n}{r_n}$$

Here, the equivalent resistance of the cell is:

$$\frac{1}{r_{eq}} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}$$

$$\frac{1}{r_{eq}} = \frac{1}{2} + \frac{1}{3} + \frac{1}{4}$$

$$\frac{1}{r_{eq}} = \frac{6+4+3}{12}$$

$$r_{eq} = \frac{12}{13}\Omega$$

Thus equivalent internal resistance of the combination is $\frac{12}{13}\Omega$

25. Correct option: (Incomplete data (B)

Solution: We know, for n number of cells connected in parallel with emf $E_1, E_2, ..., E_n$ and internal resistance $r_1, r_2, ..., r_n$, the equivalent emf E_{eq} and equivalent resistance r_{eq} is given by:

$$\frac{1}{r_{eq}} = \frac{1}{r_1} + \frac{1}{r_2} + ... + \frac{1}{r_n}$$

$$\frac{r_{eq}}{E_{eq}} = \frac{E_1}{r_1} + \frac{E_2}{r_2} + ... + \frac{E_n}{r_n}$$

Here, the equivalent resistance of the cell is:

$$\frac{1}{r_{eq}} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}$$

$$\frac{1}{r_{eq}} = \frac{1}{2} + \frac{1}{3} + \frac{1}{4}$$

$$\frac{1}{r_{eq}} = \frac{6+4+3}{12}$$

$$r_{eq} = \frac{12}{13}\Omega$$

Given, the polarity of cell of 4V is reversed

Now, the equivalent emf of the cell is:

$$\frac{r_{eq}}{E_{eq}} = \frac{E_1}{r_1} + \frac{E_2}{r_2} - \frac{E_3}{r_3}$$

$$\frac{r_{eq}}{E_{eq}} = \frac{2}{2} + \frac{3}{3} - \frac{4}{4}$$

$$\frac{r_{eq}}{E_{eq}} = 1$$

$$E_{eq} = \frac{r_{eq}}{1}$$

$$E_{eq} = \frac{12}{13}$$

Now, the equivalent emf between the terminals A and B cannot be found unless we have the value of the battery before 1 ohm resistor.

26. Correct match: a) ->Q), b) ->S), c) -> T),d) -> P)

Solution: We know the unit of resistivity is ohm m, hence the unit of reciprocal of resistivity is $ohm^{-1}m^{-1}$

We know the expression for resistivity of a metal is given by:

$$\rho = \frac{m}{ne^2\tau}$$

where n is the number of free electrons per unit volume and τ is the mean time between the collisions of an electron with the atoms of the metal, m is the mass of electron and e is the elementary charge.

We know, for a material obeying Ohm's law:

$$E = \rho J$$

where ρ is the resistivity of the material and J is the current density.

Since conductivity κ is inverse of resistivity, hence:

$$\kappa = \frac{J}{E}$$

Also, the drift velocity in a metallic wire carrying steady current i is:

$$v_d = \frac{i}{Ane}$$

where ne is the charge carrier density, and A is the area of cross section.

Therefore correct match is: a) ->Q), b) ->S), c) -> T),d) -> P)

27. Correct match: a) ->P), b) ->Q), c) -> S), d) -> R)

Solution: We know, the unit of resistance and resistivity is ohm and ohm-m respectively.

Also conductance and conductivity is defined to be the reciprocal of resistance and resistivity respectively.

Therefore, unit of conductance and conductivity is Siemen and ohm^{-1}-m^{-1} respectively.

Therefore correct match is: a) ->P), b) ->Q), c) -> S), d) -> R)

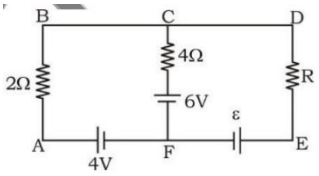
28. Correct match: a) ->P), b) ->Q), c) -> R), d) -> S)

Solution: We know, for a move through a resistance in the direction of the current, the change in potential is $-iR$; in the opposite direction it is iR.

Also, for a move through an ideal emf device in the direction of from positive terminal to negative terminal, the change in potential is E; in the opposite direction it is $-E$

We know, according to the junction rule, algebraic sum of the changes in potential encountered in a complete traversal of any loop of a circuit must be zero.

Given, a circuit is shown in fig. R is a non-zero variable with finite resistance. ε is some unknown emf with polarities as shown.



Let the current passing through 4 ohm resistor be i from F to Cand the current

passing through the 2 ohm resistor be i_1 from B to A.

Then from the junction law, the current passing through the resistance R in the direction E to D is:

$= i_1 - i$

Now, traversing the inner left loop in the counterclockwise direction, we get:

$6 - 4i - 2i_1 - 4 = 0$

$1 - 2i = i_1$

Now, traversing the inner right loop in the clockwise direction, we get:

$6 - 4i + R(i_1 - i) - \varepsilon = 0$

$6 - (R + 4)i + Ri_1 - \varepsilon = 0$

Now, substituting $1 - 2i = i_1$ into this, we get:

$6 - (R + 4)i + R(1 - 2i) - \varepsilon = 0$

$6 - Ri - 4i + R - 2Ri - \varepsilon = 0$

$6 - i(3R + 4) - \varepsilon = 0$

$i = \dfrac{\varepsilon - 6}{3R + 4}$

Hence, current passing through 4 ohm resistor can be zero if $\varepsilon = 6$.

Sine this current is positive, therefore the current passing through 4ohm resistance is from F to C if $\varepsilon > 6$

Now, if $\varepsilon < 6$, the current will be negative and will flow from C to F.

Now, current passing through the 2 ohm resistance is:

$1 - 2i = i_1$

$1 - 2\dfrac{\varepsilon - 6}{3R + 4} = i_1$

$\dfrac{3R + 4 - 2\varepsilon + 12}{3R + 4} = i_1$

$\dfrac{3R - 2\varepsilon + 16}{3R + 4} = i_1$

This current will be positive or from B to A direction when $3R-2\varepsilon+16>0$

Therefore the correct match is: a) ->P), b) ->Q), c) -> R), d) -> S)

29. Correct Matching is: a-S ,b-P , c- T , d- S

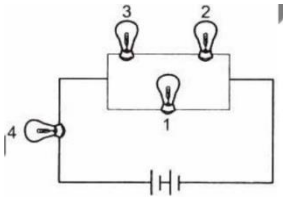
30. Correct option: (a) ->P), b) ->Q), c) -> R), d) -> S)

Solution: When two resistors R_1 and R_2 are connected in series, the equivalent resistance is the sum of the individual resistance:

$$R_{eq}=R_1+R_2$$

Also, when two resistors R_1 and R_2 are connected in parallel, the equivalent resistance is given by:

$$R_{eq}=\frac{R_1R_2}{R_1+R_2}$$



We know, the power P or rate of energy transfer, in a resistor R across which a potential difference V is maintained is

$$P=i^2R=\frac{V^2}{R}$$

Given, all bulbs consume same power and the resistance of bulb 1 is 36 ohm.

Now, as bulb 1 and series combination of bulb 2 and 3 are in parallel, the voltage across them will be same, thus:

$$\sqrt{PR_1}=V_2+V_3$$
$$\sqrt{PR_1}=\sqrt{PR_2}+\sqrt{PR_3}$$
$$\sqrt{36}=\sqrt{R_2}+\sqrt{R_3}$$
$$6=\sqrt{R_2}+\sqrt{R_3}$$

Now, since the power is $P=i^2R$, and since the current through the bulb 2 and 3 is same, it follows that their resistance is also equal, that is: $R_2=R_3$

Hence,

$$6=2\sqrt{R_2}$$
$$3=\sqrt{R_2}$$
$$R_2=9$$

Hence resistance of bulb 3 is 9 ohm.

Now, the equivalent resistance of bulb 1,2 and 3 is:

$$R_{eq}=\frac{18(36)}{18+36}$$
$$R_{eq}=\frac{18(36)}{54}$$
$$R_{eq}=12$$

Let the emf of the battery be E and the resistance of the bulb 4 be R_4. Then the current flowing through bulb 4 is:

$$i = \frac{E}{12 + R_4}$$

Now, from the current divider rule, the current across bulb 1 is:

$$i_1 = \frac{18}{36+18} i$$

$$i_1 = \frac{1}{3} i$$

Hence, the voltage across bulb 1 is:

$$V_1 = 36 i_1$$

$$V_1 = 36 \frac{i}{3}$$

Similarly, the voltage across bulb 4 is:

$$V_4 = R_4 i$$

Now, since the power is given by $P = \frac{V^2}{R}$ and since bulb 1 and bulb 4 has same power, thus:

$$\frac{V_1^2}{R_1} = \frac{V_4^2}{R_4}$$

$$\frac{\left(36 \frac{i}{3}\right)^2}{R_1} = \frac{(R_4 i)^2}{R_4}$$

$$\frac{(36)^2}{9(36)} = \frac{(R_4)^2}{R_4}$$

$$R_4 = 4\Omega$$

Hence the resistance of bulb 4 is 4 ohm.

Now, if the power of each bulb is 4 W, then the current flowing through the circuit is:

$$i = \sqrt{P / R_4}$$

$$i = \sqrt{4 / 4}$$

$$i = 1$$

Using

, the voltage across the bulb is:

$$E = (12 + R_4) i$$

$$E = (12 + 4)(1)$$

$$E = 16V$$

Finally, equivalent resistance of circuit is: $12 + R_4 = 16$

Therefore, if all bulbs consume same power and the resistance of bulb 1 is 36 ohm and are connected as shown in the figure, then resistance of bulb 3 is 9 ohm, resistance of bulb 4 is 4 ohm, voltage output of the battery if the power of each bulb is 4 W is 16 V and equivalent resistance of circuit is 16 ohm.

Therefore the correct match is: a) ->P), b) ->Q), c) -> R), d) -> S)

5. Magnetic Effects of Electric Current

Solutions

LEVEL – I

1. Correct option: (B)

Repulsion is the sure test of magnetism. Suppose we have two identical pieces; one is iron and other is a magnet. To distinguish which is which, we take another bar magnet. This bar magnet would attract the iron piece and also one end of this bar magnet would attract the opposite pole of the given magnet. However, one end of this bar magnet would repel only like pole of the given magnet.

Hence when repulsion occurs between the testing bar magnet and one of the two given pieces, that piece must be a magnet. That is why we say that repulsion is sure test of magnetism.

2. Correct option: (D)

A magnetic dipole consists of two unlike poles of equal strength and separated by a small distance.

For example, a bar magnet, a compass needle etc. are magnetic dipoles. We know that a current -loop behaves as a magnetic dipole. An atom of a magnetic material behaves as a dipole due to electrons revolving around the nucleus.

The two poles of a magnetic dipole (or a magnet), called north pole and south pole are always of equal strength, and of opposite nature. Further such two magnetic poles exist always in pairs and cannot be separated from each other. The distance between the two poles of a bar magnet is called the magnetic length of the magnet. It is

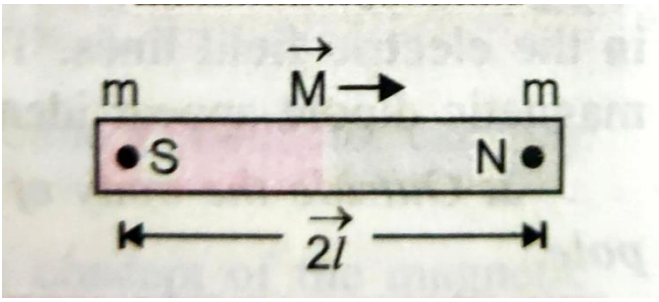
a vector directed from S-pole of magnet to its N-pole, and is represented by $2\vec{l}$.

Magnetic dipole moment is the product of strength of either pole (m) and the magnetic length ($2\vec{l}$) of the magnet.

It is represented by $\vec{M}$

Magnetic dipole moment = strength of either pole x magnetic length

$$\vec{M} = m\, 2\vec{l}$$



Magnetic dipole moment is a vector quantity directed from South to North pole of the magnet, The SI unit of M is joule/tesla or ampere metre^2

SI unit of pole strength is Am.

3. Correct option: (B)

Given: l = length of wire

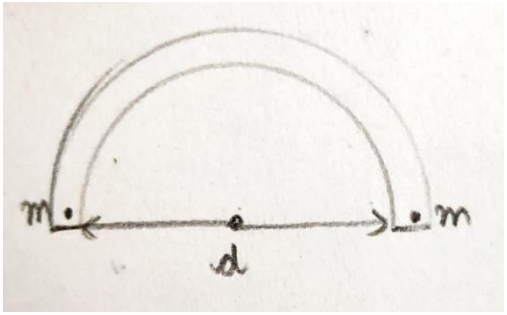
Magnetic dipole moment is the product of strength of either pole (m) and the magnetic length ($\vec{l}$) of the magnet.

It is represented by $\vec{M}$

Magnetic dipole moment = strength of either pole x magnetic length

magnetic dipole moment $M = m\, l$----------(1)

(Assume magnetic length is nearly equal to geometric length). As the magnetic wire is bent in the form of semicircular arc, then the distance between the poles is the diameter of arc. Length of the wire l = circumference of the semicircular arc



$l = \frac{\pi d}{2}$

$d = \frac{2l}{\pi}$

new magnetic dipole moment $M' = m\ d$----------(2)

$M' = m \times \frac{2l}{\pi}$

From equation 1

$M' = \frac{2M}{\pi}$

Hence, new magnetic moment = $\frac{2M}{\pi}$

4. Correct option: (D)

Magnetic moment of a magnet = m l

where

m= pole strength

l = effective distance between the two poles

Therefore if Magnetic Moment is kept constant, the pole strength is inversely proportional to the effective distance between the poles.

5. Correct option: (C)

Ratio of length of magnets= $\frac{l_1}{l_2} = \frac{2}{3}$

Ratio of pole strengths = $\frac{m_1}{m_2} = \frac{3}{4}$

Magnetic moment of a magnet M = m l

where

m= pole strength

l = effective distance between the two poles

Ratio of magnetic moments = $\frac{M_1}{M_2} = \frac{m_1 l_1}{m_2 l_2} = \frac{2x3}{3x4} = \frac{1}{2}$ = 1:2

6. Correct option: (C)

Magnetic moment of a magnet M = m l----------(1)

where m= pole strength

l = effective distance between the two poles

given: M = 5A - m^2

effective distance between the two poles l = 20 cm = 0.2 m

from equation 1

$5 = m \times 0.2 \quad \Rightarrow \quad m = \frac{5}{0.2} = \frac{5 \times 10}{2} = 25\ A - m$

Coulomb Force:

7. Correct option: (D)

The force of attraction or repulsion F between two magnetic poles of strengths $m_1$and m_2 separated by a distance d in space is directly proportional to the product of pole strengths and inversely proportional to the square of the distance between their centres.

$$magnetic\ force\ F_m = k\frac{m_1 m_2}{d^2}$$

Here, K = magnetic force constant ($K=\frac{\mu_o}{4\pi} = 10^{-7}$ Wb $A^{-1}m^{-1}$)

Hence,

$$magnetic\ force\ F_m = \frac{\mu_o}{4\pi}\frac{m_1 m_2}{d^2}$$

We can see from the above equation that magnetic force is inversely proportional to the square of the distance between their centres.

If the distance between the two magnetic poles is doubled then the force of attraction becomes $\frac{1}{2^2} = \frac{1}{4}$ one – fourth.

8. Correct option: (C)

The force of attraction or repulsion F between two magnetic poles of strengths $m_1$and m_2 separated by a distance d in space is directly proportional to the product of pole strengths and inversely proportional to the square of the distance between their centres.

$$magnetic\ force\ F_m = k\frac{m_1 m_2}{d^2}$$

Here, K = magnetic force constant ($K=\frac{\mu_o}{4\pi} = 10^{-7}$ Wb $A^{-1}m^{-1}$)

Hence,

$$magnetic\ force\ F_m = \frac{\mu_o}{4\pi}\frac{m_1 m_2}{d^2} - - - (1)$$

We can see from the above equation that magnetic force is inversely proportional to the square of the distance between their centres.

Given: pole strength is doubled $m_1'm_2' = 2\ m_1 x\ 2m_2 = 4m_1 m_2$

distance between the two magnetic poles is doubled d' = 2d

$$magnetic\ force\ F_m' = \frac{\mu_o}{4\pi}\frac{m_1' m_2'}{d'^2} = \frac{\mu_o}{4\pi}\frac{4\ m_1 m_2}{4d^2} \text{-------(2)}$$

Dividing equation 2 by 1

$$\frac{F_m'}{F_m} = \frac{\frac{\mu_o}{4\pi}\frac{4\ m_1 m_2}{4d^2}}{\frac{\mu_o}{4\pi}\frac{m_1 m_2}{d^2}} = 1$$

$F_m' = F_m$

Hence, the force remains unchanged.

9. Correct option: (B)

The force of attraction or repulsion F between two magnetic poles of strengths $m_1$and m_2 separated by a distance d in space is directly proportional to the product of pole strengths and inversely proportional to the square of the distance between their centres.

$$magnetic\ force\ F_m = k\frac{m_1 m_2}{d^2}$$

Here, K = magnetic force constant ($K=\frac{\mu_o}{4\pi} = 10^{-7}$ Wb $A^{-1}m^{-1}$)

Hence,

$$magnetic\ force\ F_m = \frac{\mu_o}{4\pi}\frac{m_1 m_2}{d^2} - - - (1)$$

Given: $F_m = 2\ x\ 10^{-7}$ N

d= 1m

pole strength of north pole $m_1 = 1\ A - m$
Putting the values in equation 1

$$2 \times 10^{-7} = 10^{-7} \frac{1 \times m_2}{1^2}$$

$$\frac{2 \times 10^{-7}}{10^{-7}} = m_2$$

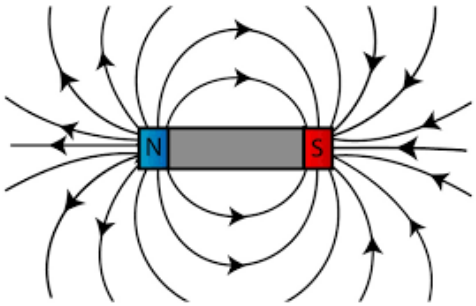
$$2 = m_2$$

$$pole\ strength\ (m) = 2A - m$$

Magnetic Field, magnetic field strength:

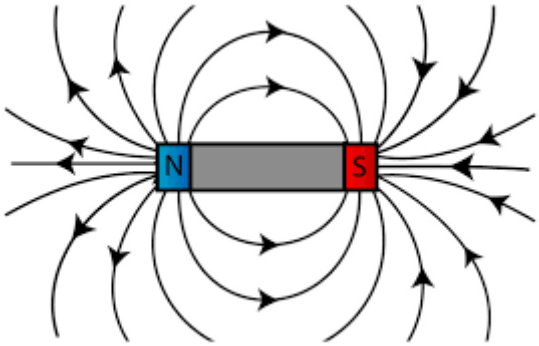
10. Correct option: (B)

The magnetic field lines of a magnet form closed continuous loops. We imagine magnetic field lines to be extending through the body of the magnet from south pole to north pole. Outside the body of the magnet, the direction of magnetic field lines is from north pole to south pole.



11. Correct option: (B)

The magnetic field lines of a magnet form closed continuous loops. We imagine magnetic field lines to be extending through the body of the magnet from south pole to north pole. Outside the body of the magnet, the direction of magnetic field lines is from north pole to south pole.



12. Correct option: (D)

in case of a short bar magnet

- *magnetic field induction* at a point on the axis of the bar magnet

$$B = \frac{\mu_o}{4\pi} \frac{2M}{d^3} - - - (1)$$

- *magnetic field induction* at a point on the equatorial line of the bar magnet

$$B = \frac{\mu_o}{4\pi} \frac{M}{d^3} - - - (2)$$

Where $\frac{\mu_o}{4\pi} = 10^{-7}\ Wb\ A^{-1}m^{-1}$

d=distance between the two magnetic poles

M= magnetic dipole moment

From equation 1 and 2 we can see that

$B \propto \frac{1}{d^3}$

13. Correct option: (B)

magnetic field induction due to the north pole

$B = \frac{\mu_o}{4\pi}\frac{m}{d^2}$----------(1)

m= pole strength

d= distance of the point from the north pole

given: m = 10 A-m

d= 5 cm = 0.05 m

$\frac{\mu_o}{4\pi} = 10^{-7}\ Wb\ A^{-1}m^{-1}$

from equation 1

$B = 10^{-7}\frac{10}{(5x\ 10^{-2})^2}$

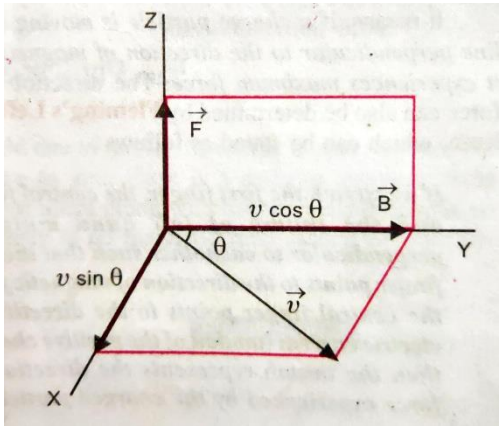
$B = 10^{-7}\frac{10}{25x\ 10^{-4}}$

$B = \frac{10^{-7+1+4}}{25} = \frac{10^{-2}}{25} = 0.04\ x\ 10^{-2}$

$B = 4\ x\ 10^{-4}$ Weber/m^2 radially outwards (because north pole is considered as hypothetical positive charge)

Motion of a Charged particle in uniform magnetic field:

14. Correct option: (D)



Consider a positive charge q moving in a uniform magnetic field B, with a velocity v. Let the angle between v and B be θ, Due to interaction between the magnetic field produced due to moving charge (ie. current) and magnetic field applied, the charge q then experiences a force, which depends on the factors given below:

(i) The magnitude of force F experienced by the moving charge is directly proportional to the magnitude of the charge i.e.

$F \propto q$

(ii) The magnitude of force F is directly proportional to the component of velocity acting perpendicular to the direction of magnetic field, i.e.

$F \propto v\sin\theta$

(iii) The magnitude of force F is directly proportional to the magnitude of the magnetic field applied i.e.

$F \propto B$

Combining the above factors, we get

$F \propto qvB\sin\theta$

$F = kqvB\sin\theta$

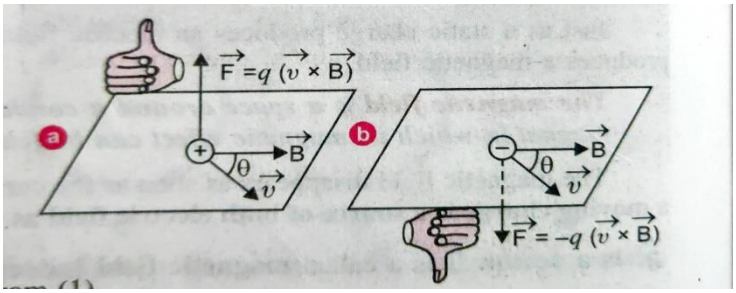
where k is a constant of proportionality. Its value is found to be one i.e. k= 1.

$F = qvB\sin\theta$

$\vec{F} = q(\vec{v} \times \vec{B})$

The direction of F is the direction of cross-product of velocity $\vec{v}$ and magnetic field $\vec{B}$, which is perpendicular to the plane containing and $\vec{v}$ *and* $\vec{B}$. It is directed as given by the Right-Handed-Screw Rule or Right-Hand Rule.

If $\vec{v}$ *and* $\vec{B}$are in the plane of paper, then according to Right-Hand Rule, the direction of $\vec{F}$ on positively charged particle will be perpendicular to the plane of paper upward and on negatively charged particle will be perpendicular to the plane of paper downwards.



There can be right angle between

$\vec{F}$ *and* $\vec{B}$

$\vec{v}$ *and* $\vec{B}$

$\vec{F}$ *and* $\vec{v}$

15. Correct option: (C)

Consider a positive charge q moving in a uniform magnetic field B, with a velocity v. Let the angle between v and B be θ, due to interaction between the magnetic field produced due to moving charge (ie. current) and magnetic field applied, the charge q then experiences a force

$F = qvB\sin\theta$

Special cases:

If

velocity of charged particle v= 0

$F = qvB\sin\theta = 0$

It means, if the charged particle is at rest in the magnetic field, it experiences no force.

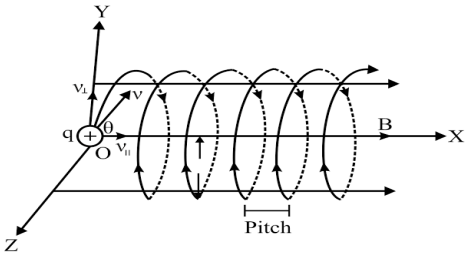
16. Correct option: (C)

Suppose a particle of mass m and charge q, entering a uniform magnetic field induction $\vec{B}$ at O, with velocity $\vec{v}$ making an angle θ with the direction of magnetic field acting in the plane of paper.

Resolving into two rectangular components, we have:

$v\cos\theta$ (= v_1) acts in the direction of the magnetic field

$v\sin\theta$ (= v_2) acts perpendicular to the direction of magnetic field.



The charged particle complete its circular path due to component velocity $v_2 = v\sin\theta$ perpendicular to the magnetic field.

For component velocity $v_1 = v\cos\theta$, there will be no force on the charged particle in the magnetic field, because the angle between $\vec{v_1}$ and $\vec{B}$ is zero. Thus the charged particle covers the linear distance in direction of the magnetic field with a constant speed $v\cos\theta$. Therefore, under the combined effect of the two component velocities, the charged particle in magnetic field will cover linear path as well as circular path i.e. the path of the charged particle will be helical, whose axis is parallel to the direction of magnetic field.

Hence, the path of the charged particle entering uniform magnetic field making an angle 45° is a helix.

17. Correct option: (C)

Just as a static charge produces an electric field, a moving charge or current through the conductor produces a magnetic field.

The magnetic field is a space around a conductor carrying current or the space around a magnet in which its magnetic effect can be felt.

The magnetic field disappears as soon as the current is switched off or charges stop moving. It means a moving charge is a source of both electric field as well as a magnetic field.

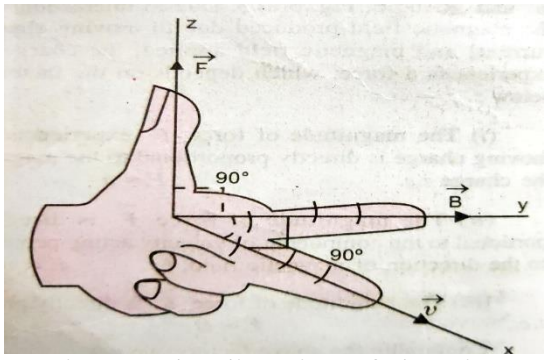
18. Correct option: (C)

A particle of charge 'q' and mass 'm' moving with uniform velocity 'v' enters normally into a magnetic field of induction 'B'. then radius 'r' of the circular path traced by it is given by $r = \frac{mv}{qB}$, hence, radius $r \propto m$.

19. Correct option: (C)

The direction of the force can also be determined by Fleming's Left Hand Rule, which can be stated as follows:

If we stretch the first finger, the central finger and the thumb of left hand mutually perpendicular to each other such that the first finger points to the direction of magnetic



field(east), the central finger points to the direction of electric current (motion of the positive charge) (south) then the thumb represents the direction of force experienced by the charged particle.Accordingly, the direction of F is perpendicular to the plane of paper directed vertically upwards.

20. Correct option: (A)

If charged particle is moving parallel to the direction of $\vec{B}$, it also does not experience any force because angle between $\vec{v}$ and $\vec{B}$ is 0° or 180° and sin 0° = sin 180° = 0. Therefore, the charged particle in this situation will continue moving along the same straight path with the same velocity.

21. Correct option: (B)

A particle of charge 'q' and mass 'm' moving with uniform velocity 'v' enters normally into a magnetic field of induction 'B'. then radius 'r' of the circular path traced by it is given by

$r = \frac{mv}{qB}$--------(1)

we know that kinetic energy of the particle $E = \frac{1}{2}mv^2$

$$v = \sqrt{\frac{2E}{m}}$$

putting in equation 1

$$r = \frac{\sqrt{2mE}}{qB}$$

as kinetic energy for electron and proton is same

$r \propto \frac{\sqrt{m}}{q}$

$q_e = q_p$

$m_p > m_e$

Hence,

$r_p > r_e$

Hence, trajectory of proton is less curved because it has larger radius.

22. Correct option: (C)

A particle of charge 'q' and mass 'm' moving with uniform velocity 'v' enters normally into a magnetic field of induction 'B'. then radius 'r' of the circular path traced by it is given by

$$r = \frac{mv}{qB}$$

hence, radius $r \propto mv$

mv = momentum of the charged particle

hence, radius of curvature is directly proportional to the momentum of the particle.

23. Correct option: (B)

Consider a positive charge q moving in a uniform magnetic field B, with a velocity v. Let the angle between v and B be θ, due to interaction between the magnetic field produced due to moving charge (ie. current) and magnetic field applied, the charge q then experiences a force

$F = qvB\sin\theta$

Given: B = 1.5 T

$v = 20 \times 10^7$ ms^{-1}

$\theta = 30^\circ$

$q = e_p = 1.6 \times 10^{-19}$ C

Putting the values in the above equation

$F = 1.6 \times 10^{-19} \times 20 \times 10^7 \times 1.5 \times \sin 30°$N

$$F = 1.6 \times 10^{-19} \times 20 \times 10^7 \times 1.5 \times \frac{1}{2}$$

Force $F = 2.4 \times 10^{-11}$N

24. Correct option: (D)

Consider a positive charge q moving in a uniform magnetic field B, with a velocity v. Let the angle between v and B be θ, due to interaction between the magnetic field produced due to moving charge (ie. current) and magnetic field applied, the charge q then experiences a force

$F = qvB\sin\theta$

Given: B = 2.5 T

Energy E = 2Mev

$1\text{ ev} = 1.6 \times 10^{-19}$ J

$\frac{1}{2}mv^2 = 2\text{Mev}$

$\frac{1}{2}mv^2 = 2 \times 10^6 \times 1.6 \times 10^{-19}$

mass of proton $= m_p = 1.67 \times 10^{-27}$kg

$\frac{1}{2} \times 1.67 \times 10^{-27} \times v^2 = 2 \times 10^6 \times 1.6 \times 10^{-19}$

$v^2 = \frac{2 \times 2 \times 10^6 \times 1.6 \times 10^{-19}}{1.67 \times 10^{-27}}$

$v^2 = 3.832 \times 10^{6-19+27}$

$v^2 = 3.832 \times 10^{14}$

$v = \sqrt{3.832 \times 10^{14}}$

$v = 1.957 \times 10^7$

$\theta = 90^\circ$

$q = e_p = 1.6 \times 10^{-19}$ C

Putting the values in the above equation

$F = 1.6 \times 10^{-19} \times 1.957 \times 10^7 \times 2.5 \times \sin 90°$N

$F = 1.6 \times 10^{-19} \times 1.957 \times 10^7 \times 2.5$ N

Force $F = 7.8 \times 10^{-12}$N $\approx 8 \times 10^{-12}$N

25. Correct option: (A)

Consider a positive charge q moving in a uniform magnetic field B, with a velocity v. Let the angle between v and B be θ, due to interaction between the magnetic field produced due to moving charge (ie. current) and magnetic field applied, the charge q then experiences a force

$\vec{F} = q(\vec{v} \times \vec{B})$

Given: $\vec{B} = 3\hat{i} + 2\hat{j}$ T

$\vec{v} = 5 \times 10^5 \hat{i}\ ms^{-1}$

$q = e_p = 1.6 \times 10^{-19}$ C

$\vec{F} = 1.6 \times 10^{-19}(5 \times 10^5 \hat{i} \times (3\hat{i} + 2\hat{j}))$ N

$\vec{F} = 1.6 \times 10^{-19}(10 \times 10^5 \hat{k})$ N

$\vec{F} = 1.6 \times 10^{-13}\ \hat{k}$ N

Hence, force acting on the particle $1.6 \times 10^{-13}\ \hat{k}$ N

26. Correct option: (C)

A particle of charge 'q' and mass 'm' moving with uniform velocity 'v' enters normally into a magnetic field of induction 'B'. then radius 'r' of the circular path traced by it is given by

$r = \frac{mv}{qB}$----------(1)

angular velocity $\omega = \frac{v}{r} = \frac{v}{\frac{mv}{qB}} = \frac{qB}{m}$

Time period $T = \frac{2\pi}{\omega} = \frac{2\pi}{\frac{qB}{m}} = \frac{2\pi m}{qB}$

Time period $T = \frac{2\pi m}{qB}$----------(2)

Given: B = 25 T

v= 10 ms^{-1}

q = 100 x 10^{-6} C

m = 10 mg = 10 x 10^{-6}kg

putting the values in equation 1

$r = \frac{10 \times 10^{-6} \times 10}{100 \times 10^{-6} \times 25} = \frac{1}{25} m = \frac{100}{25}$ cm = 4 cm

putting values in equation 2

Time period $T = \frac{2 \times 3.14 \times 10 \times 10^{-6}}{100 \times 10^{-6} \times 25} = 0.025$ sec = 25 x 10^{-3} sec

Force acting on a current carrying conductor in magnetic field

27. Correct option: (C)

The magnetic field lines around a straight conductor (straight wire) carrying current are concentric circles whose centres lie on the wire in the plane perpendicular to the conductor.

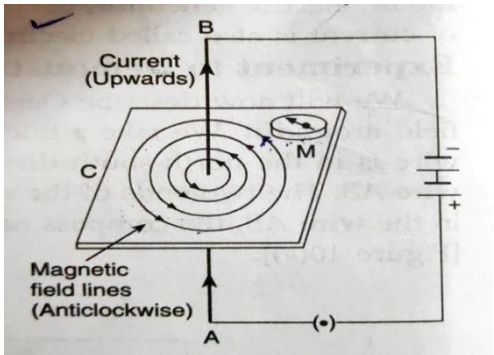


Figure 11. Magnetic field pattern due to a straight current-carrying wire.

28. Correct option: (B)

In the year 1820, Oersted realised that electricity and magnetism were intimately related. He showed experimentally that the electric current through a straight wire causes noticeable deflection of the magnetic compass needle held near the wire. He also found that the magnetic needle is aligned tangentially to an imaginary circle which has the straight current carrying wire as its centre and has its plane perpendicular to the wire.

29. Correct option: (D)

Consider a positive charge q moving in a uniform magnetic field B, with a velocity v. Let the angle between v and B be θ, due to interaction between the magnetic field produced due to moving charge (ie. current) and magnetic field applied, the charge q then experiences a force $F = qvB\sin\theta$

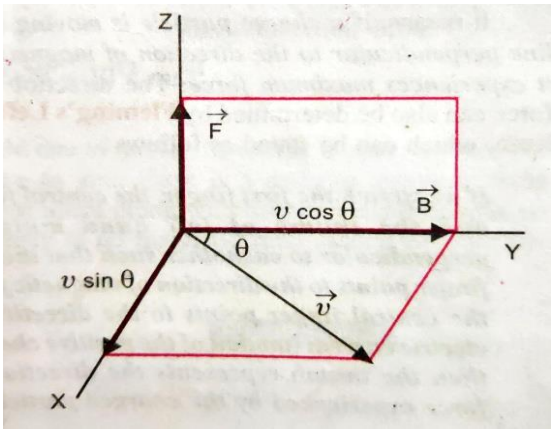
Special cases:

If

velocity of charged particle v= 0

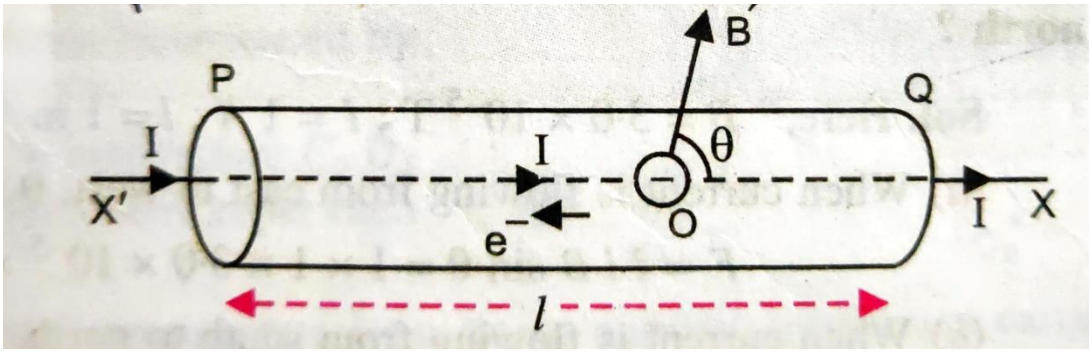
F = qvBsinθ = 0

It means, if the charged particle is at rest/stationary charge in the magnetic field, it experiences no force.



30. Correct option: (A)

We know that current through a conductor is due to the motion of free electrons in the conductor in a definite direction. When such a conductor is placed in a magnetic field, each electron moving in the magnetic field experiences a force. Hence the current carrying conductor experiences a force when placed in the magnetic field.



Expression for the force acting on the conductor carrying current placed in a magnetic field:

Consider a straight cylindrical conductor PQ of length L, area of cross-section A, carrying current I placed in a uniform magnetic field of induction $\vec{B}$. Let the conductor be placed along X-axis and magnetic field be acting in XY plane making an angle θ with X-axis. Suppose the current I flows through the conductor from the end P to Q. Since the current in a conductor is due to motion of electrons, therefore. electrons are moving from the end Q to P.

force acting on the conductor carrying current placed in a magnetic field

F = ILBsinθ

where I = current carried by the conductor

L = length of the conductor

B = magnetic field

θ = angle between the conductor and magnetic field

Special cases:

If

θ=90˚

F = ILBsin90˚ = ILB = maximum force

It means a linear conductor carrying current if placed perpendicular to the direction of magnetic field, it experiences maximum force.

31. Correct option: (C)

Force acting on the conductor carrying current placed in a magnetic field

$F = ILB\sin\theta$

where I = current carried by the conductor

L = length of the conductor

B = magnetic field

θ = angle between the conductor and magnetic field

Special cases:

If

$\theta = 90^\circ = \frac{\pi}{2}$

$F = ILB\sin\left(\frac{\pi}{2}\right) = ILB$ = maximum force

It means a linear conductor carrying current if placed at an angle of $\frac{\pi}{2}$ to the direction of magnetic field, it experiences maximum force.

32. Correct option: (A)

force acting on the conductor carrying current placed in a magnetic field

$F = ILB\sin\theta$

where I = current carried by the conductor

L = length of the conductor

B = magnetic field

θ = angle between the conductor and magnetic field

Special cases:

If

$\theta = 0^\circ$

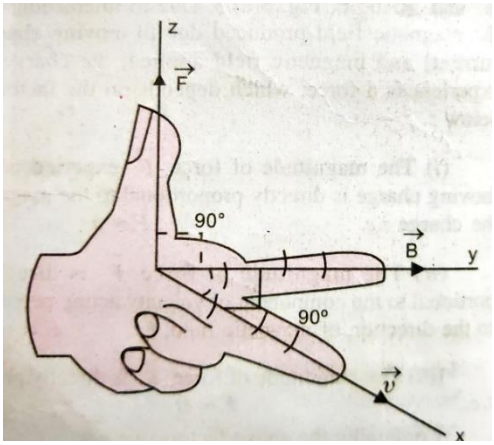
$F = ILB\sin(0°) = 0$= minimum force

It means a linear conductor carrying current if placed at an angle of parallel o the direction of magnetic field, it experiences no force.

33. Correct option: (C)

The direction of the force can also be determined by Fleming's Left Hand Rule, which can be stated as follows:

If we stretch the first finger, the central finger and the thumb of left hand mutually perpendicular to each other such that the first finger points to the direction of magnetic field(north), the central finger points to the direction of electric current (motion of the



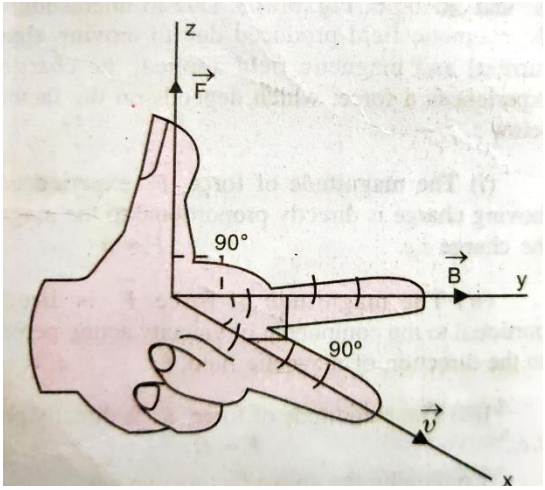
positive charge) (east) then the thumb represents the direction of force experienced by the charged particle.

Accordingly, the direction of F is perpendicular to the plane of paper directed vertically upwards.

34. Correct option: (C)

The direction of the force/motion can also be determined by Fleming's Left Hand Rule, which can be stated as follows:

If we stretch the first finger, the central finger and the thumb of left hand mutually perpendicular to each other such that the first finger points to the direction of magnetic field(right), the central finger points to the direction of electric current (motion of the positive charge) (upwards) then the thumb represents the direction of force experienced by the charged particle.



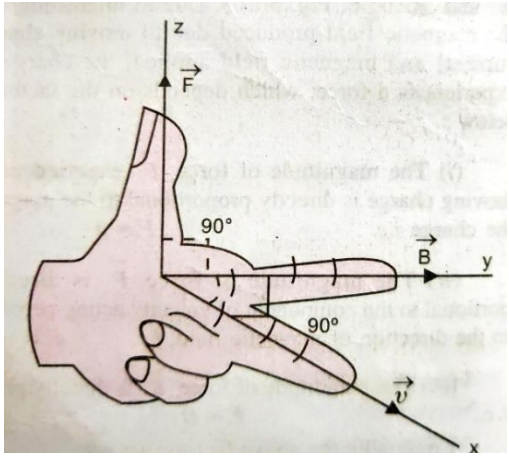
Accordingly, the direction of F is perpendicular to the plane of paper directed vertically inwards.

Hence, the wire will move inwards to the plane of paper.

35. Correct option: (D)

The direction of the force can also be determined by Fleming's Left Hand Rule, which can be stated as follows:

If we stretch the first finger, the central finger and the thumb of left hand mutually perpendicular to each other such that the first finger points to the direction of magnetic field (north to south, towards right), the central finger points to the direction of electric current (motion of the positive charge) (upwards) then the thumb represents the direction of force experienced by the charged particle.



Accordingly, the direction of F is perpendicular to the plane of paper directed vertically upwards out of the page towards you.

36. Correct option: (A)

Force acting on the conductor carrying current placed in a magnetic field

$F = ILB\sin\theta$---------(1)

where I = current carried by the conductor

L = length of the conductor

B = magnetic field

θ = angle between the conductor and magnetic field

given: I= 10 A

L= 1.5 m

B= 2 Tesla

F = 15 N

Putting values in equation 1

$15 = 10 \times 1.5 \times 2 \times \sin\theta$

$\frac{15}{10 \times 1.5 \times 2} = \sin\theta$

$\frac{1}{2} = \sin\theta$

$\sin 30° = \sin\theta$

$\theta = 30°$

hence, the angle between magnetic field and the direction of current = 30°

37. Correct option: (D)

Torque acting on a current carrying loop in magnetic field is given by

$\vec{\tau} = \vec{M} \times \vec{B}$

$\tau = MB\sin\theta$----------(1)

where M = magnetic dipole moment

B= magnetic field

θ = angle made by the normal of the loop with the magnetic field

Now

Magnetic dipole moment M = NIA

Where N = number of turns

I = current

A = area of the loop

Equation 1 becomes

$\tau = NIAB\sin\theta$----------(2)

given: angle made by the plane of the loop with the magnetic field = 90˚

$\theta = 90^\circ - 90^\circ = 0^\circ$

radius = R

current = I

area $A = \pi r^2 = \pi R^2$

N= 1

equation 2 becomes

$\tau = \pi R^2 IB \sin 0°$

$\tau = 0°$

hence, torque on the loop = 0°

38. Correct option: (C)

Force acting on the conductor carrying current placed in a magnetic field

$F = ILB\sin\theta$---------(1)

where I = current carried by the conductor
L = length of the conductor
B = magnetic field
θ = angle between the conductor and magnetic field

given: ab = l
∠abc = 45°
$bc = \frac{ab}{\cos 45°} = \frac{ab}{\frac{1}{\sqrt{2}}} = \sqrt{2}ab = \sqrt{2}\,l$
Putting values in equation 1
F_{ab} = I x ab x Bsin90° (angle between length ab and magnetic field is 90°)
$F_{ab} = IlB$-----(2)
F_{bc} = I x bc xBsin45°
$F_{bc} = I \times \sqrt{2}\,l \times B \times \frac{1}{\sqrt{2}} = IlB$ ----------(3)
Ratio of force on bc to ab
$\frac{F_{bc}}{F_{ab}} = \frac{IlB}{IlB} = 1$

LEVEL- II

1. Correct option: (B)
Magnetic moment of a magnet M = m l----------(1)
where
m= pole strength
l = effective distance between the two poles
given: m= 25 A - m
effective distance between the two poles l = 5 cm = 0.05 m
from equation 1
M= 25 x 0.05
M= 1.25 A – m^2

2. Correct option: (B)
The force of attraction or repulsion F between two magnetic poles of strengths $m_1$and m_2 separated by a distance d in space is directly proportional to the product of pole strengths and inversely proportional to the square of the distance between their centres.

$$magnetic\ force\ F_m = k\frac{m_1 m_2}{d^2}$$

Here, K = magnetic force constant ($K = \frac{\mu_o}{4\pi} = 10^{-7}$ Wb $A^{-1}m^{-1}$)
Hence,

$$magnetic\ force\ F_m = \frac{\mu_o}{4\pi}\frac{m_1 m_2}{d^2} - - - (1)$$

Given: F_m = 64 mg. wt
1mg = 10^{-6}kg
1kg-wt = 10 N
64mg. wt = 64 x 10^{-5}N
F_m = 64 mg. wt = 64 x 10^{-5}N
d= 10 cm = 0.1 m
pole strength m_1 = m

$m_2 = 4m$

Putting the values in equation 1

$64 \times 10^{-5} = 10^{-7} \frac{m \times 4m}{(0.1)^2}$

$m^2 = \frac{64 \times 10^{-5} \times 0.1 \times 0.1}{4 \times 10^{-7}}$

$m^2 = 16$

$m = \sqrt{16}$

m = 4 A-m

$m_1 = 4\ A - m$

$m_2 = 4 \times 4\ A - m = 16$ A-m

Strength of poles are 4 A-m and 16 A-m

3. Correct option: (B)

$magnetic\ force\ F_m = \frac{\mu_o}{4\pi} \frac{m_1 m_2}{d^2} - - - (1)$

Given: $F_{NS} = F$

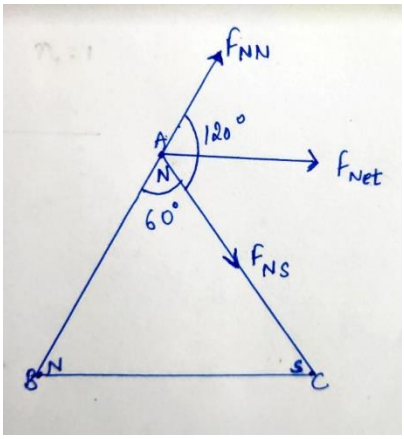
pole strength $m_1 = m_2 = m$

distance of separation = d

When the north pole is placed at the third vertex

$force\ of\ attraction\ F_{NS} = F = \frac{\mu_o}{4\pi} \frac{m^2}{d^2}$ towards AC

$force\ of\ repulsion\ F_{NN} = F = \frac{\mu_o}{4\pi} \frac{m^2}{d^2}$ towards BA



Resultant force is given by the parallelogram law of vector addition at point A

Angle between two forces =120°

resultant force $F_{Net} = \sqrt{F^2 + F^2 + 2F^2 \cos 120°}$

resultant force $F_{Net} = \sqrt{2F^2 + 2F^2 \times \left(-\frac{1}{2}\right)}$

resultant force $F_{Net} = \sqrt{2F^2 - F^2}$

resultant force $F_{Net} = \sqrt{F^2}$

resultant force $F_{Net} = F$

4. Correct option: (D)

The force of attraction or repulsion F between two magnetic poles of strengths $m_1$and m_2 separated by a distance d in space is directly proportional to the product of pole strengths and inversely proportional to the square of the distance between their centres.

$$magnetic\ force\ F_m = k\frac{m_1 m_2}{d^2}$$

Here, K = magnetic force constant ($K=\frac{\mu_o}{4\pi} = 10^{-7}\ Wb\ A^{-1}m^{-1}$)

Hence, $magnetic\ force\ F_m = \frac{\mu_o}{4\pi}\frac{m_1 m_2}{d^2}$

We can see from the above equation that magnetic force is inversely proportional to the square of the distance between their centres.

• When distance between the two magnetic poles is doubled d = 10 cm=0.1m

force of repulsion = 40N

$40N = \frac{\mu_o}{4\pi}\frac{m_1 m_2}{(0.1)^2}$----------(1)

• When distance between the two magnetic poles is doubled d = 30 cm=0.3m

$magnetic\ force\ F'_m = \frac{\mu_o}{4\pi}\frac{m_1' m_2'}{d'^2} = \frac{\mu_o}{4\pi}\frac{m_1 m_2}{(0.3)^2}$-------(2)

Dividing equation 2 by 1

$$\frac{F'_m}{40N} = \frac{\frac{\mu_o}{4\pi}\frac{m_1 m_2}{(0.3)^2}}{\frac{\mu_o}{4\pi}\frac{m_1 m_2}{(0.1)^2}} = \frac{(0.1)^2}{(0.3)^2} = \frac{1}{9}$$

$$\frac{F'_m}{40N} = 1:9$$

5. Correct option: (C)

$$magnetic\ force\ F_m = \frac{\mu_o}{4\pi}\frac{m_1 m_2}{d^2} - - - (1)$$

Given: $F_{NN} = F$

pole strength $m_1 = m_2 = m$

distance of separation = a

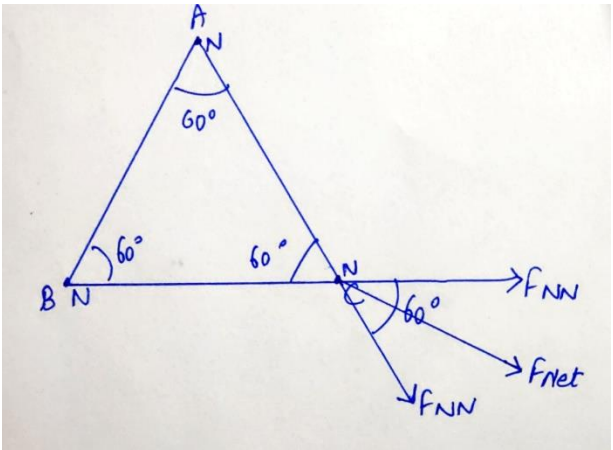
When the north pole is placed at the third vertex C

$force\ of\ repulsion\ F_{NN} = F = \frac{\mu_o}{4\pi}\frac{m^2}{a^2}$ towards AC

$force\ of\ repulsion\ F_{NN} = F = \frac{\mu_o}{4\pi}\frac{m^2}{a^2}$ towards BC

Resultant force is given by the parallelogram law of vector addition

Angle between two forces =60°



resultant force $F_{Net} = \sqrt{F^2 + F^2 + 2F^2\cos 60°}$

resultant force $F_{Net} = \sqrt{2F^2 + 2F^2\ x\ \left(\frac{1}{2}\right)}$

resultant force $F_{Net} = \sqrt{2F^2 + F^2}$

resultant force $F_{Net} = \sqrt{3F^2}$

resultant force $F_{Net} = \sqrt{3}F$

resultant force $F_{Net} = \frac{\mu_o}{4\pi}\frac{\sqrt{3}m^2}{a^2}$

6. Correct option: (A)

• No two magnetic field lines can intersect each other.

• At any given point, tangent to the magnetic field line represents the direction of net magnetic field $\vec{B}$ at that point.

• Outside the body of the magnet, the direction of magnetic field lines is from north pole to south pole.

7. Correct option: (B)

A particle of charge 'q' and mass 'm' moving with uniform velocity 'v' enters normally into a magnetic field of induction 'B'. then radius 'r' of the circular path traced by it is given by

$r = \frac{mv}{qB}$

Given: B = 0.4 T

velocity $v = 10^5\ ms^{-1}$

mass of helium $= m_{He} = 4m_p$

$q_{He} = 2q_p$

Radius for helium ion

$r_{He} = r = \frac{m_{He}v}{q_{He}B} = \frac{4m_pv}{2q_pB} = \frac{2m_pv}{q_pB}$

Radius for proton

$r_p = \frac{m_pv}{q_pB} = \frac{r}{2}$

$r_p = \frac{r}{2}$

8. Correct option: (C)

Consider a positive charge q moving in a uniform magnetic field B, with a velocity v. Let the angle between v and B be θ, due to interaction between the magnetic field produced due to moving charge (ie. current) and magnetic field applied, the charge q then experiences a force

$\vec{F} = q(\vec{v} \times \vec{B})$

Given: $\vec{B} = \hat{i} + 4\hat{j} - 3\hat{k}$ T

$\vec{v} = 2 \times 10^5\hat{i}\ ms^{-1}$

$q = e = -1.6 \times 10^{-19}$ C

$\vec{F} = -1.6 \times 10^{-19}\left(2 \times 10^5\hat{i} \times (\hat{i} + 4\hat{j} - 3\hat{k})\right)$ N

$\vec{F} = -1.6 \times 10^{-19}\left(0 + 8 \times 10^5\hat{k} - 6 \times 10^5(-\hat{j})\right)$ N

$\vec{F} = -1.6 \times 10^{-14}\left(8\hat{k} + 6\hat{j}\right)$ N

$F = 1.6 \times 10^{-14}\sqrt{(8)^2 + (6)^2}$ N

$F = 1.6 \times 10^{-14}\sqrt{64 + 36}$ N

$F = 1.6 \times 10^{-14}\sqrt{100}$ N

$F = 1.6 \times 10^{-14} \times 10$ N

$F = 1.6 \times 10^{-13}$ N

Hence, force acting on the particle 1.6×10^{-13} N

9. Correct option: (B)

In the year 1820, Oersted realised that electricity and magnetism were intimately related. He showed experimentally that the electric current through a straight wire causes

noticeable deflection of the magnetic compass needle held near the wire. He also found that the magnetic needle is aligned tangentially to an imaginary circle which has the straight current carrying wire as its centre and has its plane perpendicular to the wire.
Oersted showed that a current carrying wire produces a magnetic field around it.
The magnetic field lines around a straight conductor (straight wire) carrying current are concentric circles whose centres lie on the wire in the plane perpendicular to the conductor.

Force Acting on a charged particle:

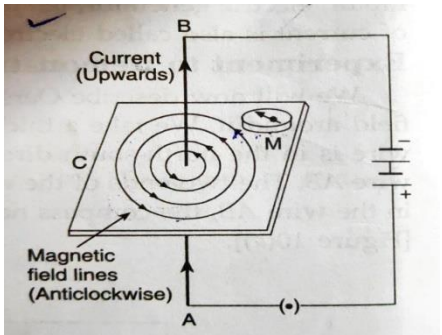


Figure 11. Magnetic field pattern due to a straight current-carrying wire.

10.Correct options(A), (B), (C), (D)

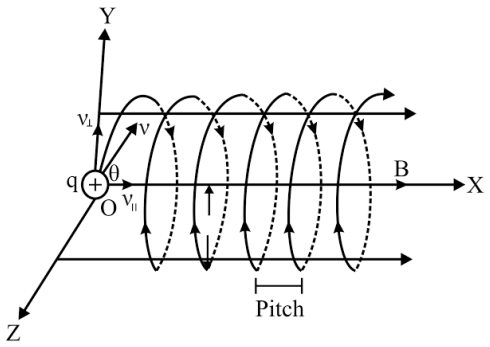
(A) as the charged particle is coming from outside, it may not complete a circular path.

(B) Suppose a particle of mass m and charge q, entering a uniform magnetic field induction $\vec{B}$ at O, with velocity $\vec{v}$ making an angle θ with the direction of magnetic field acting in the plane of paper.

Resolving into two rectangular components, we have:

v cosθ (= v_1) acts in the direction of the magnetic field

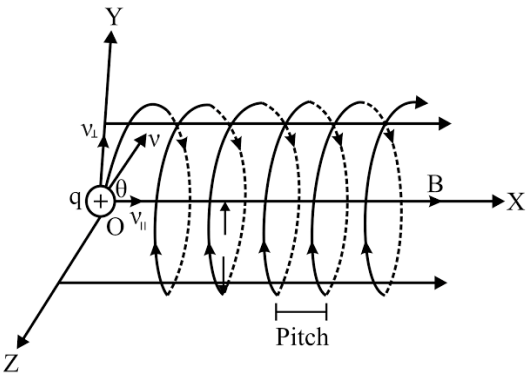
v sinθ (= v_2) acts perpendicular to the direction of magnetic field.



The charged particle complete its circular path due to component velocity $v_2 = v \sin\theta$ perpendicular to the magnetic field.

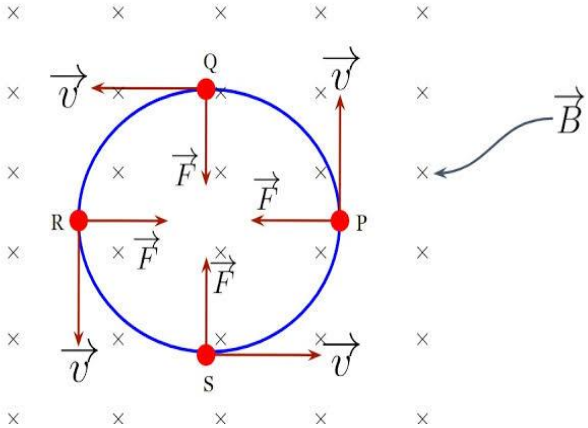
For component velocity $v_1 = v \cos\theta$, there will be no force on the charged particle in the magnetic field, because the angle between $\vec{v_1}$ and $\vec{B}$ is zero. Thus the charged particle covers the linear distance in direction of the magnetic field with a constant speed v cosθ. Therefore, under the combined effect of the two component velocities, the charged particle in magnetic field will cover linear path as well as circular path i.e. the path of the charged particle will be helical, whose axis is parallel to the direction of magnetic field.

(C) Suppose a particle of mass m and charge q, entering a uniform magnetic field induction $\vec{B}$ at O, with velocity $\vec{v}$ making an angle θ with the direction of magnetic field acting in the plane of paper.



Resolving into two rectangular components, we have:
v cosθ (= v_1) acts in the direction of the magnetic field
v sinθ (= v_2) acts perpendicular to the direction of magnetic field.
For component velocity v_2 , the force acting on the charged particle due to magnetic field is
$\vec{F} = q(\vec{v}_2 \times \vec{B})$
$F = qv_2 B\sin\theta$
$\theta = 90^\circ$
$F = qv_2 B\sin 90°$
$F = qv_2 B$
$F = qvB\sin\theta$
The direction of this force F is perpendicular to the plane containing $\vec{B}$ and $\vec{v_2}$ and is directed as given by Right hand rule**. As this force is to remain always perpendicular to $\vec{v_2}$,** it does not perform any work and hence cannot change the magnitude of velocity. It changes only the direction of motion of the particle. Due to it, the charged particle is made to move on a circular path in the magnetic field
power = $\vec{F}.\vec{v}_2 = Fv\cos\theta = 0$ ($\theta = 90°$)

Motion of Charged Particles in Uniform Magnetic Field

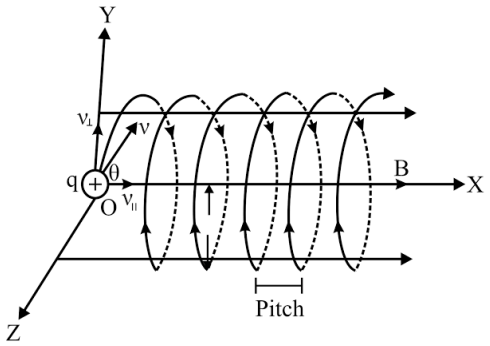


(D) if in a region a uniform magnetic field and electric field both exist, a charged particle moving in this region cannot trace a circular path as the Lorentz force will be applicable in the presence of both fields.

11.Correct options(A), (D)

Suppose a particle of mass m and charge q, entering a uniform magnetic field induction $\vec{B}$ at O, with velocity $\vec{v}$ making an angle θ with the direction of magnetic field acting in the plane of paper.
Resolving into two rectangular components, we have:
v cosθ (= v_1) acts in the direction of the magnetic field

v sinθ (= v_2) acts perpendicular to the direction of magnetic field.
The charged particle completes its circular path due to component velocity $v_2 = v \sin\theta$



perpendicular to the magnetic field.
For component velocity $v_1 = v\cos\theta$, there will be no force on the charged particle in the magnetic field, because the angle between $\vec{v_1}$ and $\vec{B}$ is zero. Thus, the charged particle covers the linear distance in direction of the magnetic field with a constant speed v cosθ. Therefore, under the combined effect of the two component velocities, the charged particle in magnetic field will cover linear path as well as circular path i.e. the path of the charged particle will be helical, whose axis is parallel to the direction of magnetic field.
Since, the trajectory will be helical, the trajectory will never be perpendicular to the direction of the magnetic field or parallel to the magnetic field. Its trajectory will be aligned with the magnetic field.

12. Correct options(A), (C)

In the absence of a magnetic field, the particle will experience gravitational force mg. As a result, the particle will not continue moving in the horizontal direction but will describe a parabolic path. So, a magnetic field must be present and its direction must be perpendicular to the direction of the velocity. The magnetic force experienced by the particle is given by
$F = qvB\sin\theta$
If the particle is to move in the horizontal direction, this force must balance the force of gravity, i.e.,
$mg = qvB\sin\theta$
The minimum value of B corresponds to $\sin\theta = 1$ or $\theta = 90°$
Thus, $mg = qvB$----------(1)
Given:
charge $q = 2.5x10^{-8}$ C
velocity $v = 6\ x10^{4}\ ms^{-1}$
$m = 0.5\ g = 0.5\ x\ 10^{-3}$ kg
$g = 9.8$ m/s
putting values in equation 1
$0.5\ x\ 10^{-3}\ x\ 9.8 = 2.5x10^{-8}\ x\ 6\ x10^{4}\ x\ B$
$\frac{0.5x10^{-3}x9.8}{2.5x10^{-8}\ x\ 6\ x10^{4}} = B$
$\frac{0.5x10^{-3}x9.8}{2.5x10^{-8}\ x\ 6\ x10^{4}} = B$
3.2 T = B
B = 3.2 T

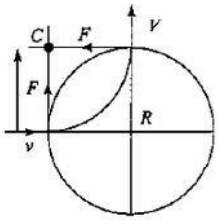
13.Correct options(A), (B), (C)

A particle of charge 'q' and mass 'm' moving with uniform velocity 'v' enters normally into a magnetic field of induction 'B'. then radius 'R' of the circular path traced by it is given by

$R = \frac{mv}{qB}$----------(1)

or

$B = \frac{mv}{qR}$



angular velocity $\omega = \frac{v}{R} = \frac{v}{\frac{mv}{qB}} = \frac{qB}{m}$----------(2)

Time period $T = \frac{\frac{2\pi}{4}}{\omega} = \frac{\pi R}{2v}$

From equation 2

$\frac{v}{R} = \frac{qB}{m}$

Time period $T = \frac{\pi R}{2v} = \frac{\pi m}{2qB}$

14. Correct option: (A)

Consider a positive charge q moving in a uniform magnetic field B, with a velocity v. Let the angle between v and B be θ, due to interaction between the magnetic field produced due to moving charge (ie. current) and magnetic field applied, the charge q then experiences a force

$F = qvB\sin\theta$

$\theta = 90^{\circ}$

$F = qvB\sin 90°$

$F = qvB$--------(1)

Given: B = 1.0 T

charge on helium ion $q = 2 \text{ x} 1.6 \text{ x } 10^{-19}$ C

velocity v= 10 kms^{-1} = 10000 ms^{-1}

Putting the values in the above equation

$F = 2 \text{ x} 1.6 \text{ x } 10^{-19} \text{x } 10000 \text{ x } 1$

$F = 3.2 \text{ x} 10^{-15}$ N

15. Correct option: (B)

A particle of charge 'q' and mass 'm' moving with uniform velocity 'v' enters normally into a magnetic field of induction 'B'. then radius 'r' of the circular path traced by it is given by

$r = \frac{mv}{qB}$----------(1)

Given: B = 1.0 T

charge on helium ion $q = 2 \text{ x} 1.6 \text{ x } 10^{-19}$ C

velocity v= 10 kms^{-1} = 10000 ms^{-1}

mass of helium = $m_{He} = 4 \text{ x} 1.67 \text{ x } 10^{-27}$kg

putting the values in equation 1

radius $r = \frac{4 \text{ x} 1.67 \text{ x } 10^{-27} \text{x } 10000}{2 \text{ x} 1.6 \text{ x } 10^{-19} \text{ x } 1} = 2.1 \text{ x } 10^{-4}$ m

16. Correct option: (D)

$r = \frac{mv}{qB}$

angular velocity $\omega = \frac{v}{r} = \frac{v}{\frac{mv}{qB}} = \frac{qB}{m}$

Time period $T = \frac{2\pi}{\omega} = \frac{2\pi}{\frac{qB}{m}} = \frac{2\pi m}{qB}$

Time period $T = \frac{2\pi m}{qB}$----------(2)

Given: B = 1.0 T

charge on helium ion $q = 2 \text{ x} 1.6 \text{ x } 10^{-19}$ C

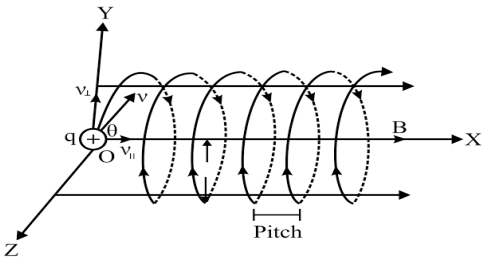
mass of helium $= m_{He} = 4 \text{ x} 1.67 \text{ x } 10^{-27}$kg

putting values in equation 2

Time period $T = \frac{2 \text{ x } 3.14 \text{ x } 4 \text{ x} 1.67 \text{ x } 10^{-27}}{2 \text{ x} 1.6 \text{ x } 10^{-19} \text{ x } 1} = 1.256 \text{ x } 10^{-7}$sec

17. Correct option: (A)

Suppose a particle of mass m and charge q, entering a uniform magnetic field induction $\vec{B}$ at O, with velocity $\vec{v}$ making an angle θ with the direction of magnetic field acting in the plane of paper.



Resolving into two rectangular components, we have:

v cosθ (= v_1) acts in the direction of the magnetic field

v sinθ (= v_2) acts perpendicular to the direction of magnetic field.

For component velocity v_2 , the force acting on the charged particle due to magnetic field is

$\vec{F} = q(\vec{v}_2 \text{ x } \vec{B})$

$F = qv_2 B\sin\theta$

$\theta = 90^{\circ}$

$F = qv_2 B\sin 90°$

$F = qv_2 B$

$F = qvB\sin\theta$

The direction of this force F is perpendicular to the plane containing $\vec{B}$ and $\overrightarrow{v_2}$ and is directed as given by Right hand rule. As this force is to remain always perpendicular to $\overrightarrow{v_2}$, it does not perform any work and hence cannot change the magnitude of velocity. It changes only the direction of motion of the particle. Due to it, the charged particle is made to move on a circular path in the magnetic field

Here, magnetic field is shown perpendicular to the plane of paper directed inwards and particle is moving in the plane of paper. When the particle is at points R,S and P the

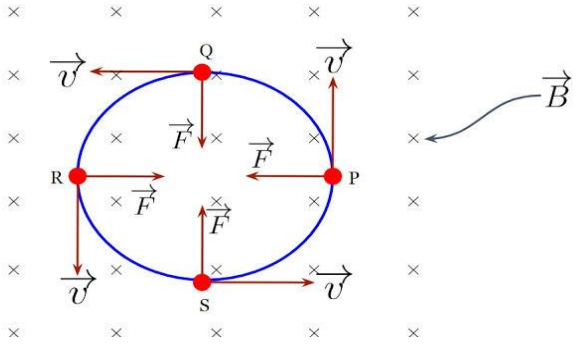
direction of magnetic force on the particle will be along RO, SO and PO respectively, ie, directed towards the center O of the circular path.

The force F on the charged particle due to magnetic field provides the required centripetal force ($\frac{mv_2^2}{r}$) necessary for motion along a circular path of radius r.

$$qv_2B = \frac{mv_2^2}{r}$$

$$r = \frac{mv_2}{qB}$$

Motion of Charged Particles in Uniform Magnetic Field



$$r = \frac{mv\sin\theta}{qB}$$

where r = radius of the helical path

The charged particle complete its circular path due to component velocity $v_2 = v\sin\theta$ perpendicular to the magnetic field.

For component velocity $v_1 = v\cos\theta$, there will be no force on the charged particle in the magnetic field, because the angle between $\vec{v_1}$ and $\vec{B}$ is zero. Thus the charged particle covers the linear distance in direction of the magnetic field with a constant speed $v\cos\theta$. Therefore, under the combined effect of the two component velocities, the charged particle in magnetic field will cover linear path as well as circular path i.e. the path of the charged particle will be helical, whose axis is parallel to the direction of magnetic field.

18. Correct option: (C)

$$r = \frac{mv_2}{qB}$$

$$\text{angular velocity } \omega = \frac{v_2}{r} = \frac{v_2}{\frac{mv_2}{qB}} = \frac{qB}{m}$$

$$\text{Time period } T = \frac{2\pi}{\omega} = \frac{2\pi}{\frac{qB}{m}} = \frac{2\pi m}{qB}$$

$$\text{Time period } T = \frac{2\pi m}{qB}$$

19. Correct option: (B)

The linear distance covered by the charged particle in the magnetic field in time equal to one revolution of its circular path (known as pitch of helix) will be

$$d = v_1T = \frac{2\pi mv\cos\theta}{qB}$$

20. (a)----(P), (R), (S) (b)----(P), (Q) (c)----(S) (d)----(P)

(a) $\vec{E} = 0, \vec{B} \neq 0$

Case1

When initial velocity $\vec{v}$ is parallel to $\vec{B}$

$\vec{F} = qvB\sin0° = 0$

So, the particle will not experience any force and will continue to move in its path without getting deflected (i.e straight).

Case 2

When initial velocity $\vec{v}$ is perpendicular to $\vec{B}$

$\vec{F} = qvBsin90° = qvB$

So, in this case the path followed by the particle is going to be circular.

Case 3

When initial velocity $\vec{v}$ is at some angle to $\vec{B}$

$\vec{F} = qvBsin\theta$

So, in this case the particle will follow a helical path.

(b) $\vec{E} \neq 0, \vec{B} = 0,$

Case1

When initial velocity $\vec{v}$ is parallel to $\vec{E}$

the particle will follow straight line path.

Case 2

When initial velocity $\vec{v}$ is at some angle to $\vec{E}$

it will follow parabolic path.

(c) $\vec{E} \neq 0, \vec{B} \neq 0$

$\vec{E}$ is parallel to $\vec{B}$

And initial velocity is perpendicular to both

Then the particle will follow helical path.

(d)

$\vec{E} \neq 0, \vec{B} \neq 0$

$\vec{E}$ is perpendicular to $\vec{B}$

And initial velocity is perpendicular to both

Then the particle will pass through the fields without any change in its velocity, and it goes undeviated.

That is particle moves along a straight line.

21. (a)----(S) (b)----(R) (c)----(Q) (d)----(P)

(a) A particle of charge 'q' and mass 'm' moving with uniform velocity 'v' enters normally into a magnetic field of induction 'B'. then radius 'r' of the circular path traced by it is given by

$r = \frac{mv}{qB}$--------(1)

we know that kinetic energy of the particle $E = \frac{1}{2}mv^2$

$$v = \sqrt{\frac{2E}{m}}$$

putting in equation 1

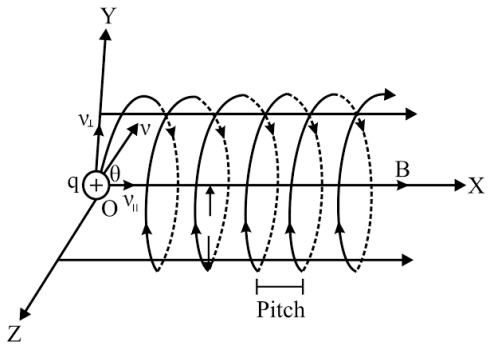
$$r = \frac{\sqrt{2mE}}{qB}$$

hence, radius is directly proportional to the square root of energy.

(b)Time period $T = \frac{2\pi m}{qB}$

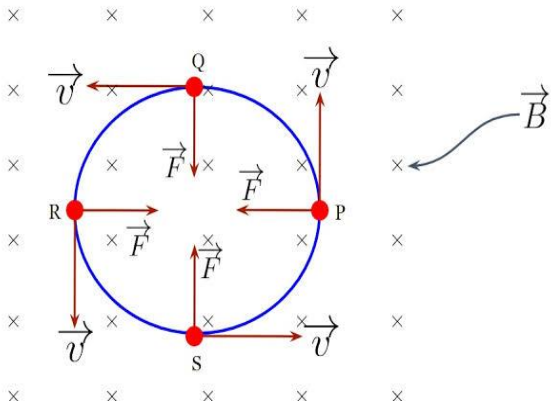
Hence, time period is independent of the velocity

(c) Suppose a particle of mass m and charge q, entering a uniform magnetic field induction $\vec{B}$ at O, with velocity $\vec{v}$ making an angle θ with the direction of magnetic field acting in the plane of paper.



Resolving into two rectangular components, we have:
v cosθ (= v_1) acts in the direction of the magnetic field
v sinθ (= v_2) acts perpendicular to the direction of magnetic field.
For component velocity v_2 , the force acting on the charged particle due to magnetic field is
$\vec{F} = q(\vec{v}_2 \times \vec{B})$
$F = qv_2 B\sin\theta$
$\theta = 90^\circ$
$F = qv_2 B\sin 90°$
$F = qv_2 B$
$F = qvB\sin\theta$
The direction of this force F is perpendicular to the plane containing $\vec{B}$ and $\overrightarrow{v_2}$ and is directed as given by Right hand rule. As this force is to remain always perpendicular to $\overrightarrow{v_2}$, it does not perform any work and hence cannot change the magnitude of velocity. It changes only the direction of motion of the particle. Due to it, the charged particle is made to move on a circular path in the magnetic field

Motion of Charged Particles in Uniform Magnetic Field



Here, magnetic field is shown perpendicular to the plane of paper directed inwards and particle is moving in the plane of paper. When the particle is at points R,S and P the direction of magnetic force on the particle will be along RO, SO and PO respectively, ie, directed towards the centre O of the circular path.
The force F on the charged particle due to magnetic field provides the required centripetal force ($\frac{mv_2^2}{r}$) necessary for motion along a circular path of radius r.
(d) specific charge is the ratio of the charge of an ion or subatomic particle to its mass (q/m)

$r = \frac{mv}{qB}$

angular frequency $\omega = \frac{v}{r} = \frac{v}{\frac{mv}{qB}} = \frac{qB}{m}$

hence, angular frequency is directly proportional to the specific charge.

22. Correct Answer is: $2\ ms^{-1}$

Given: mass of the particle $m=2 \times 10^{-5}$

Electric field $E=200\ N/C$

Magnetic field $B= 2\ T$

Charge on the particle $q= 10^{-6}$

Electric force= $F_E = qE = 10^{-6} \times 200 = 2 \times 10^{-4}\ N$

weight of the particle acting downwards= $mg = 2 \times 10^{-5} \times 9.8 = 1.96 \times 10^{-4} Kg$

as $F_E > mg$

so magnetic force has to act downward so that net force is zero

$F_m = (2 - 1.96) \times 10^{-4} = 4 \times 10^{-6} N$

Also $F_m = qvB = 4 \times 10^{-6}$

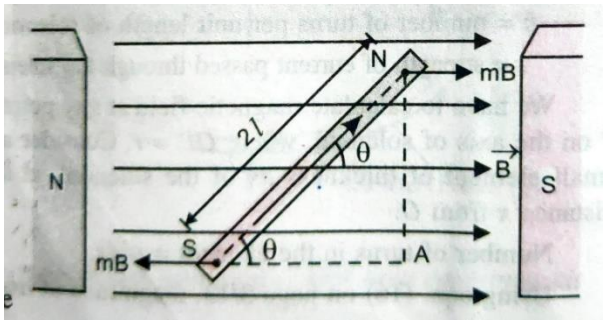
$10^{-6} \times v \times 2 = 4 \times 10^{-6}$

$v = \frac{4 \times 10^{-6}}{2 \times 10^{-6}} = 2\ ms^{-1}$

LEVEL – I

Torque:

1. Correct option: (D)



In the figure shown above a uniform magnetic field $\vec{B}$ is represented by equidistant parallel lines. NS is a bar magnet of length $2l$ and strength of each pole = m. The magnet is held at angle θ with the direction of $\vec{B}$.

Force on N-pole = m B, along $\vec{B}$

Force on S-pole = mB, opposite to $\vec{B}$

These forces being equal, unlike and parallel form a couple, which tends to rotate the magnet clockwise so as to align it along $\vec{B}$.

The bar magnet will only experience a torque, It will not experience any force.

Torque acting on the bar magnet

$\vec{\tau} = \vec{M} \times \vec{B}$

$\tau = MB \sin\theta$

where M = magnetic dipole moment

B= magnetic field

The direction of $\vec{\tau}$ is perpendicular to the plane containing $\vec{M}$ *and* $\vec{B}$ and is given by right-handed screw rule. That is along the suspension wire.

Special cases:

If

$\theta=90^{\circ}$

τ = MBsin90°= MB= maximum torque

It means a magnet if placed at an angle of 90° to the direction of magnetic field, it experiences maximum torque.

2. Correct option: (A)

Torque acting on the bar magnet/moment of the couple

$\tau = MB\sin\theta$--------(1)

where M = magnetic dipole moment

B= magnetic field

Rate of change of torque is given by differentiation equation 1 w.r.t time

$\frac{d\tau}{d\theta} = \frac{d(MB\sin\theta)}{d\theta}$

$\frac{d\tau}{d\theta} = MB\cos\theta$----------(2)

For equation 2 to be maximum

$\frac{d\tau}{d\theta}$ = maximum

$MB\cos\theta$= maximum

if

$\cos\theta = 1$

Or

$\theta = 0°$

3. Correct option: (C)

Torque acting on the bar magnet/moment of the couple

$\tau = MB\sin\theta$

where M = magnetic dipole moment

B= magnetic field

Given: B = 320 weber/m^2

$\theta=30^{\circ}$

length of the magnet l= 10cm = 0.1 m

pole strength m = 20 A-m

we know that

M = m l

τ = mlBsin30°

$\tau = 20 \times 0.1 \times 320 \times \frac{1}{2}$

$\tau = 20 \times 0.1 \times 320 \times \frac{1}{2}$= 320 Nm

$\tau = 320$ Nm

4. Correct option: (B)

Torque acting on the bar magnet/moment of the couple

$\vec{\tau} = \vec{M} \times \vec{B}$

$\tau = MB\sin\theta$

where M = magnetic dipole moment

B= magnetic field

Given: B = 3×10^{-2} T

$\theta=30^{\circ}$

M =4.8×10^{-2} J/T

$\tau = MB\sin30°$

$\tau = 4.8 \times 10^{-2} \times 3 \times 10^{-2} \times \frac{1}{2}$

$\tau = 7.2 \times 10^{-4}$ Nm

hence, torque acting on the needle = 7.2×10^{-4} Nm

5. Correct option: (B)

Torque acting on the bar magnet/moment of the couple

$\vec{\tau} = \vec{M} \times \vec{B}$

where M = magnetic dipole moment

B= magnetic field

given: $\vec{B} = 0.5\hat{i} + 3.0\hat{j}$ T

$\vec{M} = 50\hat{i}\ A - m^2$

$\vec{\tau} = 50\hat{i} \times (0.5\hat{i} + 3.0\hat{j})$

$\vec{\tau} = 50 \times 3(\hat{i} \times \hat{j})$

$\vec{\tau} = 150(\hat{k})$ N-m

6. Correct option: (A)

Torque acting on the bar magnet

$\tau = MB\sin\theta$

where M = magnetic dipole moment

B= magnetic field

If

$\theta = 90^{\circ}$

$\tau = MB\sin90°$

maximum torque= MB

half of maximum torque $\tau' = \frac{MB}{2}$

$\tau' = MB\sin\theta$

$\frac{MB}{2} = MB\sin\theta$

$\frac{1}{2} = \sin\theta$

$\sin 30° = \sin\theta$

$\theta = 30°$

angle of rotation = $90° - 30° = 60°$

7. Correct option: (B)

Torque acting on a current carrying loop in magnetic field is given by

$\vec{\tau} = \vec{M} \times \vec{B}$

$\tau = MB\sin\theta$----------(1)

where M = magnetic dipole moment

B= magnetic field

θ = angle made by the normal of the loop with the magnetic field

Now

Magnetic dipole moment M = NIA

Where N = number of turns

I = current

A = area of the loop

Equation 1 becomes

$\tau = NIAB\sin\theta$----------(2)

given:

$\theta = 30^{\circ}$
N= 100
current I = 2 A
area A= $10 \text{ x} 15 \text{ cm}^2 = 150 \text{ x } 10^{-4} \text{ m}^2$
B= $2 \text{ x } 10^{-3} \text{ Wb/m}^2$
equation 2 becomes
$\tau = 100 \text{ x } 2 \text{ x } 150 \text{ x } 10^{-4} \text{ x } 2 \text{ x } 10^{-3} \sin 30°$
$\tau = 100 \text{ x } 2 \text{ x} 150 \text{ x } 10^{-4} \text{ x } 2 \text{ x } 10^{-3} \text{ x} \frac{1}{2}$
$\tau = 30000 \text{ x } 10^{-7}$
$\tau = 3 \text{ x } 10^{-3} \text{ N} - \text{m}$

Work done:

8. Correct option: (D)

Torque acting on the bar magnet
$\tau = MB \sin\theta$
where M = magnetic dipole moment
B= magnetic field
θ= angle made by bar magnet with the magnetic field
This torque tends to align the dipole in the direction of the field. Work has to be done in rotating the dipole against the action of the torque. This work done is stored in the magnetic dipole as potential energy of the dipole.
Work done = $-MB(\cos\theta_2 - \cos\theta_1)$----------(1)
$\theta_1 = 0^{\circ}$
$\theta_2 = 180^{\circ}$
M= 1.5 J/T
B = 0.22 T
Putting the values in equation 1
Work done = $-1.5 \text{ x } 0.22(\cos 180° - \cos 0°)$
Work done = $-0.33(-1 - 1)$
Work done = $-0.33(-2)$
Work done = 0.66 J
Torque acting on the bar magnet
$\tau = MB \sin\theta$
$\tau = 1.5 \text{ x } 0.22 \text{ x } \sin 180° = 0$
$\tau = 0$ Nm

9. Correct option: (B)

Torque acting on the bar magnet
$\tau = MB \sin\theta$
where M = magnetic dipole moment
B= magnetic field
θ= angle made by bar magnet with the magnetic field
This torque tends to align the dipole in the direction of the field. Work has to be done in rotating the dipole against the action of the torque. This work done is stored in the magnetic dipole as potential energy of the dipole.
Work done = potential energy = $-MB(\cos\theta_2 - \cos\theta_1)$----------(1)
let
$\theta_1 = 90^{\circ}$

$\theta_2 = \theta$

potential energy $= -MB\cos\theta$

given: M = 5 A-m^2

B = 2T

$\theta = 60^\circ$

potential energy $= -5 \times 2 \times \cos 60°$

potential energy $= -5 \times 2 \times \frac{1}{2}$

potential energy $= -5$ J

Magnetic flux:

10. Correct option: (C)

The SI unit of magnetic flux is Weber (Wb). One weber is the amount of magnetic flux over an area of 1 metre^2 held normal to a uniform magnetic field of one tesla.

11. Correct option: (B)

The magnetic flux through any surface held in a magnetic field $\vec{B}$ is measured by the total number of magnetic lines of force crossing the surface normally.

For a uniform magnetic field $\vec{B}$ crossing the plane of area $\vec{A}$ at an angle θ with the normal to the plane, magnetic flux is

$\phi = \vec{B}.\vec{A} = BA\cos\theta$

given: A = 0.5 m^2

B = 4.0 Wb/m^2

$\theta = 60°$

$\phi = BA\cos\theta$

$\phi = 4 \times 0.5 \times \cos 60°$

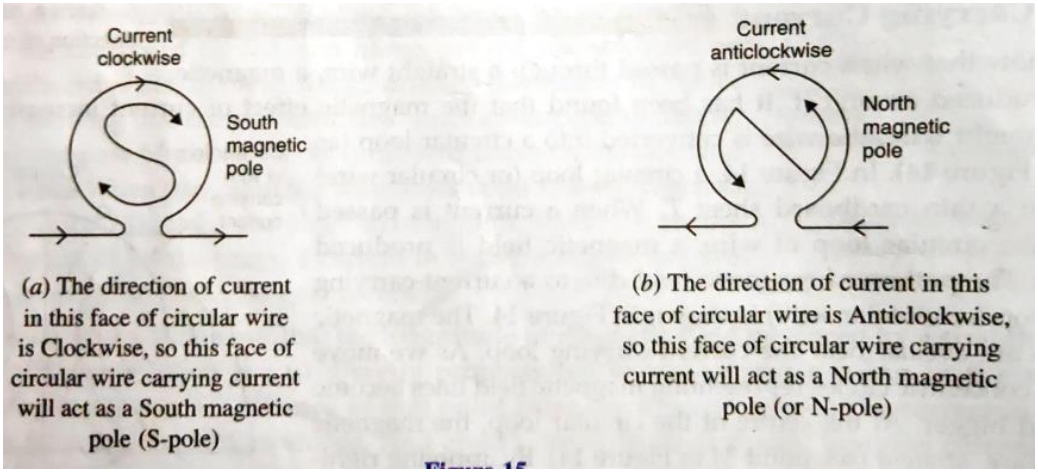
$\phi = 2 \times \frac{1}{2}$ Wb

$\phi = 1$ Wb

12. Correct option: (A)

When current is maintained in a coil, the magnetic field is produced in it, where one face behaves like a north pole(current in anticlockwise direction), and the other as the south pole(current in clockwise direction).

Thus, it behaves as a bar magnet.

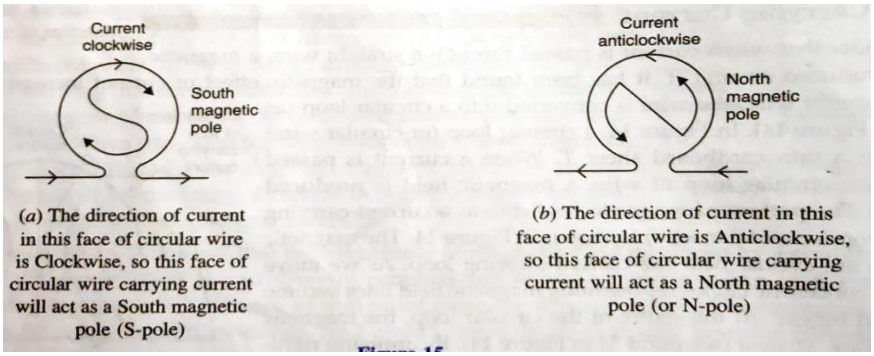


(*a*) The direction of current in this face of circular wire is Clockwise, so this face of circular wire carrying current will act as a South magnetic pole (S-pole)

(*b*) The direction of current in this face of circular wire is Anticlockwise, so this face of circular wire carrying current will act as a North magnetic pole (or N-pole)

13. Correct option: (B)

If the direction of current through the coil is anticlockwise, then that face has a north polarity.



(*a*) The direction of current in this face of circular wire is Clockwise, so this face of circular wire carrying current will act as a South magnetic pole (S-pole)

(*b*) The direction of current in this face of circular wire is Anticlockwise, so this face of circular wire carrying current will act as a North magnetic pole (or N-pole)

Faraday's law and Lenz's law:

14. Correct option: (C)

According to faraday

• Whenever the amount of magnetic flux linked with a circuit changes, an e.m.f. is induced in the circuit. The induced e.m.f. lasts so long as the change in magnetic flux continues.

• The magnitude of e.m.f. induced in a circuit is directly proportional to the rate of change of magnetic flux linked with the circuit.

$\text{induced emf (e)} = -\frac{d\phi}{dt}$

e.m.f is induced in a circuit, whenever the amount of magnetic flux linked with the circuit is changed

$\phi = BA\cos\theta$

the magnetic flux can be changed by changing the B, A or θ individually, or any two of them or changing all three of them together.

For option A, the flux changes as the magnetic field is changed, thus, emf is generated.

For option B, the flux also changes since the area of the loop is changing. When θ is changed again flux is changed so option D is incorrect.

Now option C, says the magnetic field is non-uniform in space, but we are not changing magnetic field, area of the loop or the angle with direction B.

Hence, the induced emf will be zero in case of option C

15. Correct option: (D)

According to Lenz's law, the polarity of the induced emf is such that it opposes the change in magnetic flux responsible for its production.

This law gives us the direction of current induced in a circuit.

16. Correct option: (A)

According to Lenz's law, the polarity of the induced emf is such that it opposes the change in magnetic flux responsible for its production.

Hence, the force will act on the strip to oppose the motion.

17. Correct option: (D)

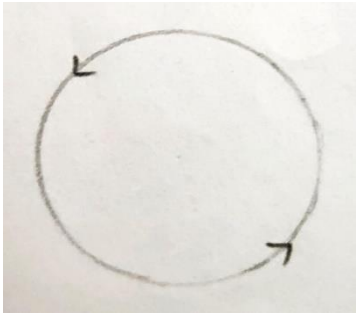
$\text{magnetic field B} \propto \frac{1}{\text{distance r}}$

If electron is moving from left to right distance decreases hence, B increases.

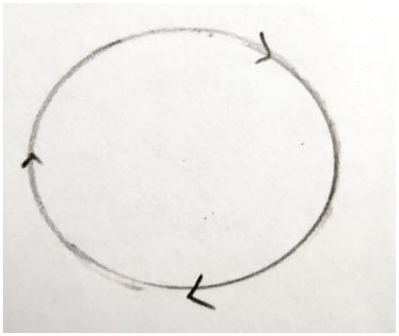
current is from right to left, the flux linked with the loop (which is into the page according to right hand rule) will first increase.

According to Lenz's law, the polarity of the induced emf is such that it opposes the change in magnetic flux responsible for its production.

So the induced current in the loop will be anticlockwise to decrease the magnetic flux in order to oppose the increase in magnetic flux.



as the electron passes by distance increases, hence, B decreases
current is from right to left, the flux linked with the loop (which is into the page according to right hand rule) will now decrease.
According to Lenz's law, the polarity of the induced emf is such that it opposes the change in magnetic flux responsible for its production.
So the induced current in the loop will be clockwise to increase the magnetic flux in order to oppose the decrease in magnetic flux.



18. Correct option: (D)

induced emf (e) $= -\frac{d\phi}{dt}$

e.m.f is induced in a circuit, whenever the amount of magnetic flux linked with the circuit is changed

magnetic flux $\phi = BA\cos\theta$

for N turns

induced emf (e) $= N\frac{d\phi}{dt} = N\frac{d(BA\cos\theta)}{dt}$

given: N = 500

side of square = 10 cm = 0.1 m

area of the square = 0.1 x 0.1 = $0.01m^2$

$\frac{dB}{dt} = 1T/s$

$\theta = 0°$

induced emf (e) = 500 x 0.01 x cos0° x1

induced emf (e) = 5 V

19. Correct option: (A)

induced emf (e) $= -\frac{\phi_2 - \phi_1}{dt}$

e.m.f is induced in a circuit, whenever the amount of magnetic flux linked with the circuit is changed.

$\phi_2 = 0$ because finally coil is removed from magnetic field

magnetic flux ϕ_1 = NBA cosθ
given:
given: N = 50
area of the coil =$100cm^2 = 100x10^{-4}m^2$
induced emf e = 0.1 V
B = $2x10^{-2}$ Tesla
θ = 0°
putting the values in above equation

$$0.1 = \frac{50\ x2x10^{-2}x100x10^{-4}\ cos0°}{t}$$

$$0.1 = \frac{10^{-2}}{t}$$

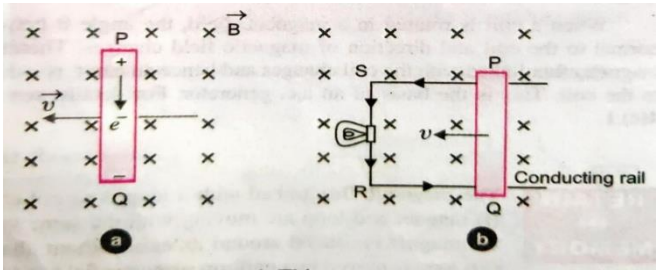
$$t = \frac{10^{-2}}{0.1}$$

t = 0.1 s

20. Correct option: (B)

When a conducting rod is moved through a constant magnetic field, an e.m.f. is induced in the rod. This is called motional electromotive force, because it arises due to motion of charges through a magnetic field.

Figure below shows a metal rod PQ of length, L being moved with a velocity $\vec{v}$ to the left in a uniform magnetic field $\vec{B}$ directed into the plane of the paper. Each charge q within the rod moves with velocity $\vec{v}$ and experiences a magnetic force, F= qvB. The mobile free electrons in the rod are driven in the rod from P to Q, according to Fleming's left hand rule. An equal amount of positive charge appears at the end P of the rod. This process goes on till the attractive electric force between the accumulated charges is balanced by the magnetic force that separates the charges.



The separated charges on the ends of rod PQ give rise to an induced e.m.f. which is called motional electromotive force. It exists as long as the rod moves. When the rod stops moving, magnetic force vanishes. The attractive electric force reunites the separated charges at the ends and motional e.m.f. disappears. The expression for motional e.m.f. can be obtained from Lorentz force equation, F = qvB. This is the force experienced by any arbitrary charge + q in the conductor, moving with speed v in magnetic field B.

Work done in moving the charge + q from Q to P
W= F x QP = (qvB)x L
e.m.f. induced= work done per unit charge.

$$e = \frac{W}{q} = \frac{qvBL}{q} = BvL$$

hence, potential difference between the ends of the rod = BvL

21. Correct option: (A)

When a conducting rod is moved through a constant magnetic field, an e.m.f. is induced in the rod. This is called motional electromotive force, because it arises due to motion of charges through a magnetic field.

In case of insulating rod there is no motion of charges, hence the induced emf = zero

22. Correct option: (B)

induced emf (e) $= \frac{d\phi}{dt}$

$\phi = 3t^2 + 4t + 9$

$e = \frac{d(3t^2+4t+9)}{dt} = 6t + 4$

at t = 2 sec

$e = 6x2 + 4 = 12 + 4 = 16V$

23. Correct option: (C)

For a uniform magnetic field $\vec{B}$ crossing the plane of area $\vec{A}$ at an angle θ with the normal to the plane, magnetic flux is

$\phi = \vec{B}.\vec{A} = BA\cos\theta$

given: radius r = 0.2 m

$A = \pi r^2 = \pi \times 0.2 \times 0.2\ m^2$

$B = \frac{1}{\pi}\ Wb/m^2$

$\theta = 60°$

$\phi = BA\cos\theta$

$\phi = \frac{1}{\pi} x\ \pi \times 0.2 \times 0.2 \times \cos 60°$

$\phi = 0.2 \times 0.2 \times \frac{1}{2}$

$\phi = 0.2 \times 0.1 = 0.02Wb$

24. Correct option: (B)

When a loop is moved in a region of uniform magnetic field, and the whole of the loop remains in this region, the magnetic flux linked with the loop does not change, therefore the induced e.m.f in this case is zero.

induced emf (e) $= -\frac{d\phi}{dt} = 0$

magnetic flux through the loop is non zero, induced emf is zero.

25. Correct option: (B)

Force acting on the conductor carrying current placed in a magnetic field

$F = ILB\sin\theta$---------(1)

where I = current carried by the conductor

L = length of the conductor

B = magnetic field

θ = angle between the conductor and magnetic field

given: θ = 90°

B= 0.4 T

F= 0.5 N

L = 0.5m

Putting the values in equation 1

$0.5 = I \times 0.4 \times 0.5 \sin 90°$

$0.5 = I \times 0.20$

$\frac{0.5}{0.20} = I$

2.5 A = I

Current = 2.5 A

LEVEL-II

1. Correct option: (A)

Torque acting on the bar magnet/moment of the couple

$\tau = MB\sin\theta$

where M = magnetic dipole moment

B= magnetic field

Given:

$\tau = 2 \text{ x } 10^{-5}\ N - m$

$\theta = 30^{\circ}$

length of the magnet l= 20cm = 0.2 m

we know that

$M = m\ l$

$\tau = ml B\sin 30°$

force = mB (pole strength x magnetic field)

$2 \text{ x } 10^{-5} = F \text{ x } 0.2 \text{ x} \frac{1}{2}$

$F = \frac{2 \text{ x } 10^{-5}}{0.1}$

$F = 2x10^{-4}\ N$

2. Correct option: (C)

Work done $= -MB(\cos\theta_2 - \cos\theta_1)$----------(1)

$\theta_1 = 0^{\circ}$

$\theta_2 = 60^{\circ}$

Work done $= 8 \text{ x} 10^{-5}$ J

Putting the values in equation 1

$8 \text{ x} 10^{-5}$= -MB($\cos 60° - \cos 0°$)

$8 \text{ x} 10^{-5}$= -MB$\left(\frac{1}{2} - 1\right)$

$8 \text{ x} 10^{-5}$= -MB$\left(-\frac{1}{2}\right)$

$2 \text{ x } 8 \text{ x} 10^{-5}$= MB

$16x10^{-5}$= MB

Now

Work done in rotating through further 30°

Work done $= -MB(\cos\theta_2 - \cos\theta_1)$----------(1)

$\theta_1 = 60^{\circ}$

$\theta_2 = 60° + 30° = 90^{\circ}$

Putting the values in equation 1

work done= $-16x10^{-5}$($\cos 90° - \cos 60°$)

work done= $-16x10^{-5}\left(0 - \frac{1}{2}\right)$

work done= $8x10^{-5}$ J

3. Correct option: (B)

Work done in rotating from east-west to north south direction

Work done $= -MB(\cos\theta_2 - \cos\theta_1)$----------(1)

$\theta_1 = 0^{\circ}$

$\theta_2 = 90^{\circ}$

$M = 40A - m^2$

$B = 30\ \mu T = 30\ x10^{-6} T$

Putting the values in equation 1

work done$= -40x30\ x10^{-6}(\cos 90° - \cos 0°)$

work done$= -40x30\ x10^{-6}(0 - 1)$

work done$= 12x10^{-4}$

work done$= 1.2x10^{-3} = 1.2mJ$

4. Correct option: (D)

induced emf (e) $= \frac{d\phi}{dt}$

$\phi = 6t^2 - 5t + 1$

$e = \frac{d(6t^2 - 5t + 1)}{dt} = 12t - 5$

at t = 0.25 sec

$e = 12x0.25 - 5 = 3 - 5 = -2V$

induced current $= \frac{emf}{R} = \frac{-2}{10} = -0.2\ A$

5. Correct option: (C)

According to faraday's laws of electromagnetic induction

• Whenever the amount of magnetic flux linked with a circuit changes, an e.m.f. is induced in the circuit. The induced e.m.f. lasts so long as the change in magnetic flux continues.

• The magnitude of e.m.f. induced in a circuit is directly proportional to the rate of change of magnetic flux linked with the circuit.

induced emf (e) $= -\frac{d\phi}{dt}$

e.m.f is induced in a circuit, whenever the amount of magnetic flux linked with the circuit is changed

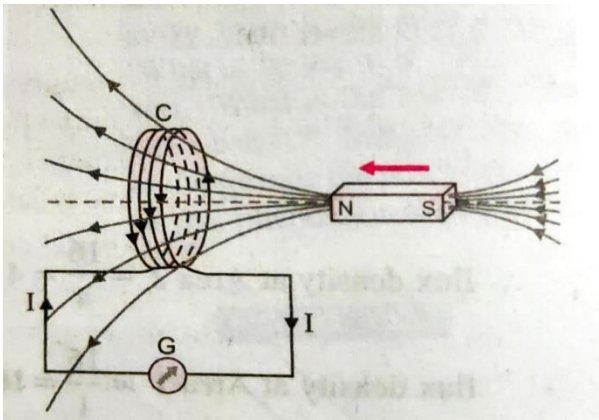
6. Correct option: (D)

When a loop is moved in a region of uniform magnetic field, and the whole of the loop remains in this region, the magnetic flux linked with the loop does not change, therefore the induced e.m.f in this case is zero.

induced emf (e) $= -\frac{d\phi}{dt} = 0$

7. Correct option: (C)

• When north pole of a bar magnet is pushed towards the coil, the galvanometer shows a



sudden deflection, indicating that current is induced in the coil.

• The galvanometer deflection is temporary. It lasts as long as the bar magnet is in motion. There is no deflection when the bar magnet is held stationary, anywhere, even within the coil.

• When the magnet is moved away from the coil, the galvanometer shows deflection in the opposite direction, indicating reversal in the direction of induced current.

8. Correct option: (D)

According to Lenz's law, the polarity of the induced emf is such that it opposes the change in magnetic flux responsible for its production.

• When the switch S is closed, a clockwise current I_P flows in P, this current is increasing with time

Due to this increasing current, the magnetic flux linked with the coil P will increase, and magnetic flux linked with coil Q will also increase. Current I_{Q_1} will be induced in the coil Q and according to lenz's law its direction will be to oppose the increase in the flux, that is anticlockwise.

• When the switch S is opened, a clockwise current I_P flows in P, this current is decreasing with time.

Due to this decreasing current, the magnetic flux linked with the coil P will decrease, and magnetic flux linked with coil Q will also decrease. Current I_{Q_2} will be induced in the coil Q and according to lenz's law its direction will be to oppose the decrease in the flux, that is clockwise.

Flux, electromagnetic induction:

9.Correct options(A), (B), (D)

(A) all the points on the ring are at same potential

(B) Given:

$\frac{dB}{dt} = \alpha$

Area $A = \pi a^2$

induced emf $(e) = \frac{d\phi}{dt} = \frac{dBA}{dt} = A\frac{dB}{dt} = \pi a^2\alpha$

(D) from the maxwell's equation

$\oint \vec{E}.\vec{dl} = -\frac{d\phi}{dt}$

$E \times 2\pi a = \pi a^2\alpha$

$E = \frac{\pi a^2\alpha}{2\pi a} = \frac{a\alpha}{2}$

10. Correct options(A), (B), (C)

$\phi = 4t^n + 6$

induced emf $(e) = -\frac{d\phi}{dt} = -\frac{d(4t^n+6)}{dt}$

$e = -4nt^{n-1}$

if n=1, |e|=4

if n>1, $|e|=4nt^{n-1}$

if $n-1>1$, then e increases

else if $n-1<1$, e decreases

11. Correct options(B), (C)

According to Lenz's law, the polarity of the induced emf is such that it opposes the change in magnetic flux responsible for its production.Since magnet is moving towards B loop magnetic flux is increasing in loop B and decreasing in loop A. so if observer is at left side

then, current induced in loop A will be in clockwise direction and in loop B in anticlockwise direction. And hence, the loops will repel each other.

12. Correct options(A), (C)

According to Lenz's law, the polarity of the induced emf is such that it opposes the change in magnetic flux responsible for its production.

A north pole will be formed at the incoming faces of both the coils with current flowing in anticlockwise direction.

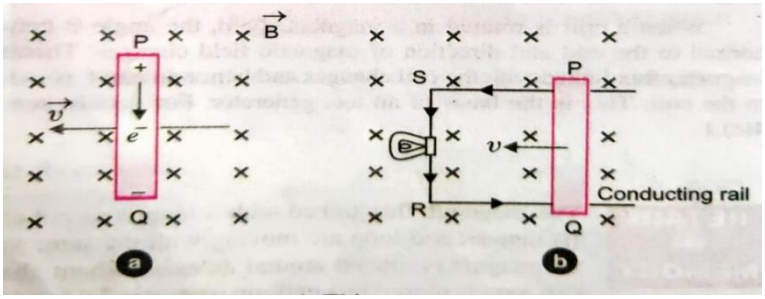
As both the coils have current flowing in the same directions, they will attract.

Also, the induced current(emf) is directly proportional to the rate of change of magnetic flux. Therefore, as change in flux associated with coil A is more (it is nearer to the magnet), induced current will be more in coil A.

13. Correct options(A), (B), (D)

When a conducting rod is moved through a constant magnetic field, an e.m.f. is induced in the rod. This is called motional electromotive force, because it arises due to motion of charges through a magnetic field.

Figure below shows a metal rod PQ of length, L being moved with a velocity $\vec{v}$ to the left in a uniform magnetic field $\vec{B}$ directed into the plane of the paper. Each charge q within the rod moves with velocity $\vec{v}$ and experiences a magnetic force, F= qvB. The mobile free electrons in the rod are driven in the rod from P to Q, according to Fleming's left hand rule. An equal amount of positive charge appears at the end P of the rod. This process goes on till the attractive electric force between the accumulated charges is balanced by the magnetic force that separates the charges.



The separated charges on the ends of rod PQ give rise to an induced e.m.f. which is called motional electromotive force. It exists as long as the rod moves. When the rod stops moving, magnetic force vanishes. The attractive electric force reunites the separated charges at the ends and motional e.m.f. disappears. The expression for motional e.m.f. can be obtained from Lorentz force equation, F = qvB. This is the force experienced by any arbitrary charge + q in the conductor, moving with speed v in magnetic field B.

Work done in moving the charge + q from Q to P

W= F x QP = (qvB)x L

e.m.f. induced= work done per unit charge.

$$e = \frac{W}{q} = \frac{qvBL}{q} = BvL$$

hence, potential difference between the ends of the rod/ emf induced in the loop = BvL

$$\text{induced current (i)} = \frac{e}{R} = \frac{BvL}{R}$$

magnitude of force required to move conducting rod at constant speed = BiL = $\frac{B^2vL^2}{R}$

Hence an external force equal to this magnitude is required to be applied for the rod to move with constant velocity.

Rate of work done by external agent is:

$\frac{dW}{dt} = \frac{dFx}{dt} = BiL\frac{dx}{dt} = BiLv$

Thermal power dissipated in the resistor =e x i=(BvL)i

clearly both are equal
Since magnetic field does not do any work on the conducting rod, from conservation of energy, the thermal power dissipated in the resistor is equal to the rate of work done by an external person pulling the rod.
(B) If applied external force is doubled, the rod will experience a net force and hence acceleration. As a result velocity increases.
(D) Thermal power dissipated in the resistor =e x i=(BvL)i
induced current (i) $= \frac{e}{R} = \frac{BvL}{R}$
power = $\frac{B^2v^2L^2}{R}$
On doubling 'R', current becomes half and hence required power becomes half.

14. Correct option: (B)

magnetic flux $\phi = BA = Blx$
where B = magnetic field
l =length of the bar
x= distance moved
given: v $= \frac{dx}{dt} = 5\ cms^{-1} = 0.05\ ms^{-1}$
$\frac{dB}{dt} = 0.2\ T/s$
$l = 5cm = 0.05\ m$
X = 5 cm = 0.05 m
induced emf (e) $= \frac{d\phi}{dt} = \frac{dBlx}{dt} = l\left(\frac{dBx}{dt}\right)$
induced emf (e) $= l\left(B\frac{dx}{dt} + x\frac{dB}{dt}\right)$
induced emf (e) $= 0.05(0.1\ x - 0.05 + 0.05\ x\ 0.2)$
induced emf (e) $= 0.05(-0.005 + 0.01)$
induced emf (e) $= 0.05(0.005)$
induced emf (e) $= 0.00025 = 250\mu V$
induced current (i) $= \frac{e}{R} = \frac{250\ x\ 10^{-6}}{10^{-4}} = 250x10^{-2} = 2.5A$

15. Correct option: (A)

Given: resistance R $=10^{-4}$
induced emf (e) $= 0.00025 = 250\ x\ 10^{-6} = 250\mu V$
induced current (i) $= \frac{e}{R} = \frac{250\ x\ 10^{-6}}{10^{-4}} = 250x10^{-2} = 2.5A$

16. Correct option: (A)

given: $l_1 = 40cm, l_2 = 20cm$
$A = 40\ x\ 20\ cm^2 = 800\ cm^2 = 0.08m^2$
$\frac{dB}{dt} = 2\ T/s$
magnetic flux $\phi = BA$
induced emf (e) $= \frac{d\phi}{dt} = \frac{dBA}{dt} = A\frac{dB}{dt} = 0.08\ x\ 2 = 0.16V$

induced current (i) $= \frac{e}{R} = \frac{0.16}{1} = 0.16$A clockwise

17. Correct option: (B)

At t= 2s

B = 4T

Distance moved = speed x time

= 5cm/s x 2

= 10 cm

$l_1 = 30\text{cm}, l_2 = 20\text{cm}$

A = 30 x 20 cm^2 = 600 cm^2 = 0.06m^2

$\frac{dB}{dt} = 2$ T/s

magnetic flux ϕ = BA

given: $\frac{dA}{dt} = \frac{dl_1 l_2}{dt} = l_2 \frac{dl_1}{dt} = 20 \text{ x } 5cm^2/s = 100cm^2/s = 0.01m^2/s$

induced emf (e) $= \frac{d\phi}{dt} = \frac{dBA}{dt}$

induced emf (e) $= \left(B\frac{dA}{dt} + A\frac{dB}{dt}\right)$

induced emf (e) $= (4x - 0.01 + 0.06x2)$

induced emf (e) $= (-0.04 + 0.12)$

induced emf (e) = 0.08V

18. Correct option: (C)

Length of the loop after t sec= 0.2 + 2(0.4 – v t)

Length of the loop after 2 sec= 0.2 + 2(0.4 – 0.05 x 2)= 0.2+ 2(0.4 – 0.1)

=0.2+ 2(0.3)

= 0.2 + 0.6

= 0.8m

Resistance = 1 ohm/m x 0.8m = 0.8 ohm

At t = 2 s

induced current (i) $= \frac{e}{R} = \frac{0.08}{0.8} = 0.1$A

magnitude of force required to move conducting rod at constant speed = BIL

= 4 X 0.1 X 0.2

=0.08 N

19. Correct Matching is: (a)----(Q), (S) (b)----(P), (R) (c)----(R) (d)----(S)

(a)If current is increased, magnetic flux in the loop will increase in inside direction, then due to Lenz's law induced emf in the loop will be in anticlockwise direction. Due to this current, the current in the nearer side of loop to the wire will be in opposite direction to that of wire. Hence, there will be repulsion.

(b) If current is decreased, magnetic flux in the loop will decrease in inside direction, then due to Lenz's law induced emf in the loop will be in clockwise direction. Due to this current, the current in the nearer side of loop to the wire will be in same direction to that of wire. Hence, there will be attraction.

(c) If loop is moved away from the wire, then flux decreases in inside direction, then due to Lenz's law induced emf in the loop will be in clockwise direction. Due to this current, the current in the nearer side of loop to the wire will be in same direction to that of wire. Hence, there will be attraction.

(d) If loop is moved towards the wire, magnetic flux in the loop will increase in inside direction, then due to Lenz's law induced emf in the loop will be in anticlockwise direction.

Due to this current, the current in the nearer side of loop to the wire will be in opposite direction to that of wire. Hence, there will be repulsion.

20. Correct Matching is: (a)----(R) (b)----(Q) (c)----(P) (d)----(S)

(a) induced emf (e) $= -\frac{d\phi}{dt}$

e x dt $= -d\phi$

emf = current x resistance

emf = I x R

R x Idt $= -d\phi$

R x dq $= -d\phi$

SI unit of R = ohm

SI unit of dq = coulomb

Unit of flux = coulomb-ohm

(b) The magnitude of e.m.f. induced in a circuit is directly proportional to the rate of change of magnetic flux linked with the circuit.

induced emf (e) $= -\frac{d\phi}{dt}$

rate of change of flux is measure in volts.

(c) SI unit of magnetic field = Tesla

(d) potential energy $= -MB(\cos\theta_2 - \cos\theta_1)$

SI unit of Magnetic moment is ampere metre^2 = Am^2

SI unit of magnetic field B is Wb/m^2

Hence SI unit of potential energy = Am^{2}x$\frac{Wb}{m^2}$ = Ampere – weber

5. Modern Physics

Solutions

LEVEL–I

Atomic Model

1. Correct option: (A)

Solution: Given, the Rutherford α-particle experiment shows that most of the alpha-particles pass through almost un scattered while some are scattered through large angles.

Since most of the α-particles pass undeflected, this hints that most of the space inside the atom is empty. Also, since very few α-particles are scattered through large angles, this suggests that the whole mass of the atom is concentrated in a small centre. This is called as nucleus. Those α-particles that pass very near to the nucleus will suffer greatest deflection.

Therefore, Rutherford α-particle experiment gives the information that the whole mass of the atom is concentrated in a small centre called nucleus.

2. Correct option: (C)

Solution: According to the Rutherford's atomic model, the electrons inside the atom revolves around the nucleus in circular orbits. The electrons are thus not stationary.

3. Correct option: (C)

Solution: We know, Rutherford α-particle experiment shows that most of the alpha-particles pass through almost un scattered while some are scattered through large angles.

Since very few α-particles are scattered through large angles, this suggests that the whole mass of the atom is concentrated in a small centre called nucleus.

Therefore, Rutherford α-particle experiment showed that the atoms have nucleus.

4. Correct option: (D)

Solution: We know, Bohr's atomic model of atom was based on the following assumptions:

- Electrons in a quantized orbit will not radiate energy.
- The nucleus is of infinite mass and is at rest.
- Mass of electron remains constant.

5. Correct option: (A)

Solution: The concept of stationary orbits was proposed by Neil Bohr. Electrons in a stationary orbit don't not radiate energy.

Structure of Nucleus

6. Correct option: (A)

Solution: We know, the size of an atom is of the order of $10^{-10}\,\text{m}$.

7. Correct option: (D)

Solution: We know, different elements are categorized by the number of protons it contains, which is also termed as principal quantum number.

Now, if in nature there may not be an element for which the principal quantum number n is greater than 4, then the total possible number of elements will be 4.

8. Correct option: (D)

Solution: We know, the kinetic energy of an electron revolving around a nucleus is half of its potential energy.

9. Correct option: (D)

Solution: Stars generates its energy by nuclear fusion of nuclei like Hydrogen and Helium. Therefore, energy generation in stars is mainly due to fusion of light nuclei.

10. Correct option: (D)

Solution: We know, the mass R of a nucleus is related to its mass number A by:

$$R = a_0 A^{1/3}$$

where a_0 is a constant.

Given, the mass number of He is 4 and that for sulphur is 32.

Hence, the radius of sulphur nucleus is larger than that of helium, by times

$$\frac{R_1}{R_2} = \frac{a_0 A_1^{1/3}}{a_0 A_2^{1/3}}$$

$$\frac{R_1}{R_2} = \frac{a_0 32^{1/3}}{a_0 4^{1/3}}$$

$$\frac{R_1}{R_2} = 2$$

Therefore, if the mass number of He is 4 and that for sulphur is 32, then the radius of sulphur nucleus is larger than that of helium, by 2 times.

11. Correct option: (A)

Solution: Atom bomb consists of two pieces of ${}_{32}U^{235}$ and a source of neutrons.

12. Correct option: (C)

Solution: We know, in the notation ${}^{A}_{Z}X$, X denotes the element, Z the number of protons it contains, which also equals the number of electrons in the neutral atom of the element, and A is the mass number, the total number of neutrons and protons in the nucleus of the element.

Therefore, the sodium nucleus ${}^{23}_{11}Na$ contains 11 protons, 11 electrons, and 23-11=12 neutrons.

13. Correct option: (A)

Solution: The particles which can be added to the nucleus of an atom without changing its chemical properties are called neutrons. Adding a proton changes the element, and hence the chemical properties. Adding an electron also changes the chemical properties. But adding neutron don't affect chemical properties.

14. Correct option: (C)

Solution: We know, nuclear fusion powers the stars. During the nuclear fusion reaction two light nuclei combine to give a heavier nucleus and possibly other products. This releases a lot of energy.

15. Correct option: (A)

Solution: Moderator is required in a nuclear reactor to slow down the neutrons produced during the fission reaction so that the chain reaction can be sustained.

Heavy water is used as a moderator in nuclear reactors. It has very high moderating ratio.

16. Correct option: (B)

Solution: We know, the mass of the nucleus is less than the sum of the mass of the constituents of the nucleus, that is protons and neutrons. Nuclear binding energy is equivalent to this mass defect of nucleus.

17. Correct option: (A)

Solution: The binding energy of nucleus is a measure of its stability. Higher the binding energy, more stable the nucleus is.

18. Correct option: (C)

Solution: We know, during the nuclear fusion reaction two light nuclei combine to give a heavier nucleus and possibly other products.

Therefore, one example of nuclear fusion reaction is:

$$_{3}^{1}H + _{2}^{1}H = _{4}^{2}He + _{1}^{0}n$$

Here, one tritium and one deuterium nuclei fuse together to give one helium nuclei and one neutron, releasing lot of energy in the process.

19. Correct option: (A)

Solution: We know, in an atomic bomb, the energy is released due to chain reaction of neutrons and $_{92}U^{235}$

20. Correct option: (B)

Solution: The principle of controlled chain reaction is used in atomic energy reactor.

21. Correct option: (B)

Solution: The binding energy per nucleon is maximum in the case of $_{26}^{56}Fe$. It is the most stable nuclei.

22. Correct option: (D)

Solution: In nuclear fission the percentage of mass converted into energy is about 0. 1%.

23. Correct option: (A)

Solution: We know, the atoms of same element having different masses but same chemical properties, are called isotope. Isotopes have same number of protons and electrons, and thus same chemical properties, but differ in the number of neutrons and hence in mass.

24. Correct option: (C)

Solution: Out of protons, neutrons, electrons and alpha-particle, neutrons are the most unstable. They have very short half-life.

25. Correct option: (D)

Solution: We know, isotones are atoms of two different elements with the same number of neutrons.

Now, number of neutrons in ${}_{34}Se^{74}$ and ${}_{31}Ca^{71}$ are both 40. Therefore, ${}_{34}Se^{74}, {}_{31}Ca^{71}$ are isotones.

26. Correct option: (C)

Solution: We know, higher the binding energy per nucleon, greater the stability of the nucleus.

Given, the curve of binding energy per nucleon as a function of atomic mass number has a sharp peak for helium nucleus.

This implies that helium is very stable.

Radioactivity

27. Correct option: (A)

Solution: We know, charged particles are deflected by magnetic field.

Further, neutron is a neutral particle, electrons carry the charge $-e$ and protons carry the charge e and alpha particle carries two unit of positive charge, $2e$.

Given, the particles emitted by radioactive decay are deflected by magnetic field. This suggests that the particle have net charge.
Therefore, the particles will be electrons, protons and α particles.

28. Correct option: (B)

Solution: We know, alpha decay, positron emission and electron capture are all modes of radioactive decay.

Fusion is not a mode of radioactive decay. During the nuclear fusion reaction two light nuclei combine to give a heavier nucleus and possibly other products.

29. Correct option: (D)

Solution: We know, when a nucleus emits an alpha particle, it mass number goes down by 4 and atomic number goes down by 2. Further, when a nucleus emits a beta particle, its atomic number increases by 1 and mass number remains unchanged.

Given, atomic mass number of the element thorium is 232 and its atomic number is 90. The end product of this radioactive element is an isotope of lead (atomic mass 208 and atomic number 82).

Now, mass number of Thorium changes from 232 to 208, the number of alpha particles emitted is therefore:

$$\frac{232-208}{4}=\frac{24}{4}$$
$$\frac{232-208}{4}=6$$

Now, atomic number of Thorium changes from 90 to 82, a change of 8 units. 6 alpha particles released should have decreased the atomic number from 90 to 90-6(2)=78, but the end product has 82 atomic number, so 4 beta particles should have been emitted.

30. Correct option: (None of the given option is correct

Solution: We know, when a nucleus emits an alpha particle, it mass number goes down by 4 and atomic number goes down by 2. Further, when a nucleus emits a beta particle, its atomic number increases by 1 and mass number remains unchanged.

Given, three α-particles and one β-particle decaying takes place in series from an isotope ${}_{88}Se^{238}$.

Now, since three α-particles are emitted from ${}_{88}Se^{238}$, the mass number should decrease by 4(3)=12. Further, since one β-particle is also emitted, the atomic number should decrease by 2(3)-1=5. Therefore, the final atomic number is 88-5=83 and the final mass number is 238-12=226.

Therefore, if three α-particles and one β-particle decaying takes place in series from an isotope ${}_{88}Se^{238}$, then finally the isotope obtained will be ${}_{83}Se^{226}$

31. Correct option: (D)

Solution: We know, when photon is emitted in a radioactive decay, neither the atomic number nor the mass number changes.

32. Correct option: (C)

Solution: Co^{60} is used for the treatment of cancer as a source of radiation to destroys cancer cells.

33. Correct option: (D)

Solution: In beta radiation, one of the neutron in the nucleus decay to give one proton and one electron.

Therefore, the electron emitted in beta radiation originates from decay of a neutron in a nucleus.

34. Correct option: (B)

Solution: We know, when a nucleus emits an alpha particle, it mass number goes down by 4 and atomic number goes down by 2.

Also, elements in the periodic table are categorized according to their atomic number.

Therefore, when a radioactive substance emits an alpha-particle, its position in the periodic table is lowered by two places.

35. Correct option: (B)

Solution: When a nucleus emits a beta particle, its atomic number increases by 1 and mass number remains unchanged.

Atoms of different nuclei having same mass number are known as isobars.

Therefore, if a radioactive nucleus emits a beta particle, then the parent and daughter nuclei are isobars.

36. Correct option: (A)

Solution: Carbon dating is best suited for determining the age of fossils if their age in years is of the order of 10^4 years.

37. Correct option: (B)

Solution: When a nucleus emits a beta particle, its atomic number increases by 1 and mass number remains unchanged.

Now, after two successive $\beta-$decay in the nucleus ${}^{115}_{48}Se$, the atomic number increases by 2, and mass number will remain same. Therefore, it will give ${}^{115}_{50}Sn$.

38. Correct option: (C)

Solution: We know, when a nucleus emits an alpha particle, it mass number goes down by 4 and atomic number goes down by 2.

Given, a radioactive nucleus ${}_{92}X^{235}$ decays to ${}_{91}X^{231}$.

We see, the atomic number changes by one unit, and the mass number changes by 4 units. This suggests, one proton and four neutrons are emitted.

39. Correct option: (B)

Solution: We know, the half life $t_{1/2}$ is related to the mean life t_{avg} by:

$$t_{1/2} = \ln 2\, t_{avg}$$

Given, the half-life of a radioactive sample is 10 hours.

Therefore, its mean life is:

$$t_{avg} = \frac{t_{1/2}}{\ln 2}$$
$$t_{avg} = \frac{10}{\ln 2}$$
$$t_{avg} = 14.4$$

Therefore, if the half-life of a radioactive sample is 10 hours, its mean life is 14. 4 hours.

40. Correct option: (C)

Solution: We know, the number N of nuclei of a radioactive sample at time t is given by:

$N = N_0 e^{-\lambda t}$

In mean life $1/\lambda$ of a radioactive sample, e^{-1} fraction of the radioactive substance remains undecayed. Therefore, about $(1-e^{-1})$ or roughly 2/3 of the substance disintegrates.

41. Correct option: (B)

Solution: We know, when a nucleus emits an alpha particle, it mass number goes down by 4 and atomic number goes down by 2. Further, when a nucleus emits a beta negative particle, its atomic number increases by 1 and mass number remains unchanged. And, when a nucleus emits a beta positive particle, its atomic number decreases by 1 and mass number remains unchanged.

Given, a radioactive decay chain starts from ${}_{93}Np^{237}$ and produces ${}_{90}Th^{229}$.

We see, the atomic number decreases by 3 units, and the mass number decreases by 8 units.

Therefore, the emitted particles can be two $\alpha-$particles, and one β^- particle.

42. Correct options: (A), B), C)

Solution: A radioactive substance emits electromagnetic radiation, charged particles like alpha, beta particles, and uncharged particles like gamma particles.

43. Correct option: (C)

Solution: An element used for radioactive carbon dating for more than 5600 years is $C-14$

44. Correct option: (A)

Solution: We know, the activity of a radioactive sample is defined as the number of disintegrations of nuclei per second. Radioactive decay is a random process. The activity of a radioactive sample is independent of physical parameter.

45. Correct option: (C)

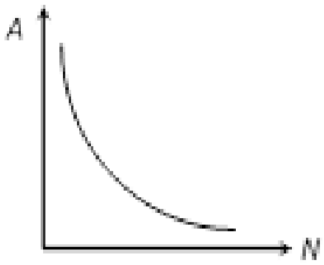
Solution: We know, when a nucleus emits a beta negative particle, its atomic number increases by 1 and mass number remains unchanged.

Therefore, ${}_Z X^A \rightarrow {}_{Z+1}Y^A + {}_{-1}e^0 + \overline{p}$ represents beta-decay.

46. Correct option: (D)

Solution: We know, the activity A of a radioactive sample at time t is given by:

$A = \lambda N_0 e^{-\lambda t}$



where λ is the decay constant, and N_0 is the initial number of nuclei in the sample.

As time increases, the activity decreases exponentially.

Therefore, the curve between the activity A of a radioactive sample and the number of active atoms N is:

47. Correct option: (D)

Solution: We know, the number N of nuclei of a radioactive sample at time t is given by:

$N = N_0 e^{-\lambda t}$

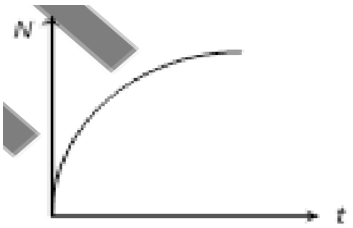
where λ is the decay constant, and N_0 is the initial number of nuclei in the sample.

Therefore, the number of decayed atoms are $N_0 - N = N_0(1 - e^{-\lambda t})$.

As time increases, more and more nuclei decay.

Hence, the graph between number of decayed atoms of a radioactive element and time t

Is:



48. Correct option: (D)

Solution: We know, when photon is emitted in a radioactive decay, neither the atomic number nor the mass number changes.

Therefore, in γ ray emission from a nucleus there is no change in the proton number and the neutron number.

LEVEL – II

Atomic Model

1. Correct option: (B)

Solution: We know, the kinetic energy of an electron revolving around a nucleus will be half of its potential energy.

But the potential energy of the electron in the orbit is negative, whereas kinetic energy is always positive.

Therefore, the electrons in the Bohr's orbit have K. E. greater than P. E.

2. Correct option: (B)

Solution: According to classical theory, the proposed path of an electron in Rutherford atom model will be spiralling inwards toward the atom. This is because, according to electromagnetic theory, an accelerated particle continuously emits radiation.

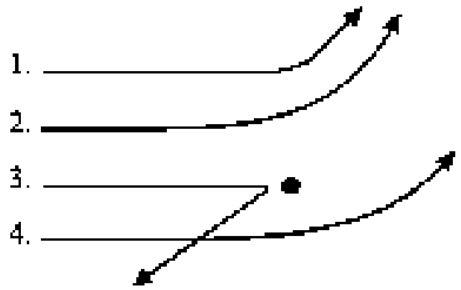
Therefore, the assertion that according to classical theory, the proposed path of an electron in Rutherford atom model will be parabolic is false but the reason according to electromagnetic theory an accelerated particle continuously emits radiation is true.

3. Correct option: (C)

Solution: We know α particle carries two units of positive charge. Nucleus also has positive charge.

We also know like charges repel and opposite charges attract.

Given, the diagram shows the path of four α particles of the same energy being scattered by the nucleus of an atom simultaneously:



Now, since both α particles and nucleus carry the positive charge, α particles should be repelled by the nucleus. We see this is the case for all the alpha particles except 4, which is attracted toward the nucleus. This cannot happen.

Therefore, 4 only is not physically possible.

4. Correct option: (D)

Solution: We know, Bohr's model is based on the postulate that the electrons in stationary orbits around the nucleus do not radiate.

According to classical physics all moving electrons, which accelerates, radiate.

Therefore, assertion is true but reason is false.

Structure Of Nucleus

5. Correct option: (D)

Solution: We know, the energy released in a nuclear reaction equals the difference of the binding energy of the reactants from the products.

Given, the binding energies per nucleon for a deuteron and an α − particle are x_1 and x_2 respectively.

Therefore, energy released in the reaction is:

$$_1H^2 + {}_1H^2 \rightarrow {}_2He^4 + Q$$

$4x_2 - 4x_1$.

6. Correct option: (B)

Solution: The assertion that isobars are the element having same mass number but different atomic number is true. The reason neutrons and protons are present inside nucleus is also true, but is not a correct explanation of the assertion.

7. Correct option: (C)

Solution: We know, the mass R of a nucleus is related to its mass number A by:

$R = a_0 A^{1/3}$

where a_0 is a constant.

Hence, ratio of the radii of the nuclei with mass numbers 8 and 27 would be

$$\frac{R_1}{R_2} = \frac{a_0 A_1^{1/3}}{a_0 A_2^{1/3}}$$
$$\frac{R_1}{R_2} = \frac{a_0 8^{1/3}}{a_0 27^{1/3}}$$
$$\frac{R_1}{R_2} = \frac{2}{3}$$

Therefore, ratio of the radii of the nuclei with mass numbers 8 and 27 would be 2:3.

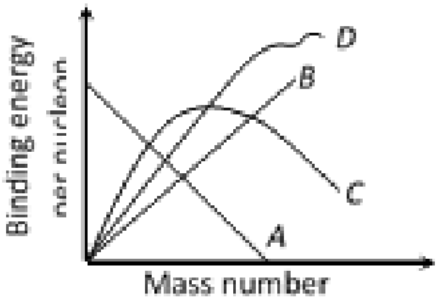
8. Correct option: (D)

Solution: The assertion that the neutrons penetrate matter more readily as compared to protons is true. The reason neutrons are slightly more massive than protons is also true but is not a correct explanation of the assertion. Neutron being neutral particle, has much penetrating power than charged particles like protons, which also undergo electromagnetic interactions in the matter.

9. Correct option: (C)

Solution: Given, binding energy per nucleon plot against the mass number for stable nuclei is shown

in the figure as:



Now, broadly speaking, as mass number increases, the binding energy per nucleon first increases and then decreases. This feature is only captured by the curve C.

Therefore, the correct plot for Binding energy per nucleon against the mass number for stable nuclei is C.

10. Correct option: (C)

Solution: We know, the mass R of a nucleus is related to its mass number A by:

$R = a_0 A^{1/3}$

where a_0 is a constant.

Given, the ratio of radii of nuclei ${}_{13}Al^{27}$ and ${}_{52}X^{A}$ is 3: 5.

Hence, mass number of X is:

$$\frac{R_1}{R_2} = \frac{a_0 A_1^{1/3}}{a_0 A_2^{1/3}}$$

$$\frac{3}{5} = \frac{a_0 27^{1/3}}{a_0 A^{1/3}}$$

$$\frac{27}{125} = \frac{27}{A}$$

$$A = 125$$

The number of neutrons in the nuclei of X are 125-52=73

Therefore, if the ratio of radii of nuclei ${}_{13}Al^{27}$ and ${}_{52}X^{A}$ is 3: 5, then the number of neutrons in the nuclei of X will be 73.

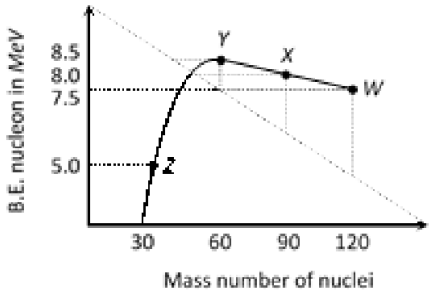
11. Correct option: (C)

Solution: The assertion that a free neutron decays to a proton but a free proton does not decay to a neutron is true.

Neutron has larger rest mass than the proton. It can thus decay to proton and a neutron and an antineutrino. Therefore, the reason neutron has larger rest mass than the proton is a correct explanation of the assertion.

12. Correct option: (C)

Solution: Given, binding energy per nucleon verses mass number curve for nuclei is shown in the figure:



W, X, Y and Z are four nuclei indicated on the curve.

Now, binding energy of X is : $90(8) = 720$ MeV

Binding energy of Y is : $60(8.5) = 510$ MeV

Binding energy of W is : $120(7.5) = 900$ MeV

Binding energy of Z is : $30(5) = 150$ MeV

Now, energy released in the reaction W-> 2Y:

$2(510) - 900 = 120$

Since the energy released is positive, therefore this reaction will release energy.

Therefore, the process that would release energy is W-> 2Y.

Radioactivity

13. Correct option: (B)

Solution: We know, when a nucleus emits an alpha particle, it mass number goes down by 4 and atomic number goes down by 2. Further, when a nucleus emits a beta particle, its atomic number increases by 1 and mass number remains unchanged. Also, when gamma particle is emitted in a radioactive decay, neither the atomic number nor the mass number changes.

Given, a radioactive nucleus undergoes a series of decay according to the scheme:

$$A \xrightarrow{\alpha} A_1 \xrightarrow{\beta} A_2 \xrightarrow{\alpha} A_3 \xrightarrow{\gamma} A_4$$

Also, the mass number and atomic number of A are 180 and 72 respectively.

Now, since there are two alpha particles emitted, one beta particle and one gamma particle, the mass number changes to 180-2(4)=172 and the atomic number changes to 72-2(2)+1=69.

Therefore, if a radioactive nucleus undergoes a series of decay according to the scheme and the mass number and atomic number of A are 180 and 72 respectively, then these number for A_4 are 172 and 69 respectively.

14. Correct option: (A)

Solution: We know, when a nucleus emits an alpha particle, it mass number goes down by 4 and atomic number goes down by 2. Further, when a nucleus emits a beta particle, its atomic number increases by 1 and mass number remains unchanged.

$$^{238}_{92}U \xrightarrow{\alpha} X \xrightarrow{\beta-} {}^{A}_{Z}Y$$

Given, a disintegration series:

Now, there is one alpha particle and one beta particle emitted.

The mass number thus changes to 238-4=234 and the atomic number changes to 92-2+1=91.

The values of Z and A respectively will be 91,234.

15. Correct option: (D)

Solution: We know, from Einstein's theory of relativity, the mass of a substance increases, as its speed increases.

The assertion that the mass of beta-particles when they are emitted is higher than the mass of electrons obtained by other means is true.

The reason given beta-particle and electron, both are similar particles is also a correct statement but is not the explanation of the assertion.

The reason that the mass of beta-particles when they are emitted is higher than the mass of electrons obtained by other means is that the beta-particles are emitted with very high

speed from the nucleus, comparable to that of speed of light. From Einstein' s theory of relativity, they thus have larger mass than electrons emitted by other means.

16. Correct option: (B)

Solution: We know, in a β^+ decay, a proton in the nuclei decays to neutron and a beta particle and a neutrino, decreasing the atomic number by one unit.

Given, a nuclear reaction:

$$^{11}_{6}C \longrightarrow {}^{11}_{5}B + \beta^+ + X$$

This is an example of a β^+ decay. Therefore, X stands for neutrino.

17. Correct option: (A)

Solution: We know, the half life $t_{1/2}$ is related to the mean life t_{avg} by:

$$t_{1/2} = \ln 2\, t_{avg}$$

Or,

$$t_{1/2} = 0.693 t_{avg}$$

18. Correct option: (A)

Solution: We know, when gamma particle is emitted in a radioactive decay, neither the atomic number nor the mass number changes.

In all options except a), the atomic number changes.

Therefore, the following processes represents a gamma-decay

$$^{A}X_{Z} \rightarrow {}^{A}X_{Z} + f$$

19. Correct option: (B)

Solution: We know, the amount N of a substance left after time t is:

$$N = N_0 \left(\frac{1}{2}\right)^{t/t_{1/2}}$$

where $t_{1/2}$ is the half-life period, and N_0 is the original amount of the substance.

Given, N_0 is the original mass of the substance of half-life period $t_{1/2} = 5$.

Then, the amount of substance left after 15 years is:

$$N = N_0 \left(\frac{1}{2}\right)^{t/t_{1/2}}$$

$$N = N_0 \left(\frac{1}{2}\right)^{15/5}$$

$$N = \frac{N_0}{8}$$

Therefore, if N_0 is the original mass of the substance of half-life period $t_{1/2} = 5$, then the amount of substance left after 15 years is $\frac{N_0}{8}$.

20. Correct option: (B)

Solution: We know, the amount N of a substance left after time t is:

$$N = N_0\left(\frac{1}{2}\right)^{t/t_{1/2}}$$

where $t_{1/2}$ is the half-life period, and N_0 is the original amount of the substance.

Given, half-life of a radioactive element is 10 days.

Now, the time during which quantity remains $1/10$ of initial mass will be:

$$\frac{N_0}{10} = N_0\left(\frac{1}{2}\right)^{t/t_{1/2}}$$
$$\frac{1}{10} = \left(\frac{1}{2}\right)^{t/10}$$
$$ln(10) = \frac{t}{10} ln\, 2$$
$$t = \frac{10\, ln\, 10}{ln\, 2}$$
$$t = 33.22$$

Therefore, if half-life of a radioactive element is 10 days, the time during which quantity remains $1/10$ of initial mass will be roughly 33 days.

21. Correct option: (B)

Solution: We know, the number N of nuclei of a radioactive sample at time t is given by:

$$N = N_0 e^{-\lambda t}$$

Hence, all radioactive elements decay exponentially with time.

Half lifetime of a radioactive element is time required for one half of the radioactive atoms to disintegrate.

Age of earth can be determined with the help of radioactive dating. As the uranium in rocks decays, it emits subatomic particles and turns into lead at a constant rate. Measuring the uranium-to-lead ratios in the oldest rocks on Earth can gave scientists an estimate for the age of the planet.

We know, the half life $t_{1/2}$ is related to the mean life t_{avg} by:

$$t_{1/2} = 0.693 t_{avg}$$

Hence, half lifetime of a radioactive element is about 70% of its average life period.

22. Correct option: (A)

Solution: We know, the amount N of a substance left after time t is:

$$N = N_0\left(\frac{1}{2}\right)^{t/t_{1/2}}$$

Given, C^{14} has half-life 5700 years.

Therefore, the amount at the end of 11400 years is:

$$N = N_0\left(\frac{1}{2}\right)^{t/t_{1/2}}$$
$$N = N_0\left(\frac{1}{2}\right)^{11400/5700}$$
$$N = N_0\left(\frac{1}{2}\right)^{2}$$
$$N = 0.25N_0$$

Therefore, if C^{14} has half-life 5700 years, then the amount at the end of 11400 years is 0. 25 of original amount.

23. Correct option: (A)

Solution: We know, the amount N of a substance left after time t is:

$$N = N_0\left(\frac{1}{2}\right)^{t/t_{1/2}}$$

Also, the decay constant λ is related to the half-life $t_{1/2}$ by:

$$t_{1/2} = \frac{\ln 2}{\lambda}$$

Given, the half-life of a radioactive substance is 40 years.

Now, the time it take to reduce to one fourth of its original amount is:

$$\frac{N_0}{4} = N_0\left(\frac{1}{2}\right)^{t/t_{1/2}}$$
$$\left(\frac{1}{2}\right)^{2} = \left(\frac{1}{2}\right)^{t/40}$$
$$2 = \frac{t}{40}$$
$$t = 80$$

And the value of decay constant is:

$$\lambda = \frac{\ln 2}{t_{1/2}}$$
$$\lambda = \frac{0.693}{40}$$
$$\lambda = 0.017$$

Therefore, if the half-life of a radioactive substance is 40 years, then time it take to reduce to one fourth of its original amount and the value of decay constant is 80-year, 0. 0173-year respectively.

24. Correct option: (D)

Solution: The assertion that amongst alpha, beta and gamma rays, alpha-particle has maximum penetrating power is false.

Amongst alpha, beta and gamma rays, gamma-rays has the maximum penetration power.

The reason the alpha particle is heavier than beta and gamma rays is also true but is not a correct explanation of the assertion.

The reason for gamma-rays maximum penetration power is their very high speed and zero mass character.

25. Correct option: (C)

Solution: We know, the half life $t_{1/2}$ is related to the mean life t_{avg} by:

$$t_{1/2} = \ln 2\, t_{avg}$$

Also, the decay constant λ is related to the half-life $t_{1/2}$ by:

$$t_{1/2} = \frac{\ln 2}{\lambda}$$

Given, the half-life period of a radioactive element X is same as the mean lifetime of another radioactive element Y.

Let the decay constant of radioactive element X be λ.

Then, we have:

$$t_{1/2,X} = t_{avg,Y}$$
$$\frac{\ln 2}{\lambda} = \frac{t_{1/2,Y}}{\ln 2}$$
$$\frac{\ln 2}{\lambda} = \frac{\ln 2 / \lambda_Y}{\ln 2}$$
$$\lambda_Y = \frac{\lambda}{\ln 2}$$

Now, higher the decay constant λ, greater the decay rate.

We see, the decay constant of X is greater than the decay constant of Y.

Therefore, if the half-life period of a radioactive element X is same as the mean lifetime of another radioactive element Y, X will decay at a faster rate than Y.

26. Correct option: (C)

Solution: The assertion that all the radioactive elements are ultimately converted into lead is false. Radioactive elements may be converted to other stable elements other than lead.

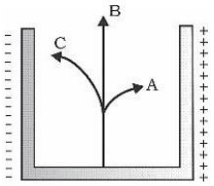
The reason all the elements above lead are unstable is false as well. Iron for example has the most stable nuclei and it comes before lead.

Therefore, both assertion and reason are false and the reason is not the correct explanation of the assertion.

27. Correct option: (C)

Solution: We know, α particle carries two units of positive charge, β particle carries one unit of negative charge, and γ particles carries no charge.

Further, like charges repel and opposite charges attract.



Given, a radioactive source is kept in a uniform electric field and α, β, γ particles are

emitting, respectively as shown in the figure on the left.

Since A is attracted towards the positive plate, A should carry negative charge. Therefore,

A should be beam of β particles.

Since C is attracted towards the negative plate, C should carry positive charge. Therefore, C should be beam of α particles.

Since B passes undeflected, B should carry no charge. Therefore, B should be beam of γ particles.

Therefore, if a radioactive source is kept in a uniform electric field and α, β, γ particles are emitting, respectively C,A,B.

28. Correct option: (D)

Solution: We know, the amount N of a substance left after time t is:

$$N = N_0 \left(\frac{1}{2}\right)^{t/t_{1/2}}$$

Given, the half-life period of a radioactive element is 4 days and 200g of the substance was

present initially.

Therefore, the amount (in g) will be left after 16 days:

$$N = N_0 \left(\frac{1}{2}\right)^{t/t_{1/2}}$$

$$N = 200\left(\frac{1}{2}\right)^{16/4}$$

$$N = 200\left(\frac{1}{2}\right)^{4}$$

$$N = 12.5$$

Therefore, the half-life period of a radioactive element is 4 days and 200g of the substance was present initially, then the amount (in g) will be left after 16 days is 12. 5.

29. Correct match: B)

Solution: We know, J. J. Thomson gave plum pudding model of the atom.

E. Rutherford gave nuclear model of the atom.

Franck-Hertz discovery of EM radiation suggested the existence of discrete energy levels in an atom

Neils Bohr provides explanation of the hydrogen spectrum.

Therefore, the correct match is: (A) -> (2), (B) -> (1), (C) -> (4), (D)-> (3)

30. Correct option: (B)

Solution: We know, the mass-energy equivalence is $E = mc^2$

Nuclear fission is generally possible for nuclei with higher atomic number.

Nuclear fusion is generally possible for nuclei with low atomic number.

β decay essentially proceeds by weak reaction nuclear forces.

Therefore, the correct match is: (A) -> (2), (B) -> (3), (C) -> (4), (D)-> (1)